MADFU

Merchant Onboarding Wireframe System

Complete, annotated low-fidelity desktop wireframes, journey decision maps and business rules for Registered Business (CR) and Freelancer onboarding.

VERSION

v1.0 · 15 October 2026
Generated from the implemented prototype (single source of truth)

INTERACTIVE PROTOTYPE

https://wearemishkat.github.io/madfu-onboarding-wireframes/
https://github.com/WeAreMishkat/madfu-onboarding-wireframes

STATUS

Wireframe prototype — not a production application. Synthetic data only. All integrations simulated. Provisional items are listed in section 16.

Contents

3. Scope and design principles 3

4. Master journey overview 4

5. Entry, OTP and eligibility 5

6. Registered Business journey map 13

7. Registered Business wireframes 15

8. Freelancer journey map 18

9. Freelancer wireframes 20

10. Manual-review journeys 23

11. Contract and submission outcomes 25

12. Validation and error-state examples 32

13. Business decision tables 56

14. UX/UI handover notes 61

15. Scenario coverage (QA) 65

16. Assumptions and pending decisions 66

Screen index

S00	Resume application	6
S01	Start application (shared entry)	7
S01-X	Existing account found	8
S01-F	Existing freelancer — add another certificate (future flow)	9
S02	Mobile OTP verification	10
S02-L	Mobile verification locked	11
S03	Business category	12
S04	Business verification (CR lookup)	15
S05	Authorized person selection	16
S06	Nafath identity verification	17
S07	Freelancing certificate selection	21
S07-M	Manual certificate entry	22
S08	Business details (shared)	26
S09	Merchant agreement (contract)	27
S09-D	Agreement not accepted	28
O-A	Outcome A — Application submitted	29
O-B	Outcome B — Application received (manual verification)	30
O-C	Outcome C — Interest received (sales lead)	31

Journey maps

Map A — Master overview 4

Map B — Entry, duplicate checks, OTP and revenue routing 5

Map C1 — Registered Business: category, CR lookup, authorized person 13

Map C2 — Registered Business: Nafath attempts 14

Map D1 — Freelancer: category and Nafath 18

Map D2 — Freelancer: certificates and reconciliation 19

Map E — Manual-review routes 23

Map F — Business details, contract and outcomes 25

Scope and design principles

Purpose
Show Product, Engineering, UX/UI, Risk, Compliance and Sales exactly how a merchant moves through onboarding — every branch, decision and fallback — as a working journey rather than disconnected screens.

In scope
- Shared entry, duplicate detection, OTP, revenue routing, category screening.
- Registered Business: government CR lookup, authorized-person selection, Nafath (2 shared attempts).
- Freelancer: Nafath on the entry National ID, certificate retrieval, selection, reconciliation, manual entry.
- Shared business details with IBAN ownership verification, contract eligibility, simulated acceptance, three neutral outcomes.
- Resume, refresh-safe persistence, invalidation of dependent verifications.

Out of scope
Merchant activation and post-onboarding journeys; sign-in to existing accounts; the add-certificate flow (marked FUTURE); real integrations, real contract text and legally binding signatures.

Design principles
- Desktop-first (1440 px), black and white, low fidelity, English only, no marketing imagery.
- Consistent structure: header, progress, single-column form, Back bottom-left, primary action bottom-right.
- Avoid unnecessary third-party cost: category screening happens before CR lookup and Nafath.
- Internal statuses are separate from applicant-facing messages.
- Every screen page shows: screen ID, wireframe, entry condition, next and alternative destinations, backend behaviour, business rules.

How this PDF was produced
Rendered by Playwright from the prototype's own print route: the wireframes are the live React screens driven into each state by the journey engine; tables are generated from the rule configuration. Screen IDs match the prototype and gallery.

G-UX · GLOBAL UX/UI HANDOVER NOTE

All screens require complete UX/UI validation, error-state, empty-state, loading-state, retry, and edge-case designs. The absence of a specific state from the wireframes does not mean it is excluded from the final product requirements. The UX/UI team is responsible for designing comprehensive error and validation experiences for every field, step, and integration.

Referenced from every screen annotation. Full handover in section 14.

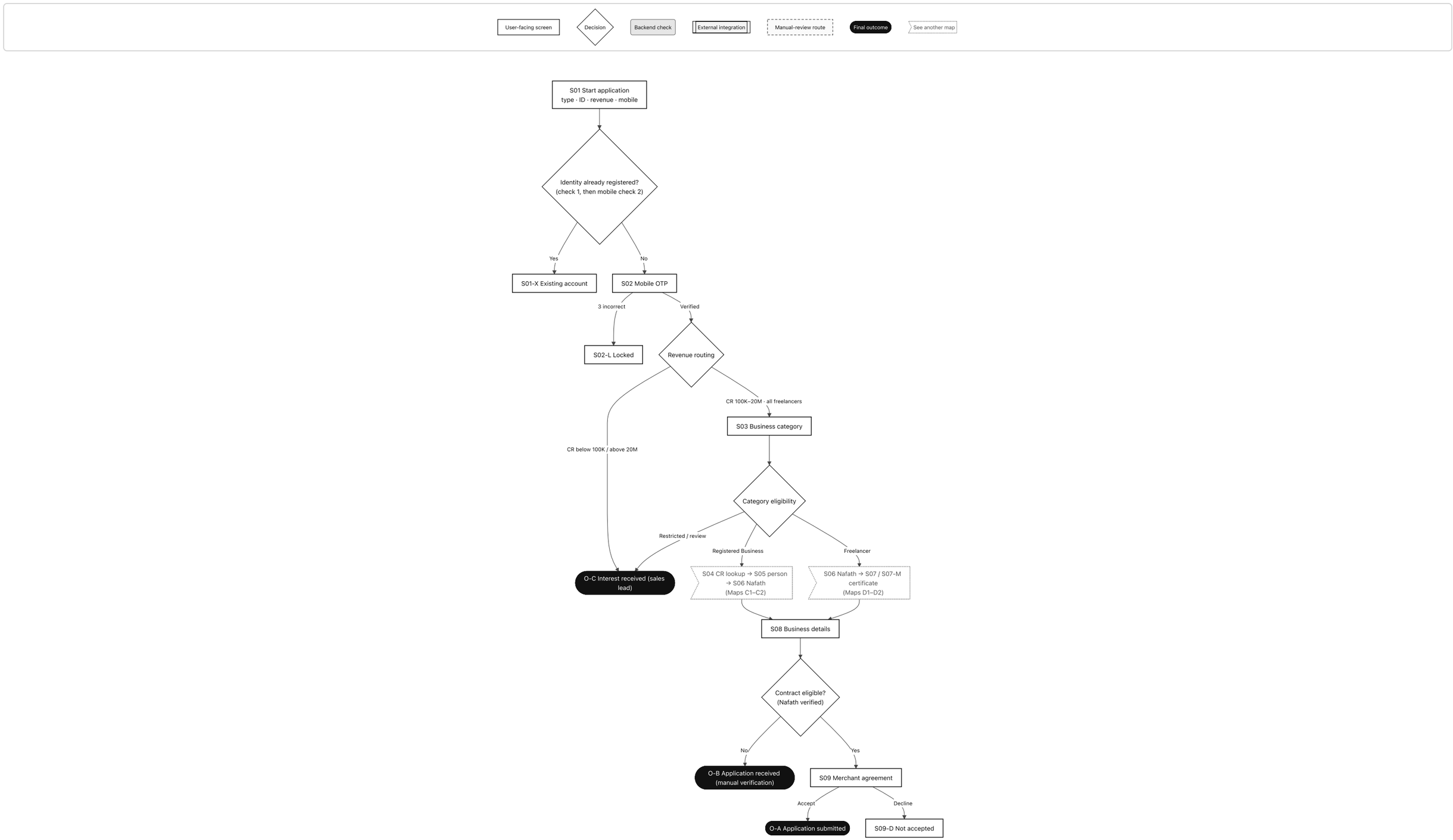
Applicant journeys at a glance

	Registered Business	Freelancer
Identification at entry	CR number	National ID
Revenue routing	Only SAR 100K–20M self-onboards	All bands self-onboard
Government data	CR lookup + related parties	Certificates (after Nafath)
Nafath subject	Selected authorized individual	Entry National ID
IBAN ownership match	Registered business	Individual National ID
Without Nafath	Details → manual verification	Manual certificate → details → manual verification

Screen count
18 screens (14 journey screens, 3 outcomes, 1 future-flow placeholder) · 67 documented states · 8 journey maps.

Map A — Master overview

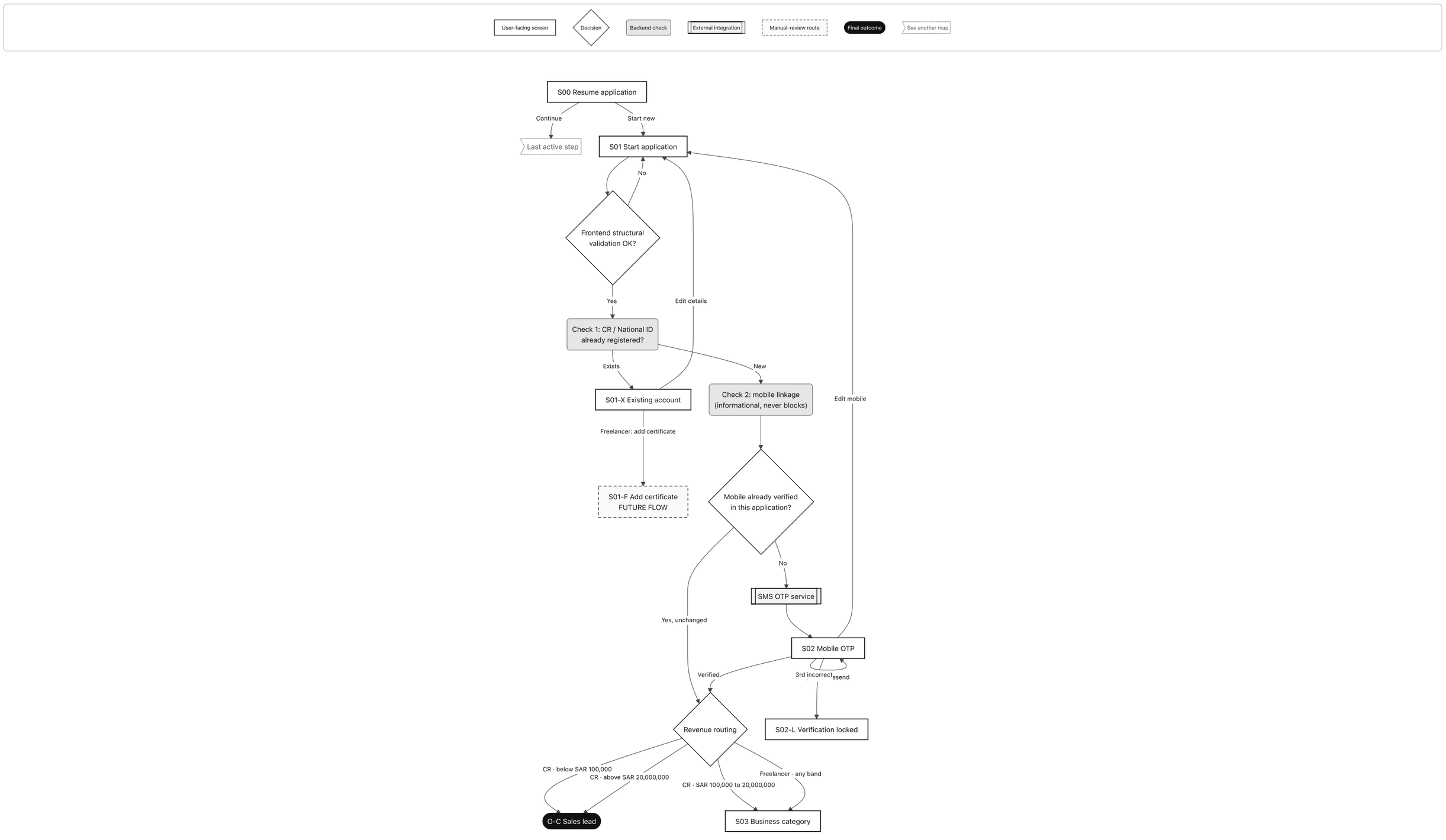
End-to-end routing for both applicant types, with continuation references to detailed maps.



Screens in this map: S01 Start application (shared entry) (p. 7) · S01-X Existing account found (p. 8) · S02 Mobile OTP verification (p. 10) · S02-L Mobile verification locked (p. 11) · S03 Business category (p. 12) · S08 Business details (shared) (p. 26) · S09 Merchant agreement (contract) (p. 27) · S09-D Agreement not accepted (p. 28) · O-A Outcome A — Application submitted (p. 29) · O-B Outcome B — Application received (manual verification) (p. 30) · O-C Outcome C — Interest received (sales lead) (p. 31)

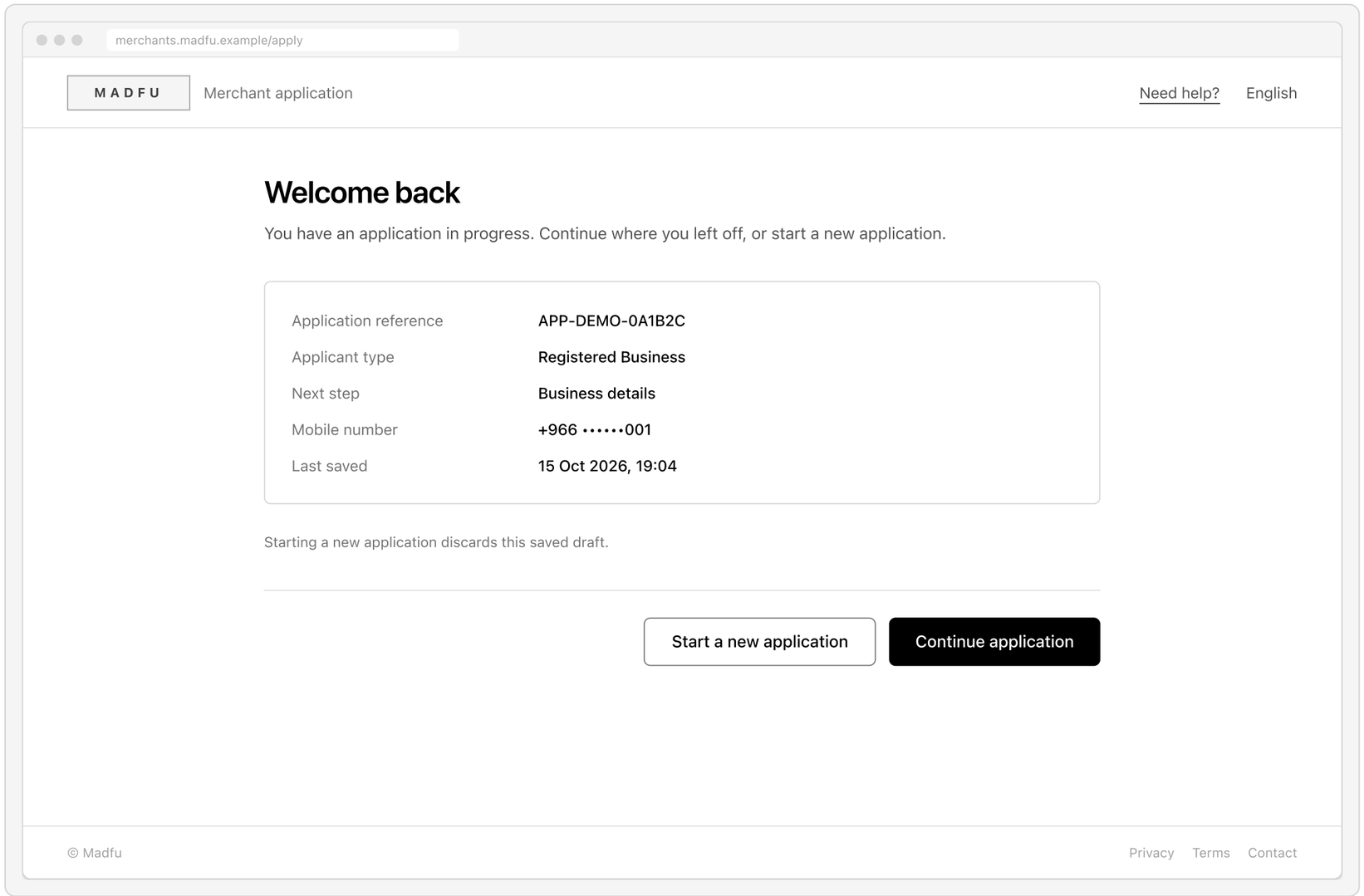
Map B — Entry, duplicate checks, OTP and revenue routing

Shared steps up to eligibility routing. Identity check always runs before the mobile check.



Screens in this map: S00 Resume application (p. 6) · S01 Start application (shared entry) (p. 7) · S01-X Existing account found (p. 8) · S01-F Existing freelancer — add another certificate (future flow) (p. 9) · S02 Mobile OTP verification (p. 10) · S02-L Mobile verification locked (p. 11) · S03 Business category (p. 12) · O-C Outcome C — Interest received (sales lead) (p. 31)

S00 Resume application · Saved application found



S00 Resume application

Both

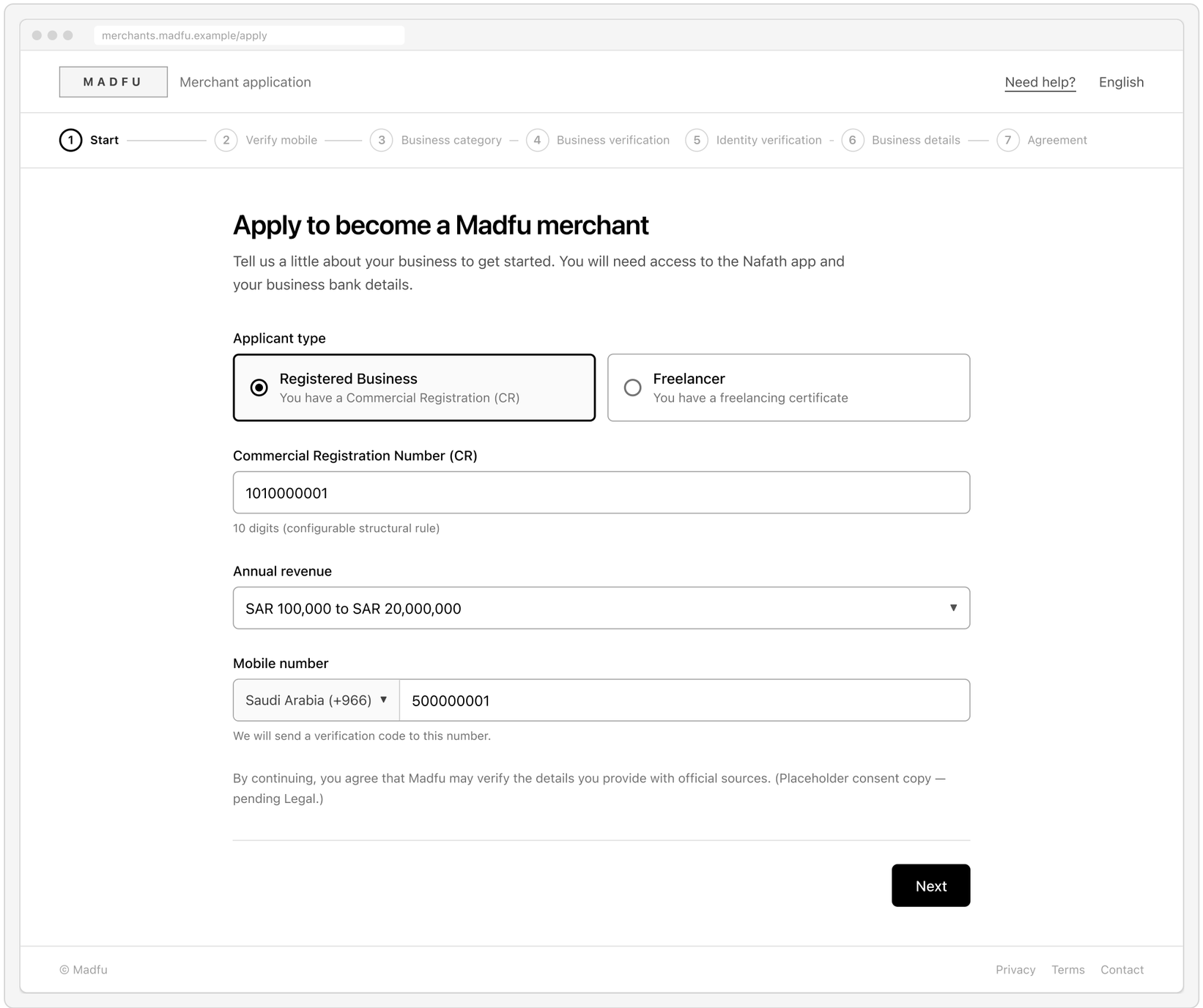
PURPOSE	Let an applicant return to an incomplete application without losing progress.
WHO SEES IT	Any applicant with a saved, unsubmitted application.
ENTRY CONDITION	Saved application exists, not submitted; applicant opens the onboarding entry point.
REQUIRED INPUTS	• None (choice only)
PRIMARY ACTION	Continue application
NEXT SCREEN	The last active step (e.g. S08 Business details).
ALTERNATIVE PATHS	• Start a new application → S01 (previous draft discarded in this demo).
BACKEND BEHAVIOUR	• Demo: browser storage. Production: server-side draft keyed by application ID. • Production should re-authenticate with OTP before showing saved data (assumption D-15).
BUSINESS RULES	• Submitted applications cannot be resumed or resubmitted.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map B (p. 5)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S00--resume

S01 Start application (shared entry) · Registered Business — completed



S01 Start application (shared entry) — Both

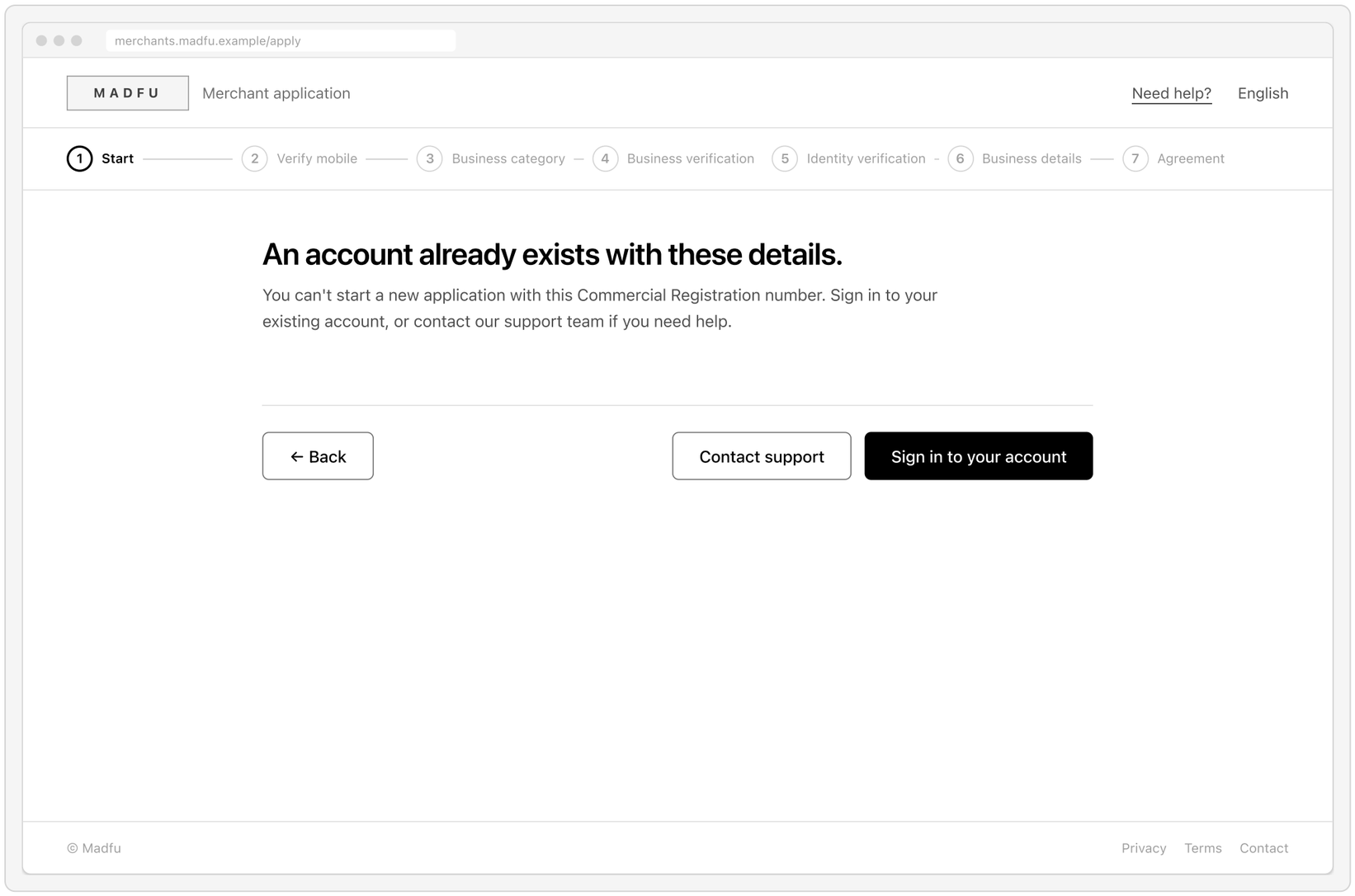
PURPOSE	Capture applicant type, identification, annual revenue and mobile on one page.
WHO SEES IT	Every new applicant (Registered Business and Freelancer).
ENTRY CONDITION	Applicant opens onboarding, or returns via "Edit details" / "Edit mobile number".
REQUIRED INPUTS	• Applicant type (Registered Business / Freelancer) • CR number (business) or National ID (freelancer) — field switches with type • Annual revenue (3 bands) • Mobile number (+966 default)
PRIMARY ACTION	Next
NEXT SCREEN	S02 OTP verification (after backend duplicate checks).
ALTERNATIVE PATHS	• Identity already registered → S01-X Existing account. • Mobile already verified in this application and unchanged → revenue routing (OTP skipped).
BACKEND BEHAVIOUR	• Check 1: identity (CR / National ID) duplicate — runs first. • Check 2: mobile linkage — only if identity is new; informational, never blocks. • Changing type resets incompatible data; changing ID resets identity-dependent verifications; changing mobile resets OTP.
BUSINESS RULES	• All fields mandatory. • Freelancing certificate number is NOT requested here. • Exactly SAR 100,000 falls in the middle band (provisional D-01). • Format rules are configurable structural checks, not government validation.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map A (p. 4) · Map B (p. 5) · Map C1 (p. 13)

Other states: Empty form (p. 32) · Freelancer — National ID field (p. 32) · Validation — required fields missing (p. 33) · Validation — invalid

S01-X Existing account found · Existing CR



S01-X Existing account found — Both

PURPOSE	Stop a duplicate application for an identity that is already registered.
WHO SEES IT	Applicants whose CR or National ID already exists.
ENTRY CONDITION	Duplicate check 1 (identity) returned "already registered".
REQUIRED INPUTS	• None
PRIMARY ACTION	Sign in to your account (outside prototype scope)
NEXT SCREEN	Existing-account sign-in (not designed here).
ALTERNATIVE PATHS	• Contact support. • Edit details → S01. • Freelancer: "Add another certificate" → S01-F (future flow).
BACKEND BEHAVIOUR	• Mobile check is skipped when identity already exists. • No new application record is created.
BUSINESS RULES	• Same CR / National ID can never be registered twice. • Mobile reuse alone is never a reason to show this screen.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

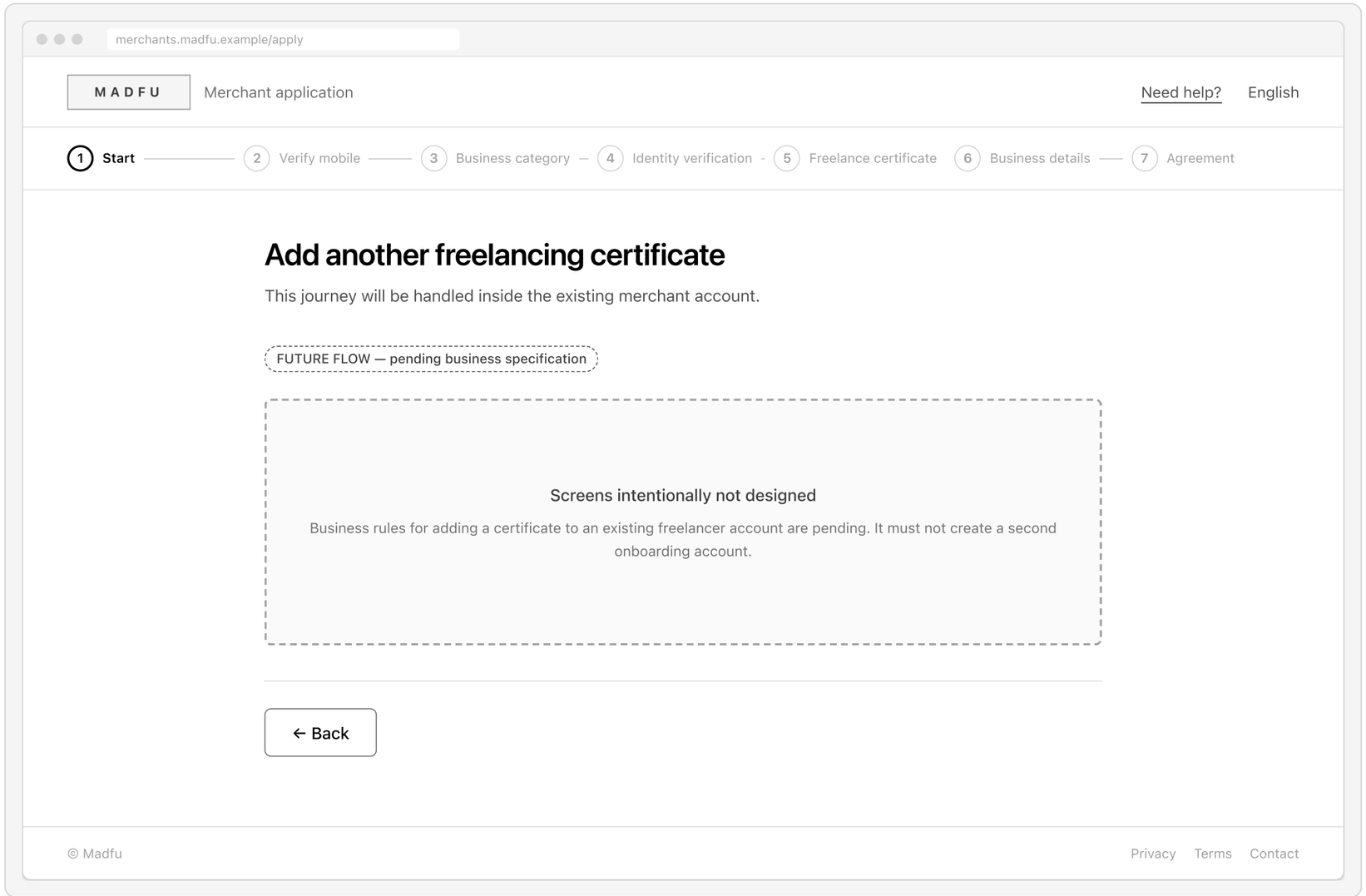
Cross-references

Journey maps: Map A (p. 4) · Map B (p. 5)

Other states: Existing National ID (with future add-certificate branch) (p. 34)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S01-X--cr

S01-F Existing freelancer — add another certificate (future flow) · Future flow placeholder



S01-F Existing freelancer — add another certificate (future flow) — Freelancer

PURPOSE	Placeholder marking a documented future branch.
WHO SEES IT	Existing freelancer who wants to add another freelancing certificate.
ENTRY CONDITION	S01-X for a Freelancer → "Add another freelancing certificate".
REQUIRED INPUTS	• To be specified
PRIMARY ACTION	None (placeholder)
NEXT SCREEN	To be handled inside the existing account — pending business specification.
ALTERNATIVE PATHS	• Back to S01-X.
BACKEND BEHAVIOUR	• Not implemented. Must not create a second onboarding account.
BUSINESS RULES	• FUTURE FLOW — pending business specification (D-07).
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map B (p. 5)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S01-F--future

S02 Mobile OTP verification · Code sent

merchants.madfu.example/apply

MADFU Merchant application

Need help? English

Start — 2 Verify mobile — 3 Business category — 4 Business verification — 5 Identity verification — 6 Business details — 7 Agreement

Verify your mobile number

We sent a 6-digit code to **+966 ••••••001**. Enter it below to continue.

Verification code

• • • • • •

Code expires in 2:00

Didn't receive a code? Resend code

Wrong number? Edit mobile number

DEV DEVELOPMENT SIMULATOR — PROTOTYPE ONLY, NOT PART OF THE PRODUCT UI

Simulates SMS outcomes. No SMS is sent; the code is never shown in the customer UI.

Simulate correct code | Simulate incorrect code | Simulate code expiry

Resend now (skip cooldown)

← Back

Verify

© Madfu

Privacy Terms Contact

S02 Mobile OTP verification — Both

PURPOSE	Verify the applicant controls the mobile number.
WHO SEES IT	All applicants who passed the duplicate checks.
ENTRY CONDITION	S01 submitted, identity not already registered, OTP sent.
REQUIRED INPUTS	• 6-digit SMS code
PRIMARY ACTION	Verify
NEXT SCREEN	Revenue routing → S03 Category (self-onboarding) or O-C (CR sales lead).
ALTERNATIVE PATHS	• Incorrect code → inline error with remaining attempts. • Expired code → request new code (no attempt consumed). • 3 incorrect attempts → S02-L. • Edit mobile number → S01. • Resend code (cooldown + max resends, configurable).
BACKEND BEHAVIOUR	• Simulated SMS only — real code never shown in the customer UI. • Development simulator strip (prototype only) triggers success / failure / expiry.
BUSINESS RULES	• Max 3 incorrect attempts. • Resend cooldown/limit and expiry are provisional (D-04).
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

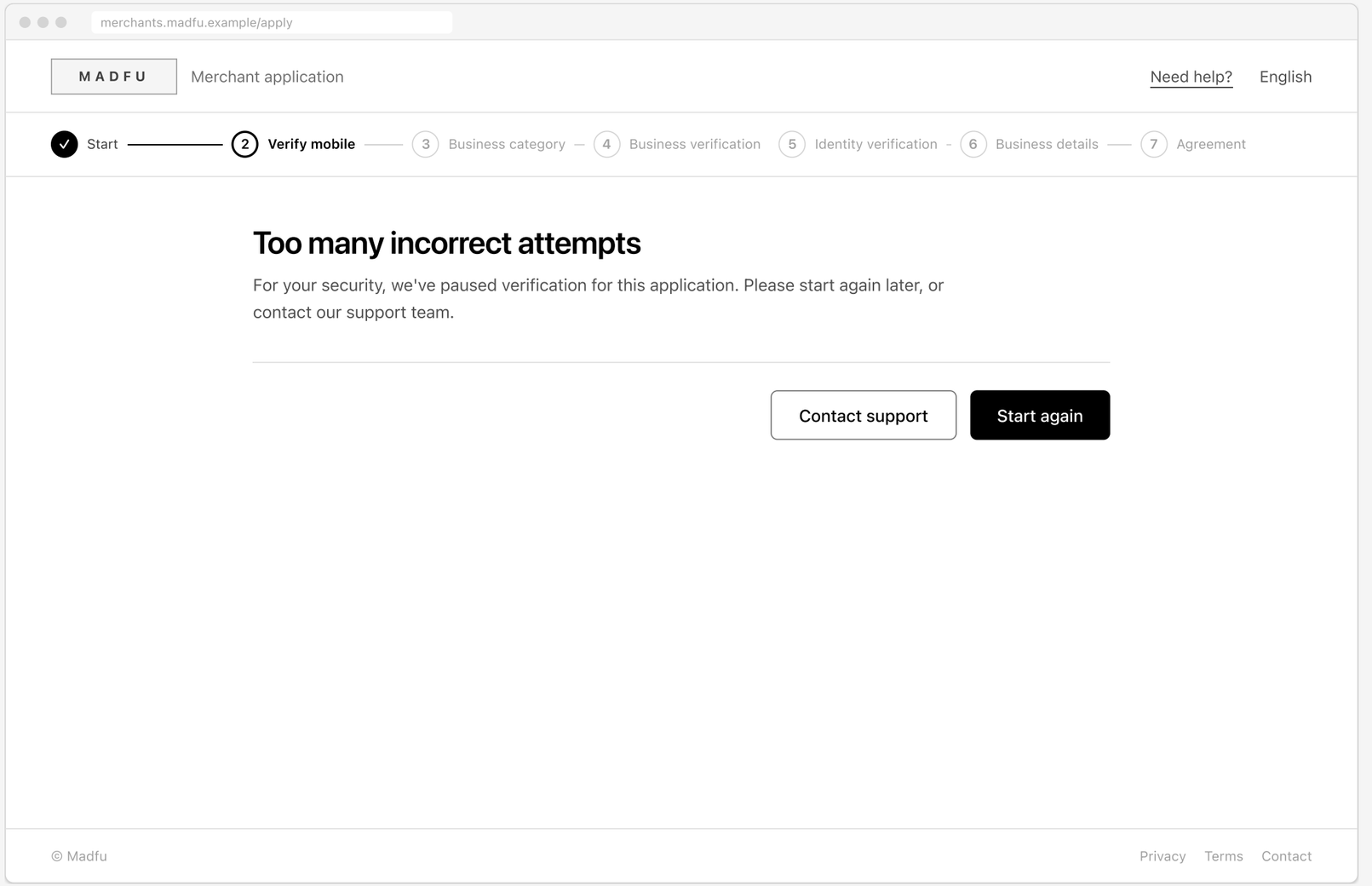
Cross-references

Journey maps: Map A (p. 4) · Map B (p. 5)

Other states: Incorrect code (2 attempts remaining) (p. 35) · Expired code (no attempt consumed) (p. 35) · New code sent (p. 36)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S02--default

S02-L Mobile verification locked · Locked after 3 incorrect codes



S02-L Mobile verification locked — Both

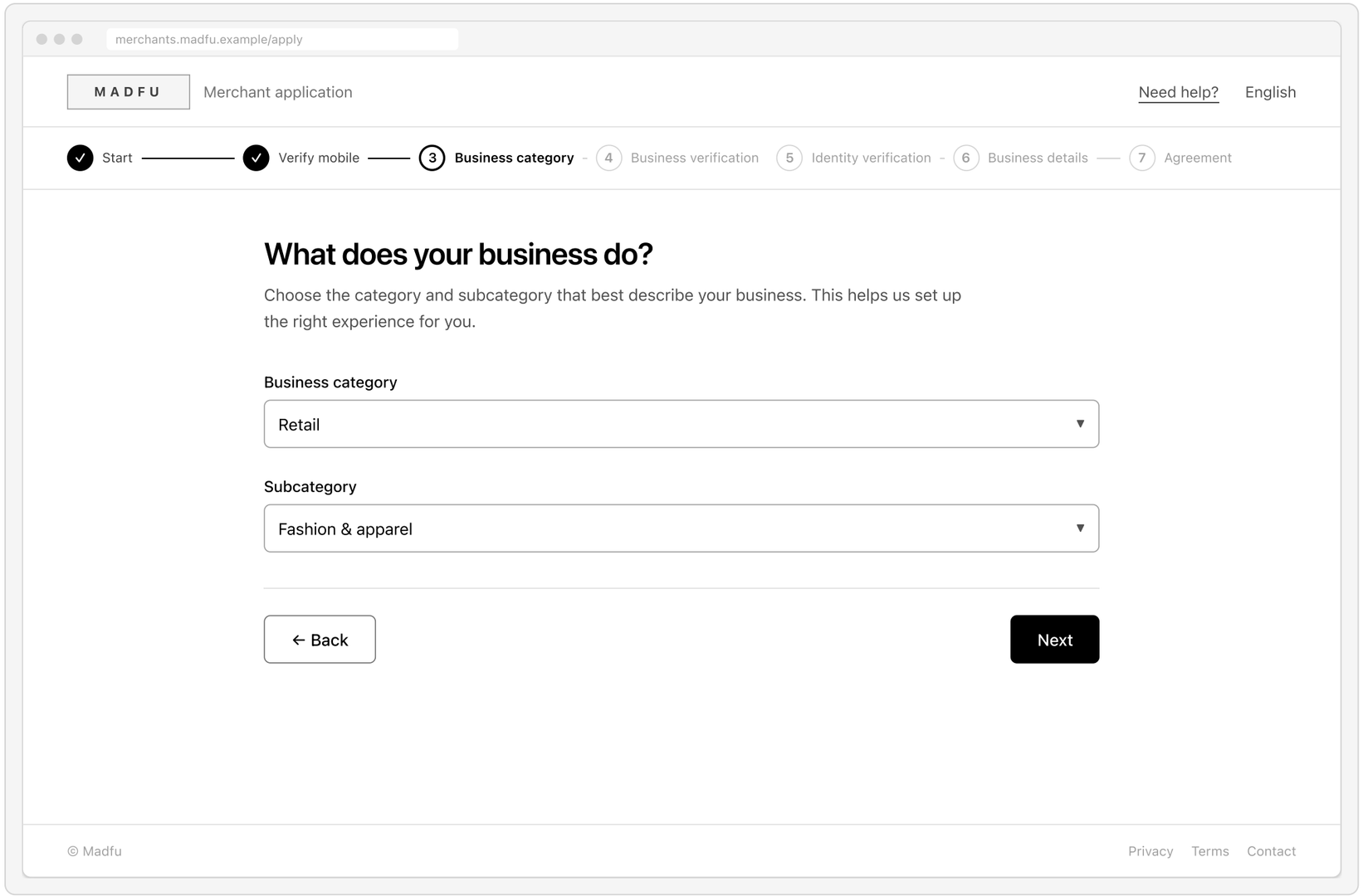
PURPOSE	Stop verification after too many incorrect codes.
WHO SEES IT	Applicants who entered 3 incorrect codes.
ENTRY CONDITION	Third incorrect OTP.
REQUIRED INPUTS	• None
PRIMARY ACTION	Start again
NEXT SCREEN	S01 with a new application (demo).
ALTERNATIVE PATHS	• Contact support.
BACKEND BEHAVIOUR	• Lockout duration / cool-off policy to be defined (D-04).
BUSINESS RULES	• No further OTP attempts on this application.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map A (p. 4) · Map B (p. 5)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S02-L--locked

S03 Business category · Category and subcategory selected



S03 Business category — Both

PURPOSE	Screen eligibility using the Madfu category catalogue BEFORE any paid verification.
WHO SEES IT	Self-onboarding applicants (CR in range, all freelancers).
ENTRY CONDITION	OTP verified and revenue routing = self-onboarding.
REQUIRED INPUTS	• Category • Subcategory
PRIMARY ACTION	Next
NEXT SCREEN	CR → S04 CR verification. Freelancer → S06 Nafath.
ALTERNATIVE PATHS	• Restricted subcategory → O-C (lead, internal CATEGORY_RESTRICTED). • Pending-review subcategory → O-C (configurable, D-13). • Freelancer + CR-only subcategory → continue with internal flag.
BACKEND BEHAVIOUR	• Category API returns category/subcategory IDs, eligibility, permitted applicant types, manual-review flag. • No CR lookup or Nafath call happens for restricted categories (cost avoidance).
BUSINESS RULES	• Madfu taxonomy ≠ government CR activity list. • Freelancer selection is preliminary; reconciled against the certificate later. • Fixture catalogue — not the real restriction list.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

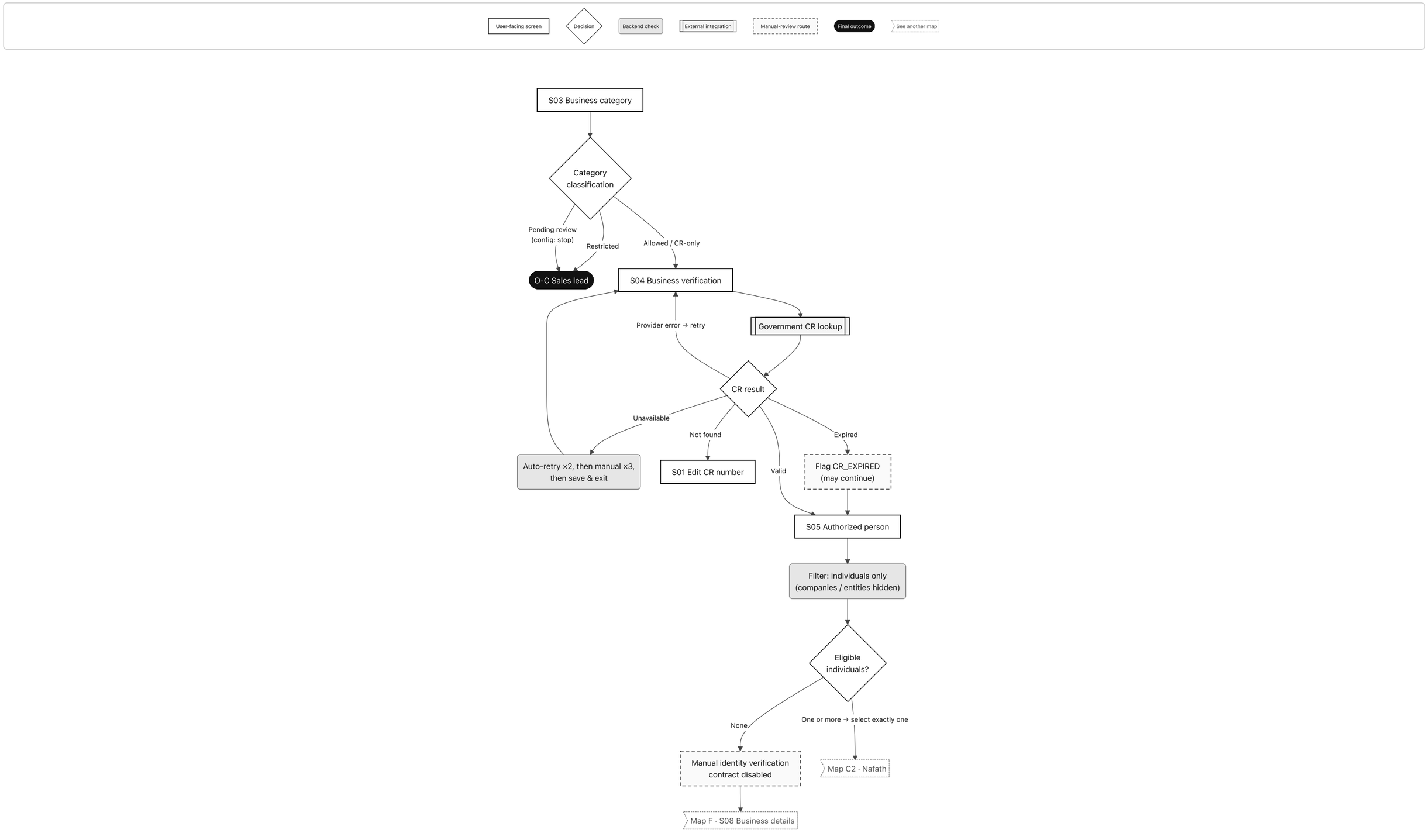
Journey maps: Map A (p. 4) · Map B (p. 5) · Map C1 (p. 13) · Map D1 (p. 18)

Other states: No selection (p. 36) · Validation — nothing selected (p. 37)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S03--selected

Map C1 — Registered Business: category, CR lookup, authorized person

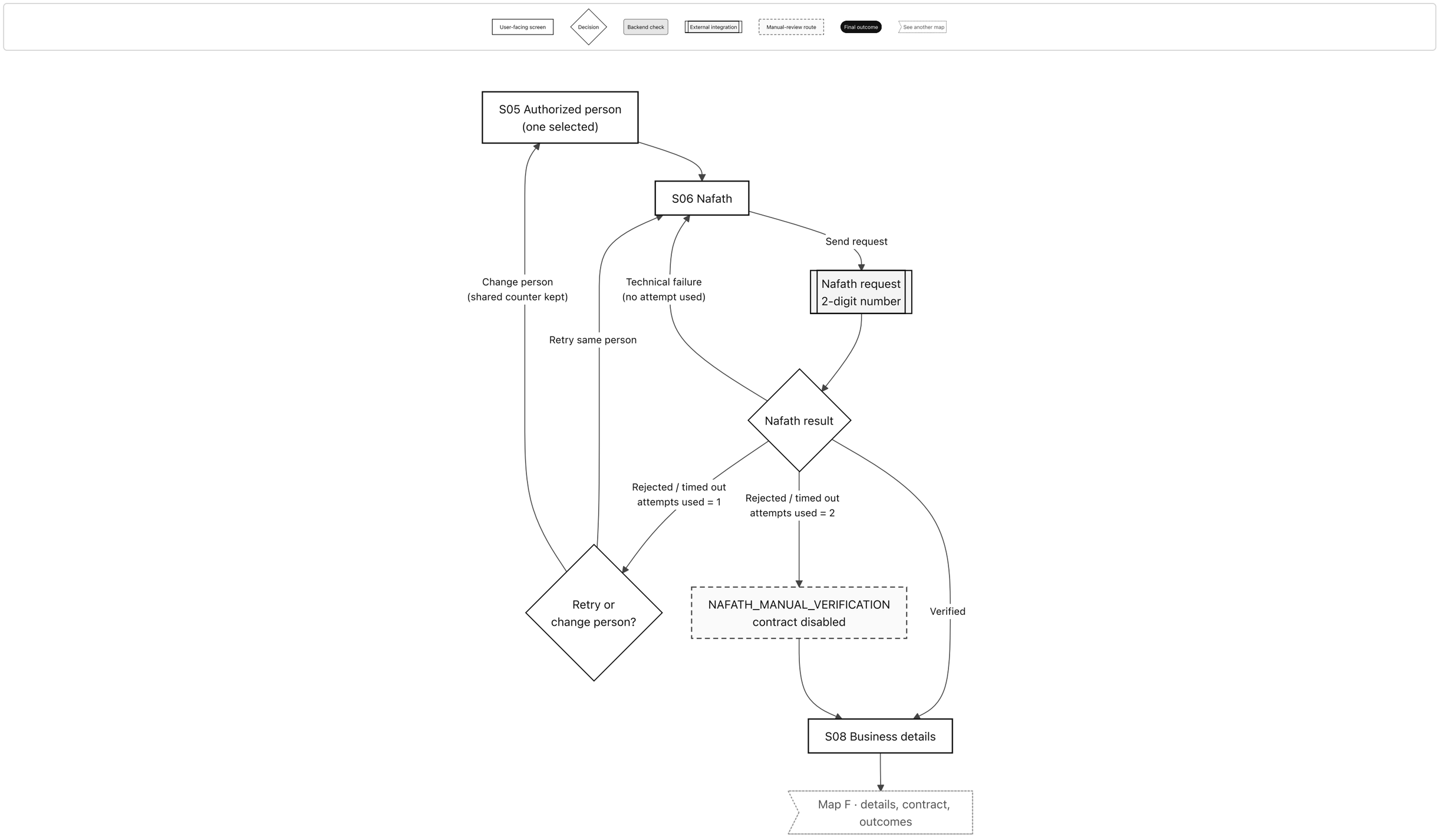
Category screening before paid calls, government CR verification outcomes, individuals-only person selection.



Screens in this map: S03 Business category (p. 12) · S04 Business verification (CR lookup) (p. 15) · S05 Authorized person selection (p. 16) · S01 Start application (shared entry) (p. 7) · O-C Outcome C — Interest received (sales lead) (p. 31)

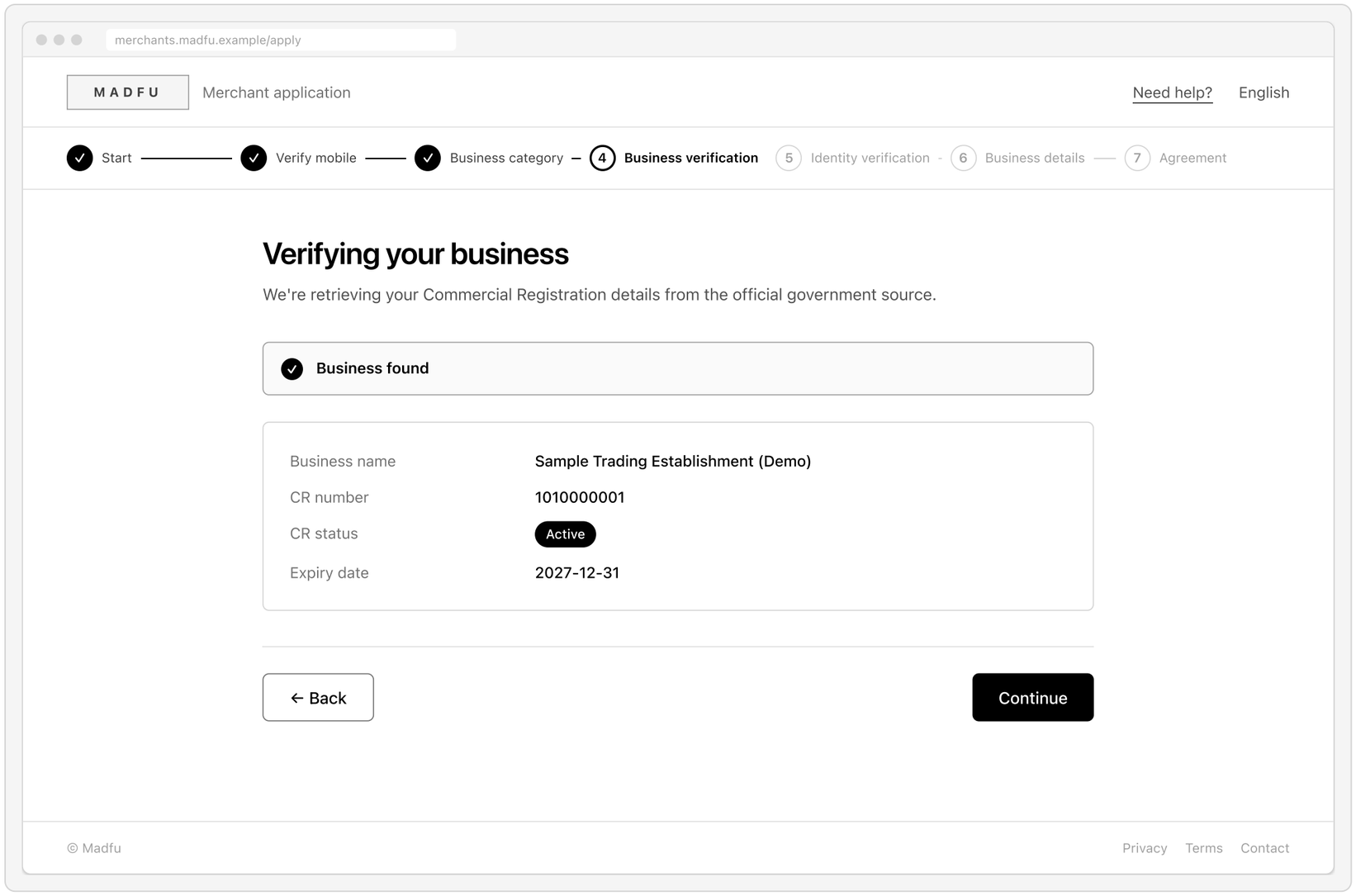
Map C2 — Registered Business: Nafath attempts

Two Nafath attempts per application, shared across authorized individuals.



Screens in this map: S05 Authorized person selection (p. 16) · S06 Nafath identity verification (p. 17) · S08 Business details (shared) (p. 26)

S04 Business verification (CR lookup) · Valid CR



S04 Business verification (CR lookup) — Registered Business

PURPOSE	Retrieve CR data from the government source and show the result.
WHO SEES IT	Registered Business applicants after category screening.
ENTRY CONDITION	Category decision = continue.
REQUIRED INPUTS	• None (automatic lookup)
PRIMARY ACTION	Continue
NEXT SCREEN	S05 Authorized person selection.
ALTERNATIVE PATHS	• Expired CR → notice, may continue (internal CR_EXPIRED). • Not found → Edit CR number → S01. • Unavailable → auto-retries, then "Try again" (bounded), then save-and-exit. • Provider error → generic error + retry.
BACKEND BEHAVIOUR	• Bounded automatic retries (2) then manual retries (3) — provisional D-08. • Each call recorded in the cost ledger. • Never fabricate authorized-person data when the API fails.
BUSINESS RULES	• Invalid / non-existent CR cannot proceed.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

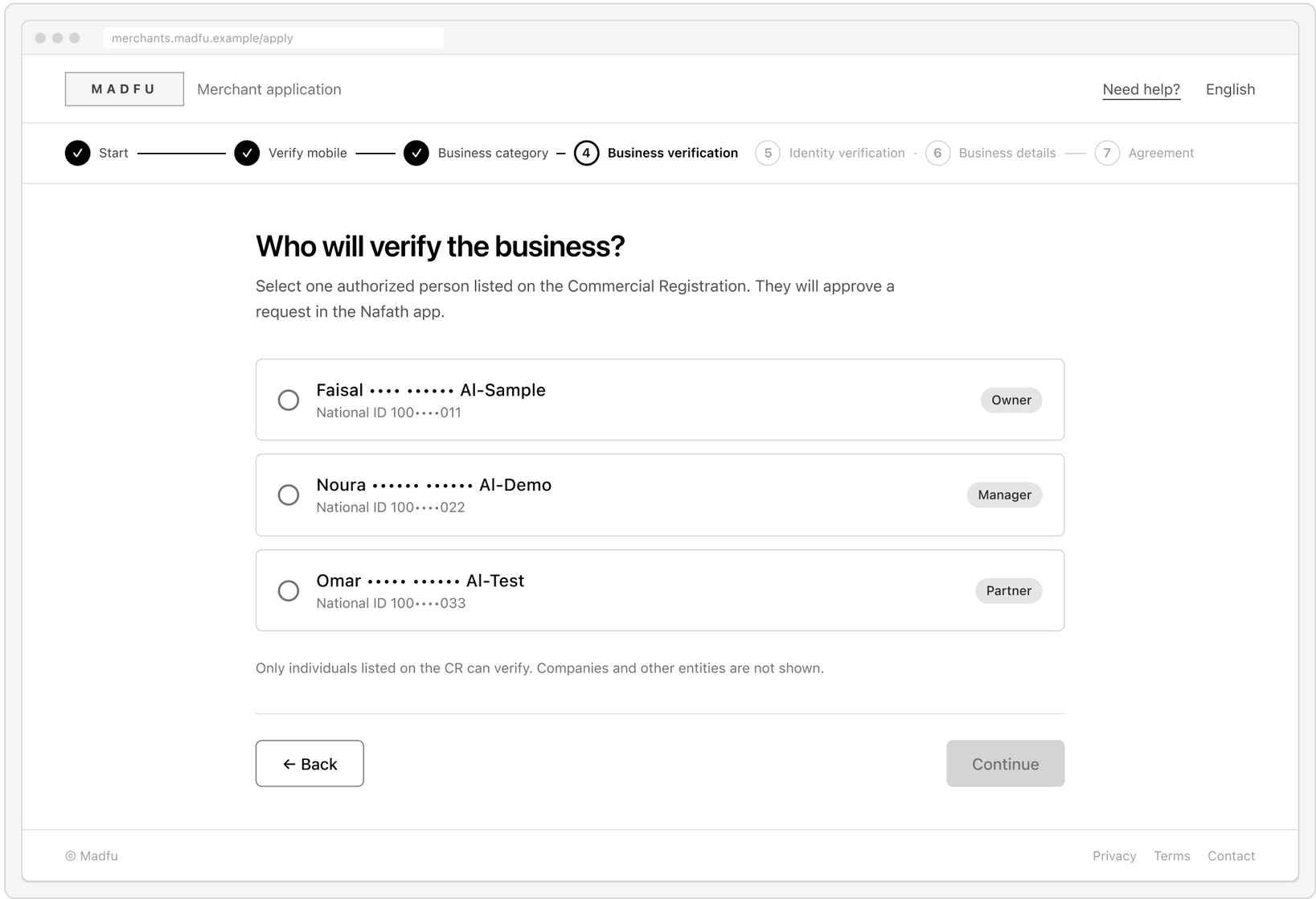
Cross-references

Journey maps: Map C1 (p. 13)

Other states: Loading — retrieving CR (p. 37) · Expired CR (may continue, flagged) (p. 38) · CR not found (p. 38) · Service unavailable after automatic retries (p. 39) · Unexpected provider error (p. 39) · Retry limit reached — progress saved (p. 40)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S04--valid

S05 Authorized person selection · Multiple individuals — none selected



S05 Authorized person selection — Registered Business

PURPOSE	Choose exactly one individual on the CR to verify with Nafath.
WHO SEES IT	Registered Business applicants with CR data.
ENTRY CONDITION	S04 result valid or expired.
REQUIRED INPUTS	• Single selection of one individual
PRIMARY ACTION	Continue (disabled until a selection is made)
NEXT SCREEN	S06 Nafath verification.
ALTERNATIVE PATHS	• No eligible individual → continue to S08 via manual-review path (contract disabled). • Related-party data unavailable → retry. • Back → S04.
BACKEND BEHAVIOUR	• Companies and other entities are filtered out — individuals only. • Selected person's National ID initiates Nafath.
BUSINESS RULES	• Single-select only. • Name: first + last visible, 2nd/3rd masked. ID: first 3 + last 3 visible. • Changing the person never resets the Nafath attempt counter.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

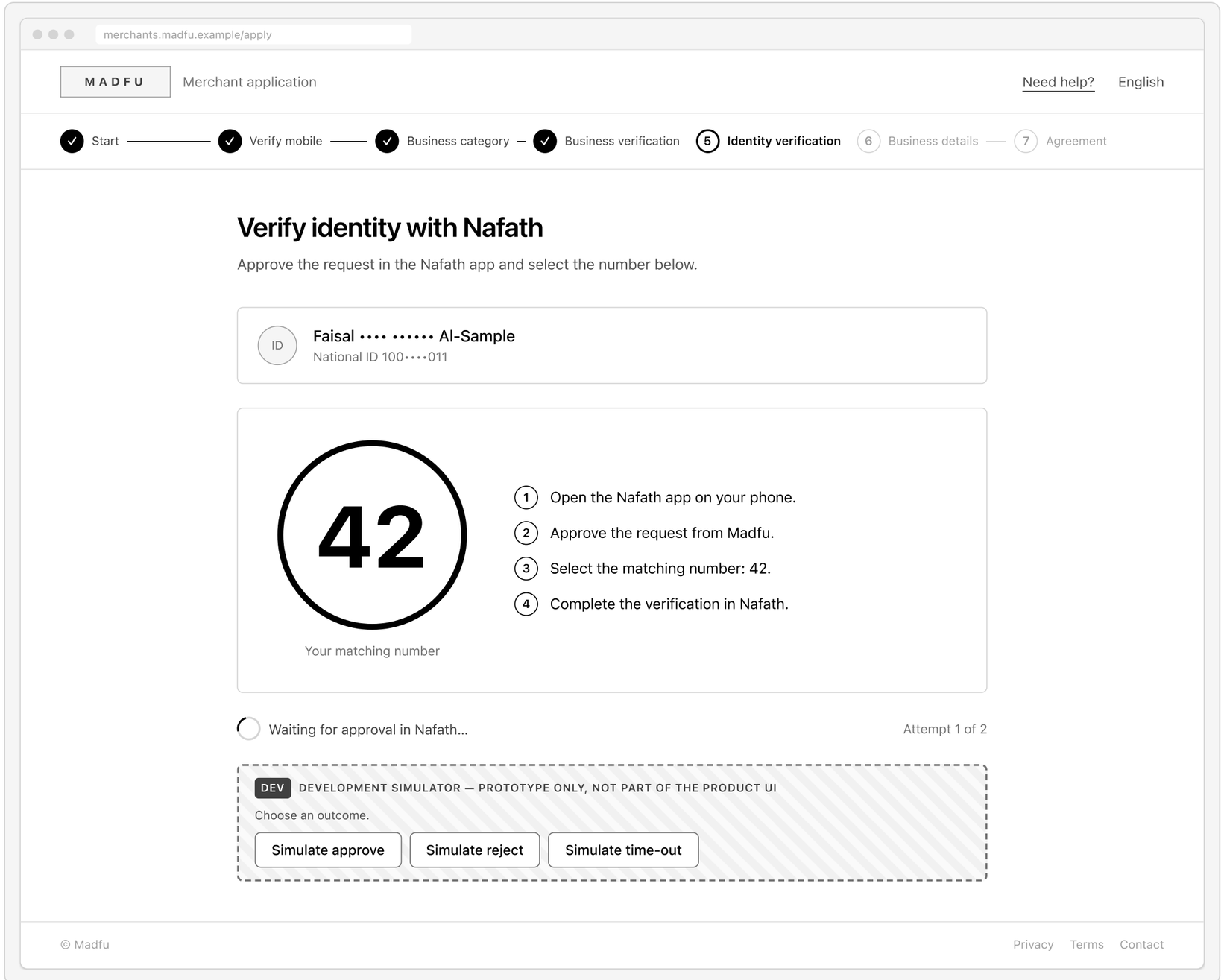
Cross-references

Journey maps: Map C1 (p. 13) · Map C2 (p. 14)

Other states: One individual selected (p. 40) · Single individual (pre-selected) (p. 41) · No eligible individual (entities only) (p. 41) · Related-party data unavailable (p. 42) · Changing person after a failed attempt (counter preserved) (p. 42)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S05--multiple

S06 Nafath identity verification · Waiting — large matching number



S06 Nafath identity verification — Both

PURPOSE	Verify identity via a Nafath request with a two-digit matching number.
WHO SEES IT	CR: the selected authorized person. Freelancer: the National ID holder from S01.
ENTRY CONDITION	CR: person selected (S05). Freelancer: category decision = continue.
REQUIRED INPUTS	• None in Madfu — the user approves in the Nafath app
PRIMARY ACTION	Send verification request → (wait) → Continue
NEXT SCREEN	Verified: CR → S08; Freelancer → S07 certificate selection.
ALTERNATIVE PATHS	• 1st failure: Try again, or (CR) choose another person. • 2nd failure: CR → S08 manual path; Freelancer → S07-M manual certificate. • Technical failure: retry without consuming an attempt (provisional D-11).
BACKEND BEHAVIOUR	• Max 2 attempts per application — shared across individuals. • Outcomes: verified, rejected, timed out, technical failure. • Each request recorded in the cost ledger.
BUSINESS RULES	• Unverified Nafath → contract signing disabled.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

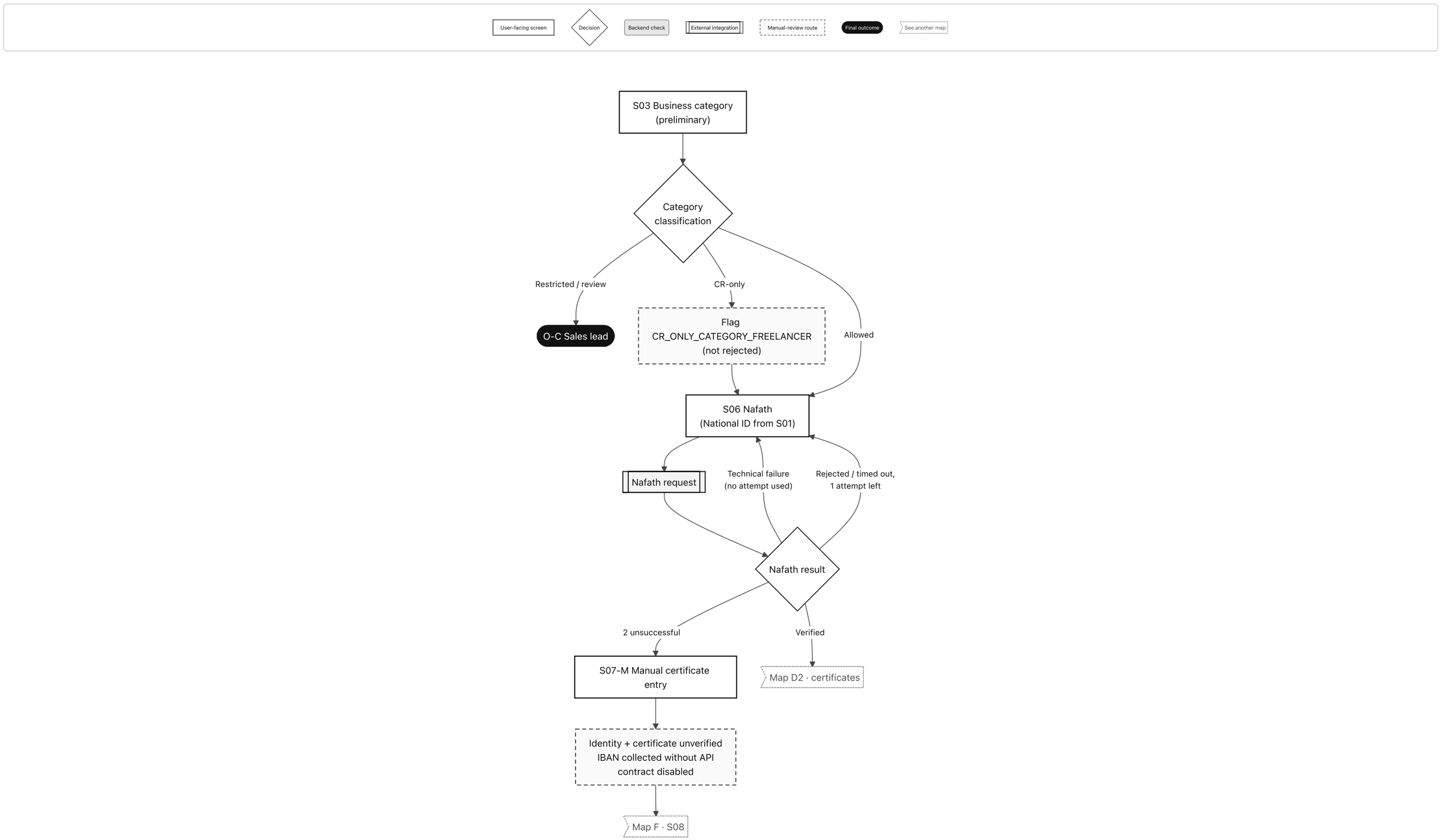
Journey maps: Map C2 (p. 14) · Map D1 (p. 18) · Map D2 (p. 19)

Other states: Ready to send (CR, selected person) (p. 43) · Rejected — 1 attempt remaining (CR) (p. 43) · Timed out — 1 attempt remaining (Freelancer) (p. 44) · Technical failure — no attempt consumed (p. 44) · Verified (p. 45) · Two unsuccessful attempts — manual path (p. 45) · Waiting — Freelancer (National ID from entry) (p. 46)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S06--waiting

Map D1 — Freelancer: category and Nafath

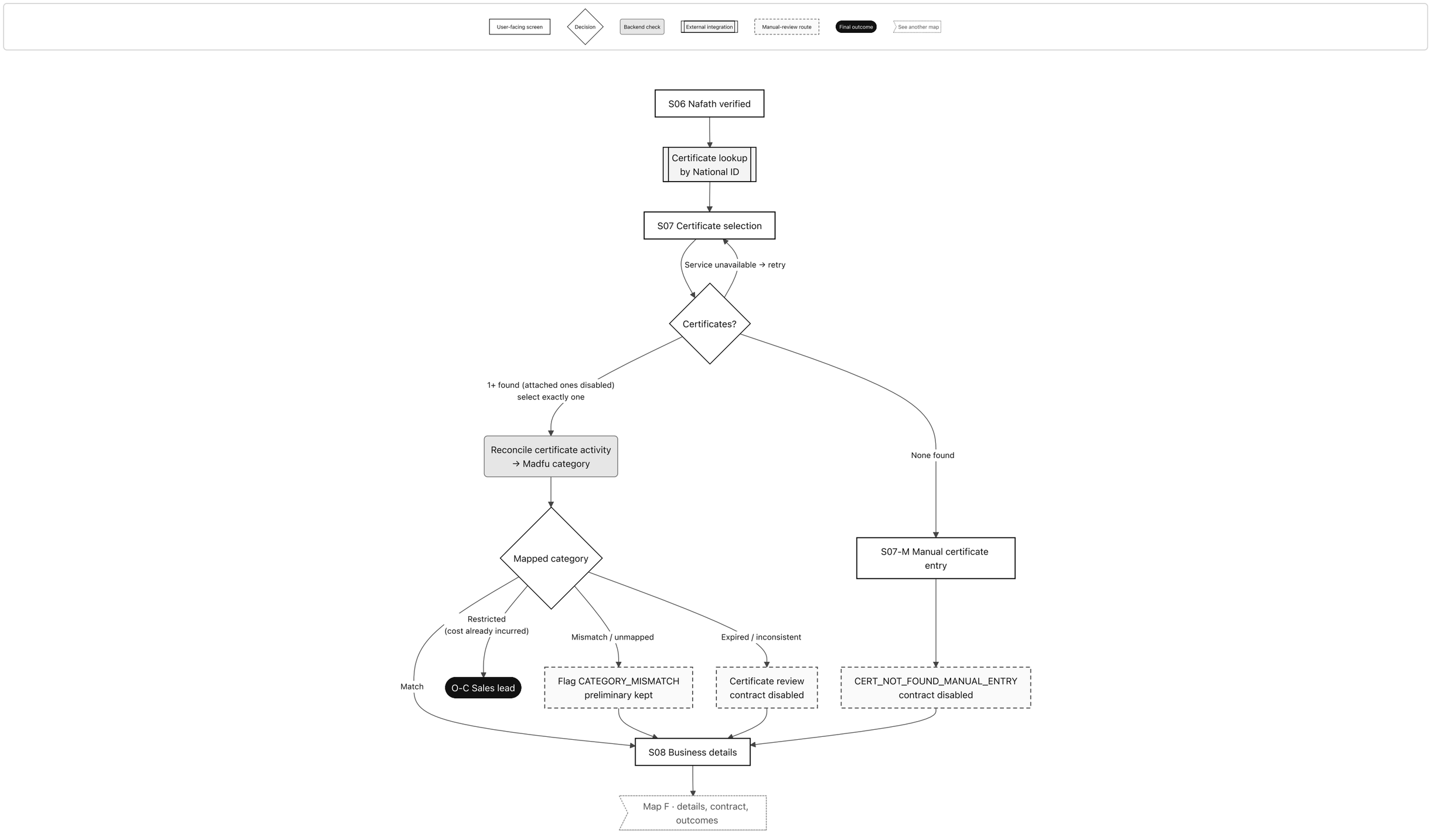
Preliminary category screening and Nafath on the National ID entered at the start.



Screens in this map: S03 Business category (p. 12) · S06 Nafath identity verification (p. 17) · S07-M Manual certificate entry (p. 22) · O-C Outcome C — Interest received (sales lead) (p. 31)

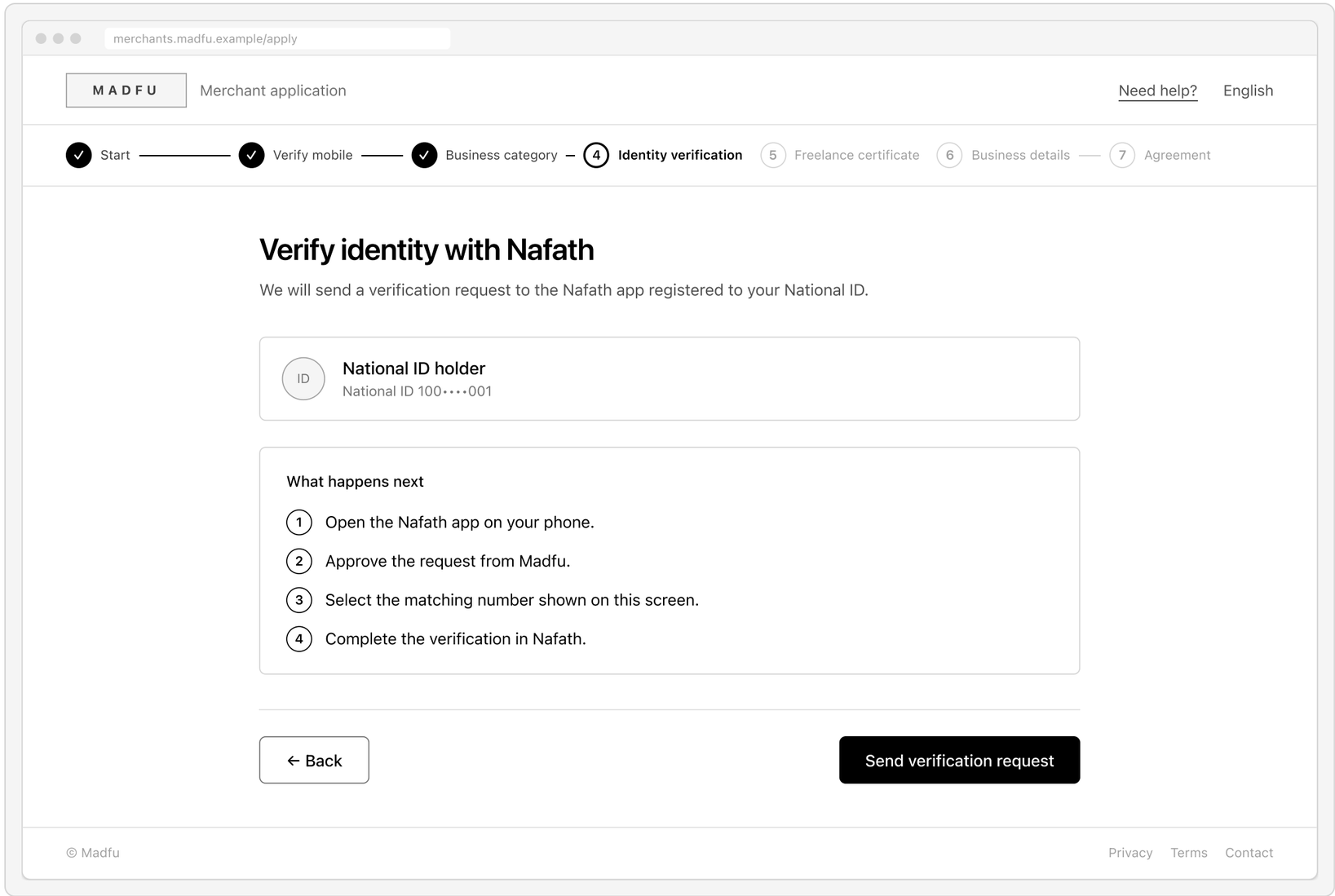
Map D2 — Freelancer: certificates and reconciliation

Certificate retrieval, single selection and reconciliation with the preliminary Madfu category.



Screens in this map: S06 Nafath identity verification (p. 17) · S07 Freelancing certificate selection (p. 21) · S07-M Manual certificate entry (p. 22) · S08 Business details (shared) (p. 26) · O-C Outcome C — Interest received (sales lead) (p. 31)

S06 Nafath identity verification · Ready to send (Freelancer, National ID from entry)



S06 Nafath identity verification — Both

PURPOSE	Verify identity via a Nafath request with a two-digit matching number.
WHO SEES IT	CR: the selected authorized person. Freelancer: the National ID holder from S01.
ENTRY CONDITION	CR: person selected (S05). Freelancer: category decision = continue.
REQUIRED INPUTS	• None in Madfu — the user approves in the Nafath app
PRIMARY ACTION	Send verification request → (wait) → Continue
NEXT SCREEN	Verified: CR → S08; Freelancer → S07 certificate selection.
ALTERNATIVE PATHS	• 1st failure: Try again, or (CR) choose another person. • 2nd failure: CR → S08 manual path; Freelancer → S07-M manual certificate. • Technical failure: retry without consuming an attempt (provisional D-11).
BACKEND BEHAVIOUR	• Max 2 attempts per application — shared across individuals. • Outcomes: verified, rejected, timed out, technical failure. • Each request recorded in the cost ledger.
BUSINESS RULES	• Unverified Nafath → contract signing disabled.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

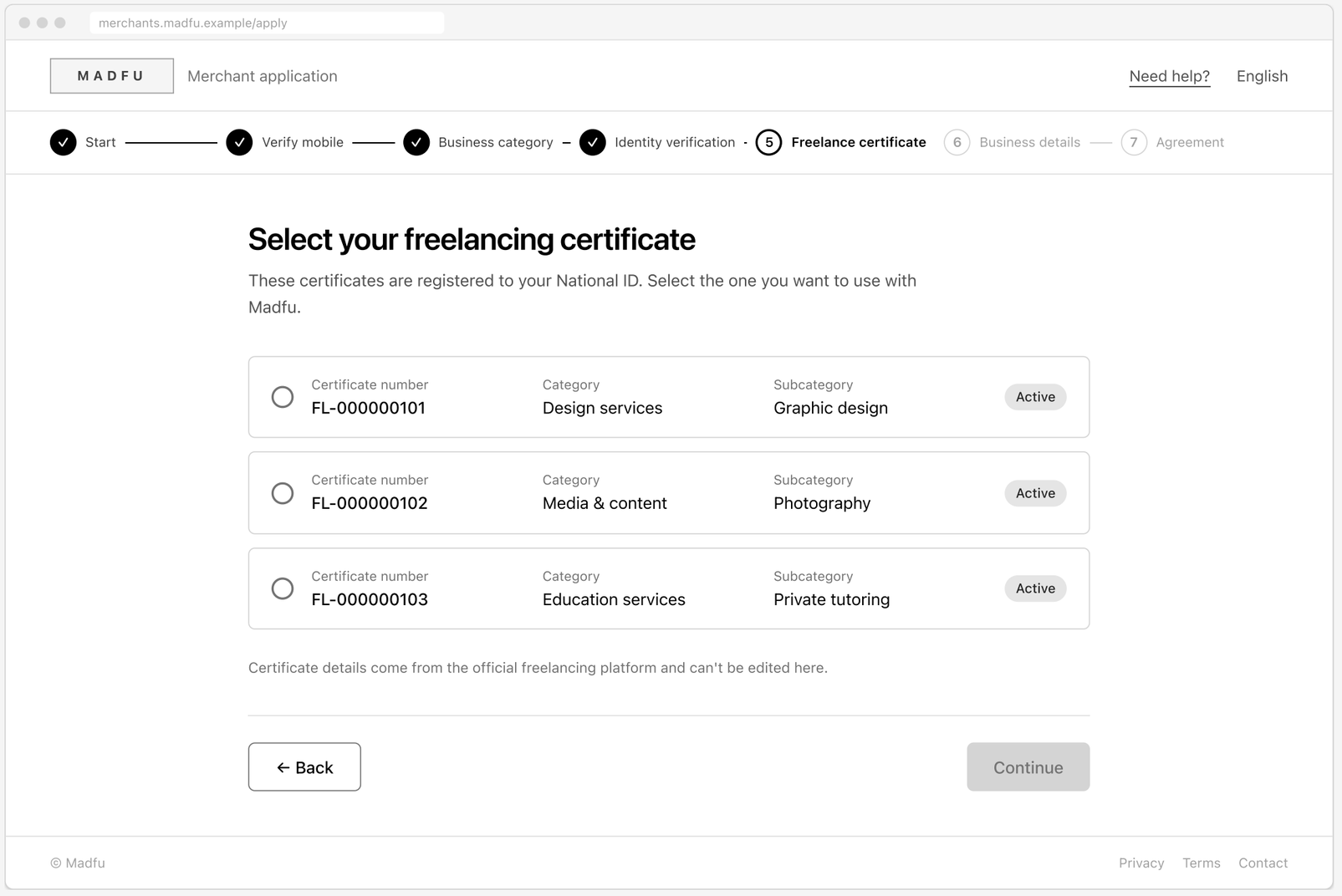
Cross-references

Journey maps: Map C2 (p. 14) · Map D1 (p. 18) · Map D2 (p. 19)

Other states: Ready to send (CR, selected person) (p. 43) · Rejected — 1 attempt remaining (CR) (p. 43) · Timed out — 1 attempt remaining (Freelancer) (p. 44) · Technical failure — no attempt consumed (p. 44) · Verified (p. 45) · Two unsuccessful attempts — manual path (p. 45) · Waiting — Freelancer (National ID from entry) (p. 46)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S06--fl-ready

S07 Freelancing certificate selection · Multiple certificates



S07 Freelancing certificate selection — Freelancer

PURPOSE	Associate exactly one government freelancing certificate with the application.
WHO SEES IT	Freelancers with successful Nafath only.
ENTRY CONDITION	Nafath verified.
REQUIRED INPUTS	• Single selection of one certificate
PRIMARY ACTION	Continue
NEXT SCREEN	S08 Business details.
ALTERNATIVE PATHS	• No certificates → manual entry S07-M (manual review). • Service unavailable → retry. • Certificate already attached to an account → not selectable. • Certificate maps to a restricted category → O-C (cost already incurred, no further paid checks).
BACKEND BEHAVIOUR	• Reconcile certificate activity → Madfu category via mapping table. • Preliminary category preserved; mismatches flagged (CATEGORY_MISMATCH). • Expired / inconsistent certificates flagged for manual review (contract disabled, D-06).
BUSINESS RULES	• Never invent missing certificates. • Taxonomy IDs are not assumed to match.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

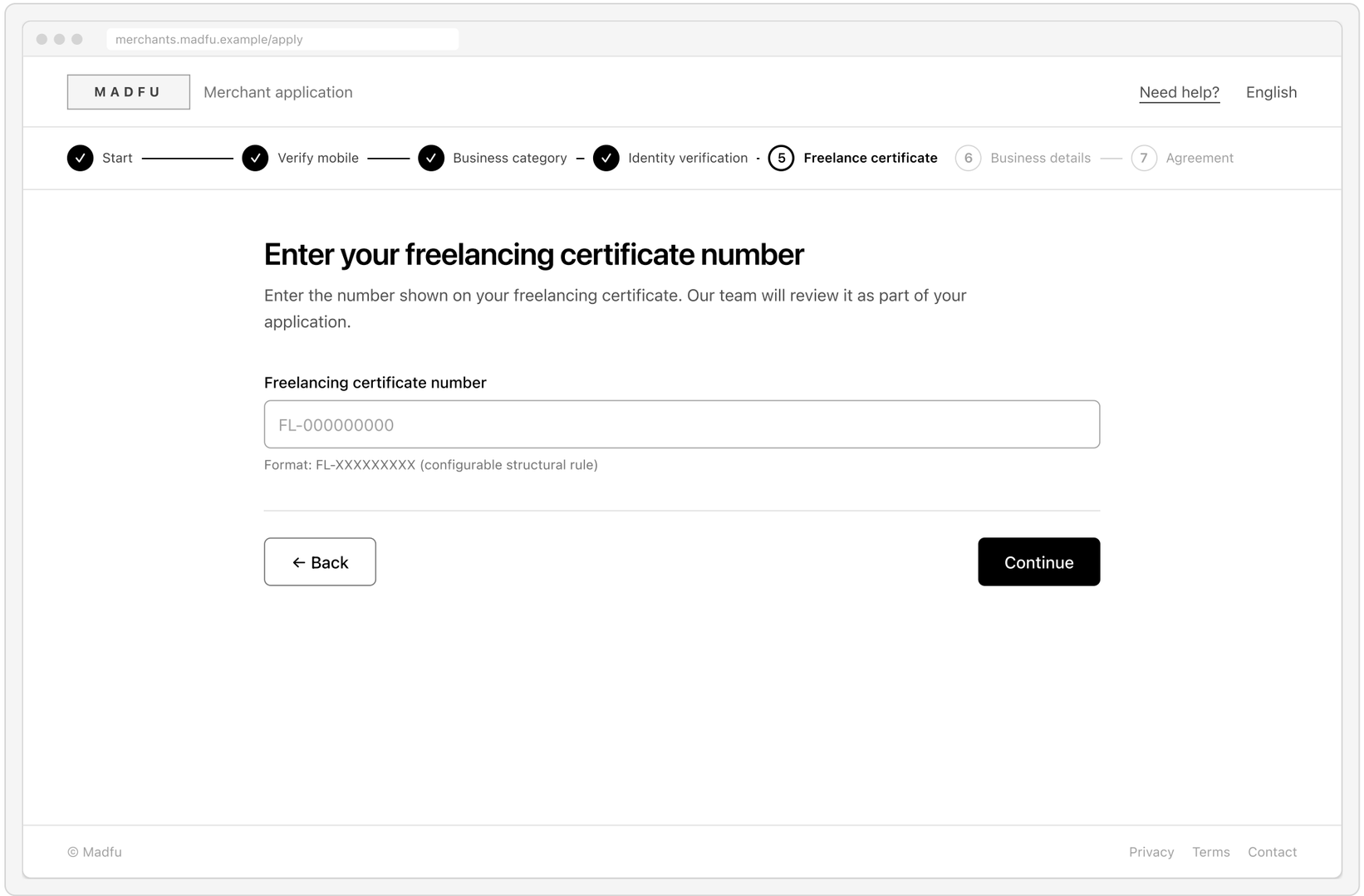
Cross-references

Journey maps: Map D2 (p. 19)

Other states: One certificate (pre-selected) (p. 46) · No certificates found (p. 47) · Certificate service unavailable (p. 47) · Certificate already attached to an account (p. 48) · Expired certificate (p. 48)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S07--multiple

S07-M Manual certificate entry · After two unsuccessful Nafath attempts



S07-M Manual certificate entry

Freelancer

PURPOSE	Collect a certificate number when it cannot be retrieved after verification.
WHO SEES IT	Freelancers whose Nafath failed twice, or whose verified ID returned no certificate.
ENTRY CONDITION	Nafath unverified (2 failures) or "Enter certificate manually" from S07.
REQUIRED INPUTS	• Freelancing certificate number
PRIMARY ACTION	Continue
NEXT SCREEN	S08 Business details (IBAN collected without ownership check when Nafath failed).
ALTERNATIVE PATHS	• Certificate already attached → inline error.
BACKEND BEHAVIOUR	• Internal: identity unverified, certificate association unverified, manual review required, contract disabled.
BUSINESS RULES	• No automatic contract signing without successful Nafath.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map D1 (p. 18) · Map D2 (p. 19)

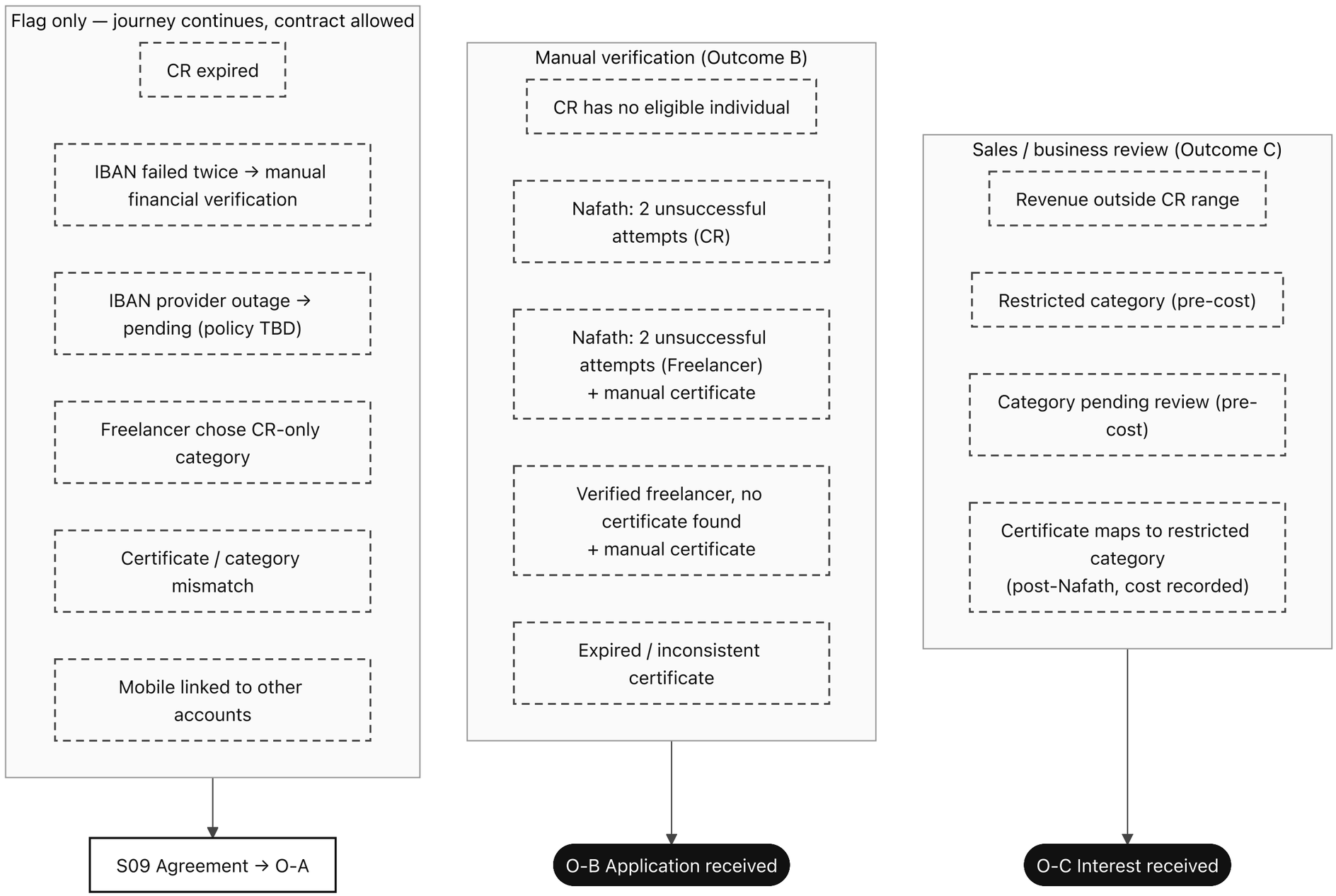
Other states: Certificate already linked to an account (p. 49)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S07-M--manual

Map E — Manual-review routes

Every route that ends in a lead, a manual verification, or a flag-only review.





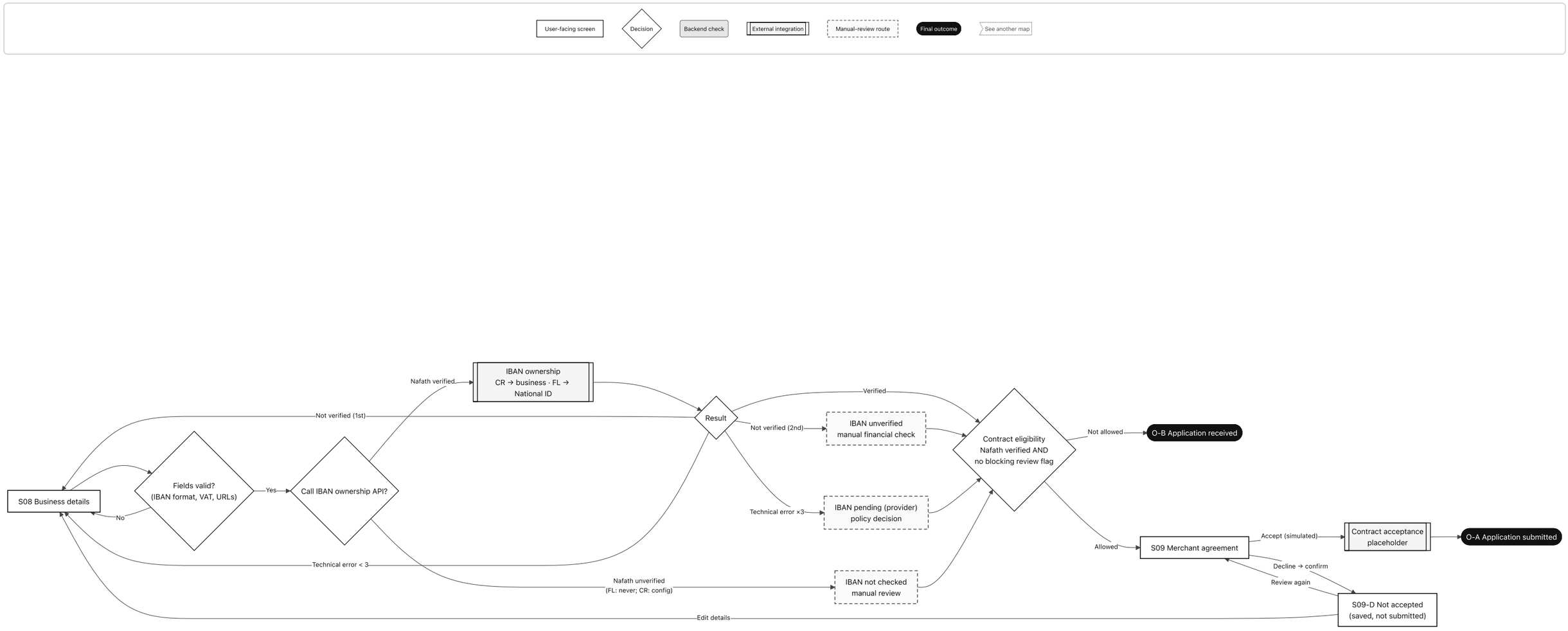
Screens in this map: S09 Merchant agreement (contract) (p. 27) · O-B Outcome B — Application received (manual verification) (p. 30) · O-C Outcome C — Interest received (sales lead) (p. 31)

Manual-review routes — detail

Route	Trigger	Paid calls already made	Contract	Applicant outcome	Internal flags / status
Revenue lead (CR)	Revenue below SAR 100K or above SAR 20M	None	—	O-C	SALES_LEAD_REVENUE · SALES_LEAD
Restricted category	Fixture BLOCKED subcategory	None	—	O-C	CATEGORY_RESTRICTED · SALES_LEAD
Category review	MANUAL_REVIEW subcategory (config: stop)	None	—	O-C	CATEGORY_REVIEW_REQUIRED · SALES_LEAD
Restricted certificate	Certificate activity maps to BLOCKED	Nafath + certificate lookup (recorded)	—	O-C	CATEGORY_RESTRICTED_POST_NAFATH
No eligible individual (CR)	CR lists only companies / entities	CR lookup	Disabled	O-B	NO_ELIGIBLE_AUTHORISED_PERSON · NAFATH_MANUAL_VERIFICATION
Nafath exhausted (CR)	2 unsuccessful attempts (any persons)	CR lookup + 2 Nafath	Disabled	O-B	NAFATH_MANUAL_VERIFICATION · IBAN not checked (config)
Nafath exhausted (Freelancer)	2 unsuccessful attempts	2 Nafath	Disabled	O-B	NAFATH_MANUAL_VERIFICATION · CERT_MANUAL_ENTRY_UNVERIFIED · IBAN_NOT_CHECKED
No certificate (verified freelancer)	Certificate lookup returns none	Nafath + lookup	Disabled (provisional)	O-B	CERT_NOT_FOUND_MANUAL_ENTRY
Expired / inconsistent certificate	Certificate status / holder mismatch	Nafath + lookup	Disabled (provisional)	O-B	CERT_EXPIRED / CERT_INCONSISTENT
IBAN failed twice	2 not-verified ownership results	+ 2 IBAN checks	Allowed	O-A	IBAN_UNVERIFIED_MANUAL_FINANCIAL_REVIEW
IBAN provider outage	3 technical errors	+ IBAN calls	Allowed (policy pending)	O-A	IBAN_VERIFICATION_PENDING_PROVIDER
CR expired	CR lookup EXPIRED	CR lookup	Allowed	O-A	CR_EXPIRED
CR-only category (freelancer)	Freelancer picks CR-only subcategory	—	Allowed	O-A	CR_ONLY_CATEGORY_FREELANCER
Category mismatch	Certificate maps to a different / no subcategory	Nafath + lookup	Allowed	O-A	CATEGORY_MISMATCH
Contract declined	Applicant declines on S09	All	Pending	S09-D (not submitted)	CONTRACT_DECLINED · CONTRACT_PENDING

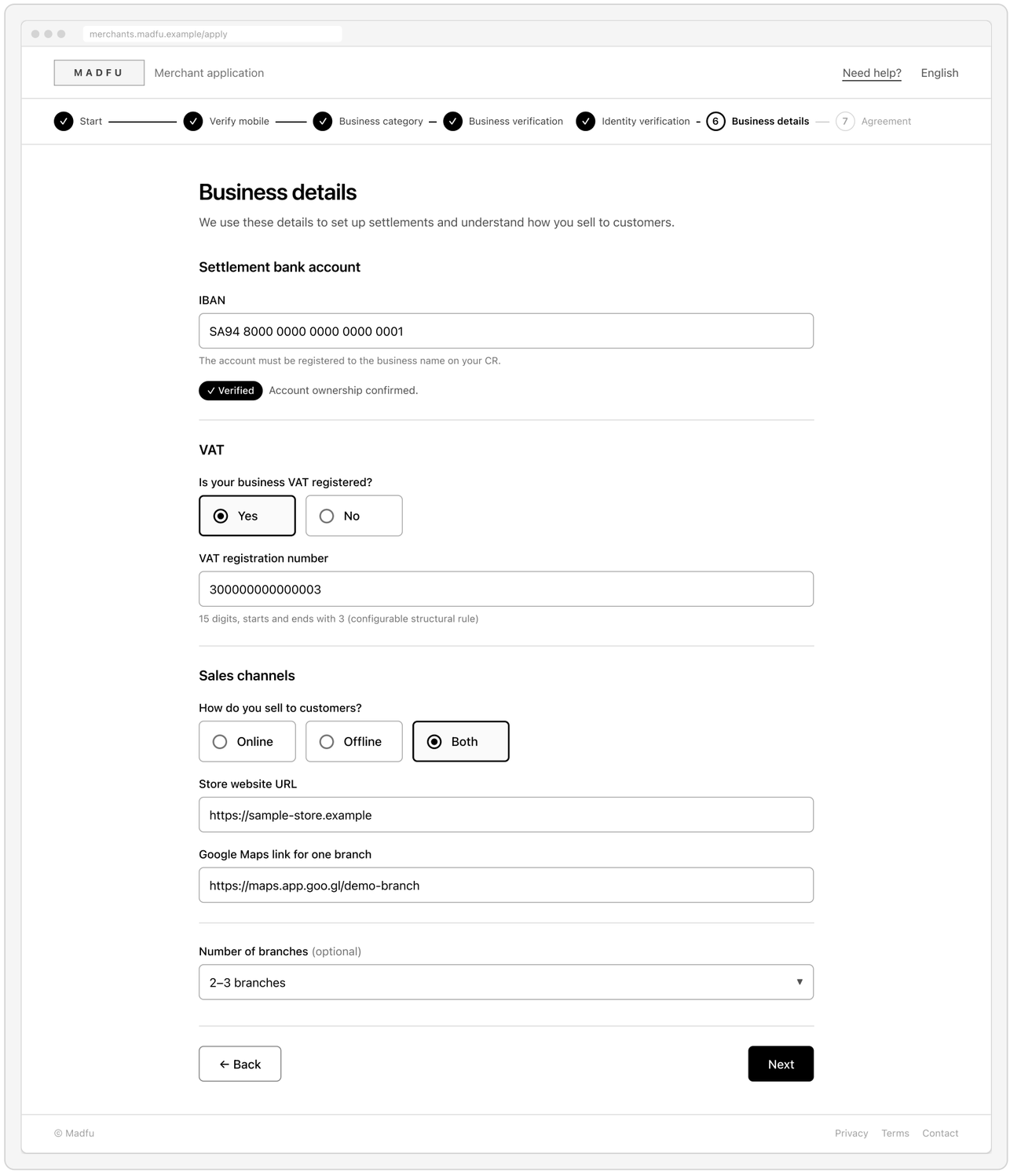
Map F — Business details, contract and outcomes

IBAN verification, contract eligibility matrix and final outcomes.



Screens in this map: S08 Business details (shared) (p. 26) · S09 Merchant agreement (contract) (p. 27) · S09-D Agreement not accepted (p. 28) · O-A Outcome A — Application submitted (p. 29) · O-B Outcome B — Application received (manual verification) (p. 30)

S08 Business details (shared) · IBAN verified, VAT yes, both channels



S08 Business details (shared) — Both

PURPOSE	Collect settlement IBAN, VAT, sales channels and branches.
WHO SEES IT	All self-onboarding applicants who reached this step (verified or manual path).
ENTRY CONDITION	CR: Nafath finished (verified or exhausted) or no eligible person. Freelancer: certificate selected or entered.
REQUIRED INPUTS	• IBAN (required) • VAT registered? Yes/No (+ VAT number if Yes) • Sales channel: Online / Offline / Both (+ website and/or Google Maps URL) • Number of branches (optional)
PRIMARY ACTION	Next
NEXT SCREEN	Contract allowed → S09. Not allowed (Nafath unverified / blocking review flag) → O-B.
ALTERNATIVE PATHS	• IBAN not verified (1st) → correct or retry. • IBAN not verified (2nd) → continue, flagged for manual financial verification. • Technical error → retry; persistent → pending (separate from failure).
BACKEND BEHAVIOUR	• Format check (SA + 22 digits, mod-97) ≠ ownership check. • Ownership API on field exit; never repeated for an unchanged value. • CR: match to registered business. Freelancer: match to National ID. Freelancer with failed Nafath: no API call. • Editing IBAN invalidates its verification result.
BUSINESS RULES	• Max 2 unsuccessful ownership attempts per application. • IBAN failure never blocks the contract when Nafath succeeded.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

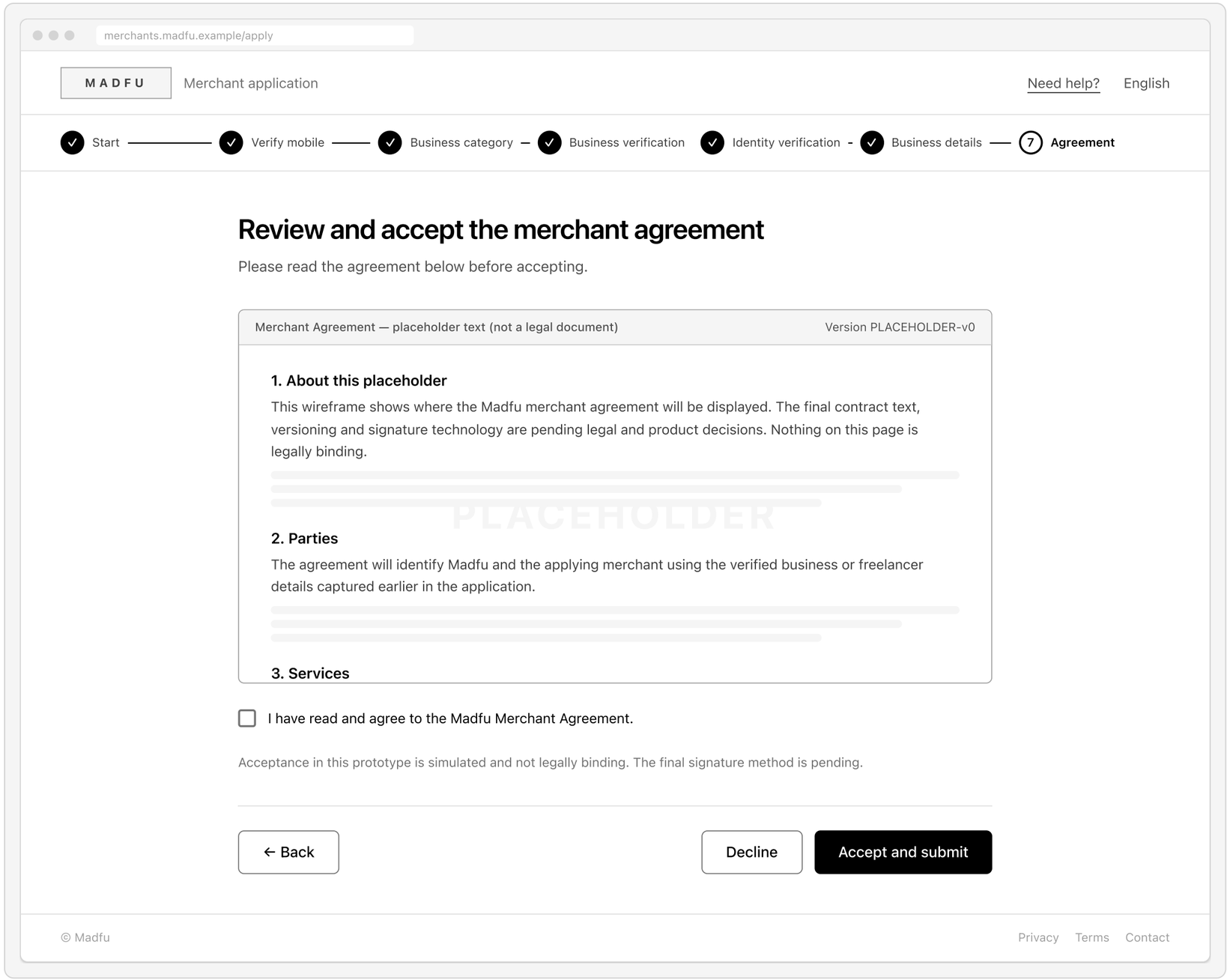
Cross-references

Journey maps: Map A (p. 4) · Map C2 (p. 14) · Map D2 (p. 19) · Map F (p. 25)

Other states: Empty form (p. 49) · Freelancer — IBAN verified against National ID (p. 50) · IBAN verifying (p. 50) · IBAN not verified (1 of 2) (p. 51) · IBAN not verified twice — may continue (p. 51) · IBAN technical error (p. 52) · IBAN provider outage — pending (not a failure) (p. 52) · Freelancer, Nafath unverified — IBAN collected only (p. 53) · Validation errors (p. 53)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S08--verified

S09 Merchant agreement (contract) · Agreement viewer



S09 Merchant agreement (contract) — Both

PURPOSE	Show the agreement and capture simulated acceptance.
WHO SEES IT	Only applicants with successful Nafath and no blocking review flag.
ENTRY CONDITION	S08 submitted and contract eligibility = allowed.
REQUIRED INPUTS	• Agreement confirmation checkbox
PRIMARY ACTION	Accept and submit (simulated)
NEXT SCREEN	O-A Application submitted.
ALTERNATIVE PATHS	• Decline → confirmation → S09-D. • Back → S08.
BACKEND BEHAVIOUR	• Generic acceptance placeholder; signature technology not decided (D-02). • Acceptance recorded with contract version.
BUSINESS RULES	• Not legally binding. No real contract text or signature credentials.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

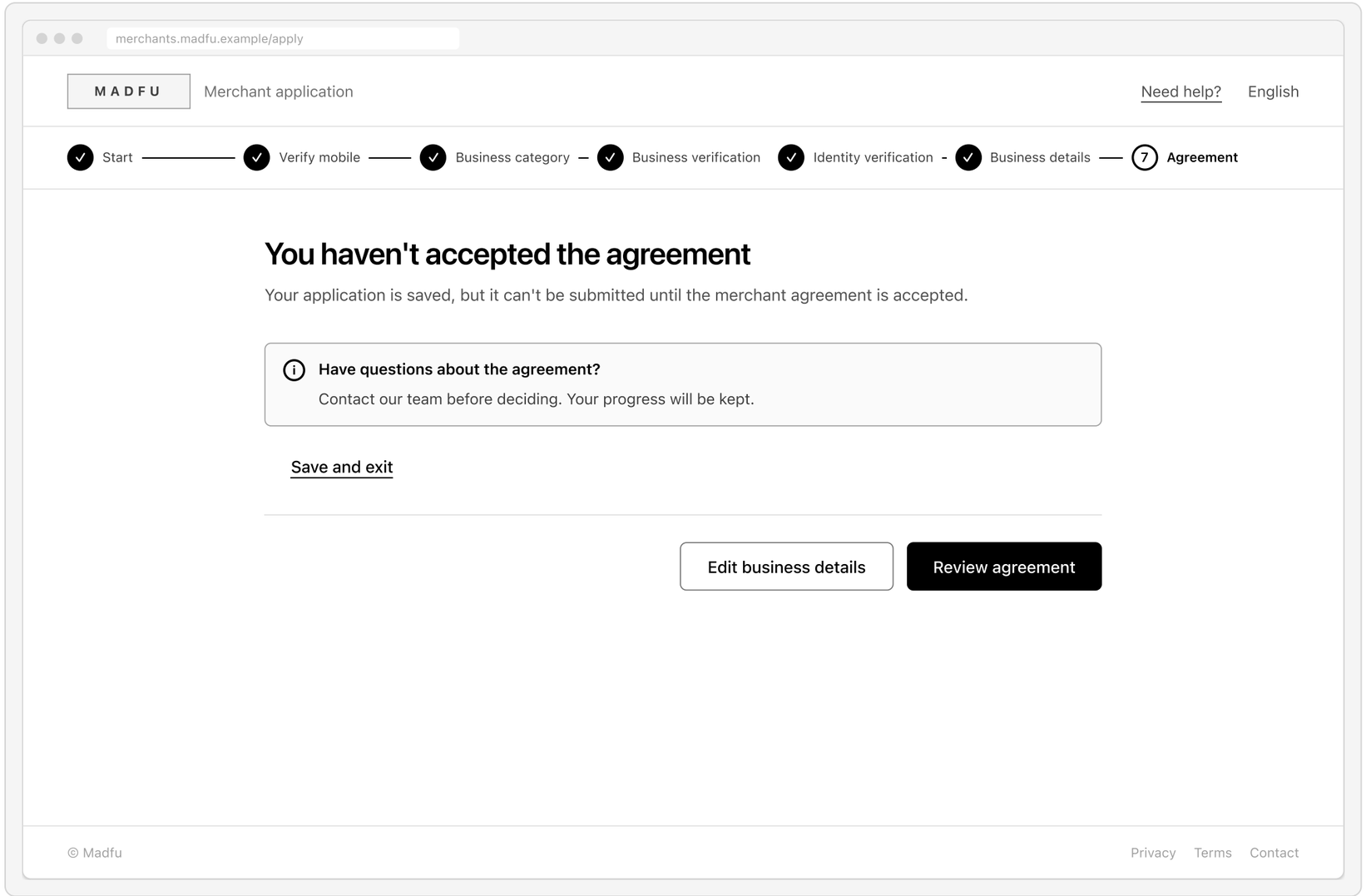
Cross-references

Journey maps: Map A (p. 4) · Map E (p. 23) · Map F (p. 25)

Other states: Agreement — Freelancer (p. 54) · Acceptance without confirmation (p. 54) · Decline confirmation (p. 55)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S09--default

S09-D Agreement not accepted · Agreement declined — saved, not submitted



S09-D Agreement not accepted — Both

PURPOSE	Hold a declined application without reporting completion.
WHO SEES IT	Applicants who declined the agreement.
ENTRY CONDITION	Decline confirmed on S09.
REQUIRED INPUTS	• None
PRIMARY ACTION	Review agreement
NEXT SCREEN	S09 Contract.
ALTERNATIVE PATHS	• Edit business details → S08. • Save and exit (resume later).
BACKEND BEHAVIOUR	• Internal CONTRACT_DECLINED flag; status CONTRACT_PENDING; not submitted.
BUSINESS RULES	• Final decline handling pending (D-09).
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

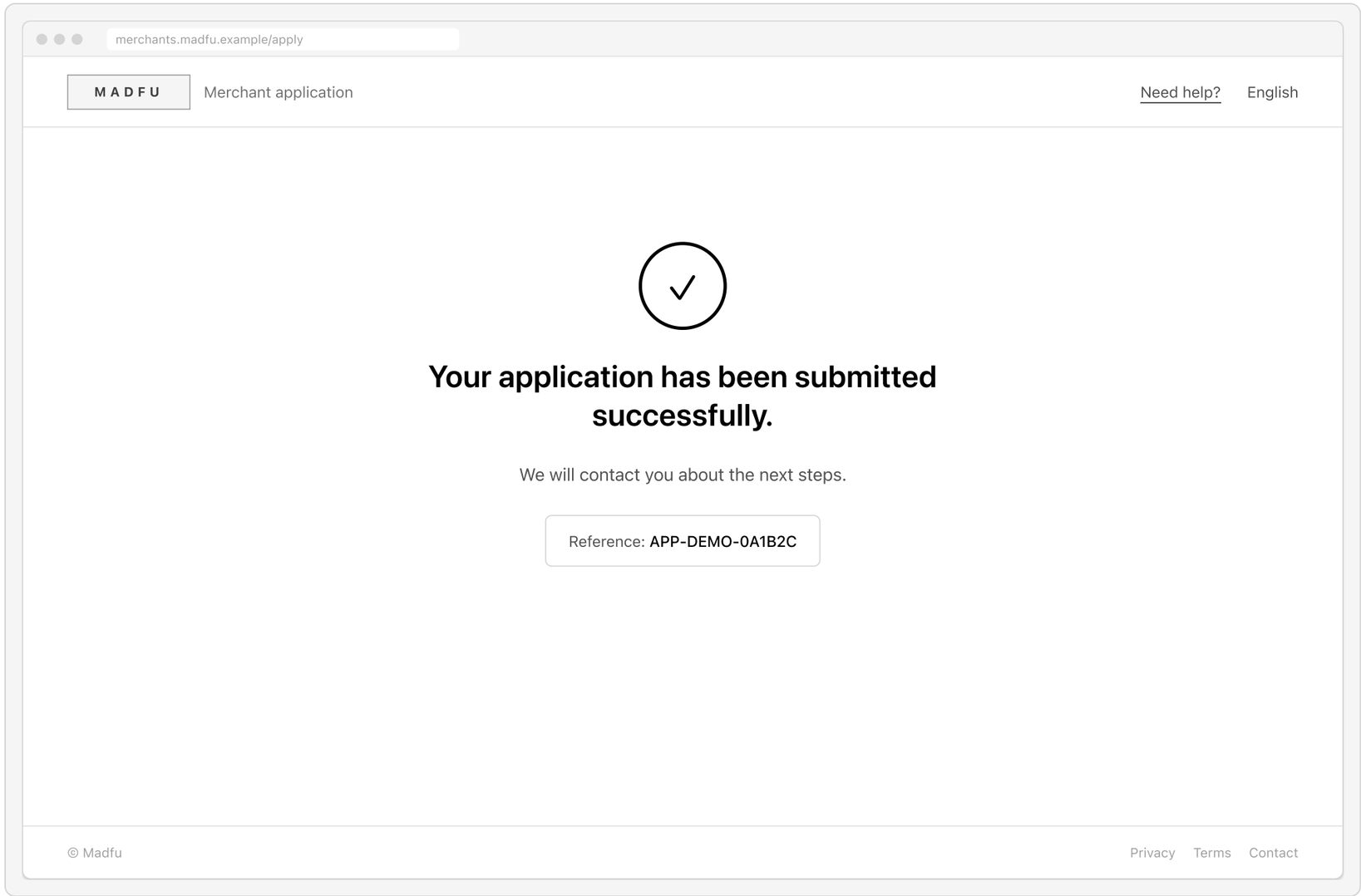
Cross-references

Journey maps: Map A (p. 4) · Map F (p. 25)

Other states: Agreement declined — Freelancer (p. 55)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/S09-D--declined

O-A Outcome A — Application submitted · Submitted



O-A Outcome A — Application submitted — Both

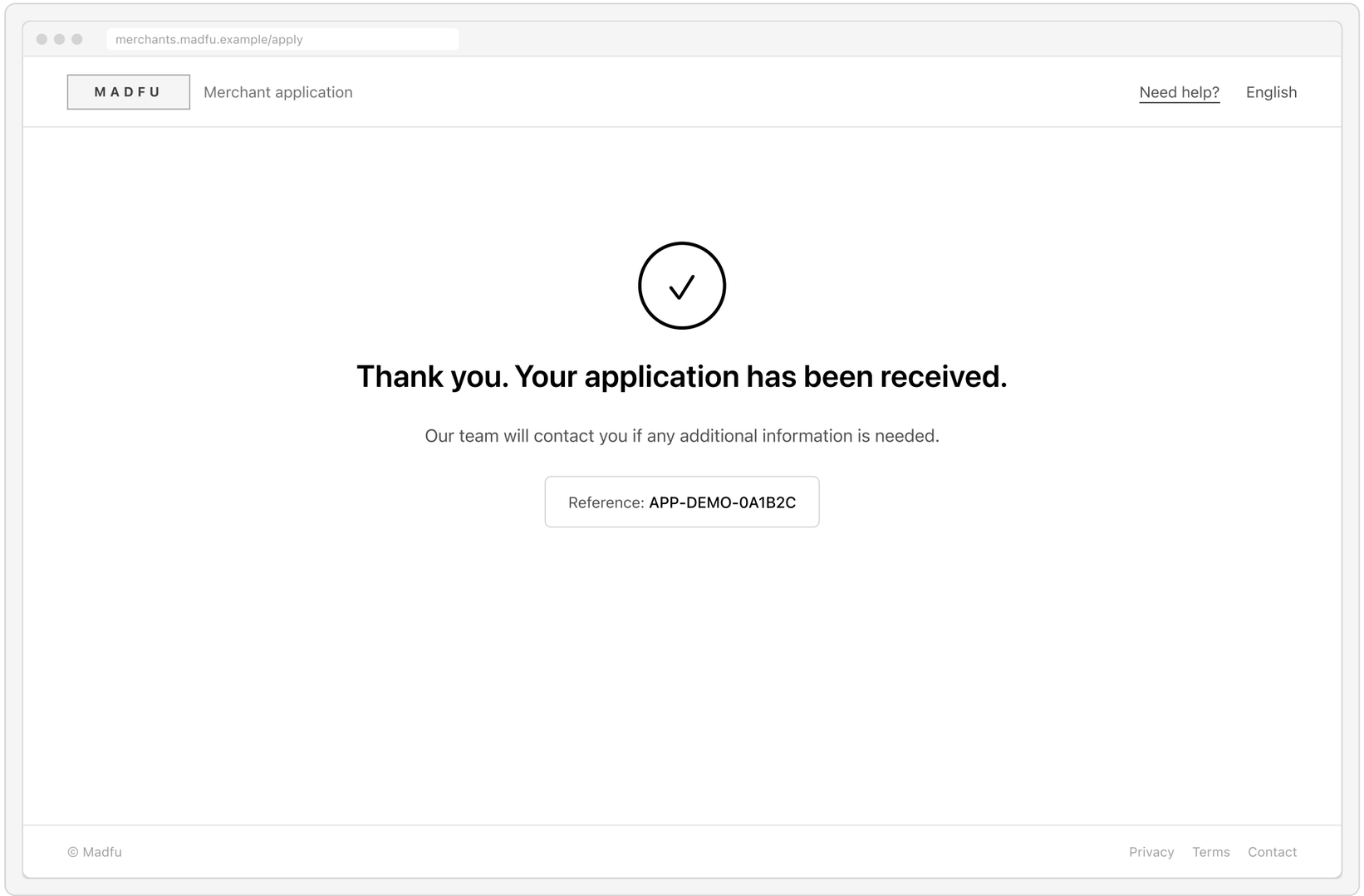
PURPOSE	Confirm full submission.
WHO SEES IT	Self-onboarding applicants with verified Nafath and simulated contract acceptance.
ENTRY CONDITION	Contract accepted.
REQUIRED INPUTS	• None
PRIMARY ACTION	None (final state)
NEXT SCREEN	End of journey.
ALTERNATIVE PATHS	—
BACKEND BEHAVIOUR	• Internal: CONTRACT_ACCEPTED, APPLICATION_SUBMITTED (+ any review flags e.g. CR_EXPIRED, IBAN pending).
BUSINESS RULES	• Does not state approval or merchant activation.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map A (p. 4) · Map F (p. 25)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/O-A--submitted

O-B Outcome B — Application received (manual verification) · Received — manual verification



O-B Outcome B — Application received (manual verification)

Both

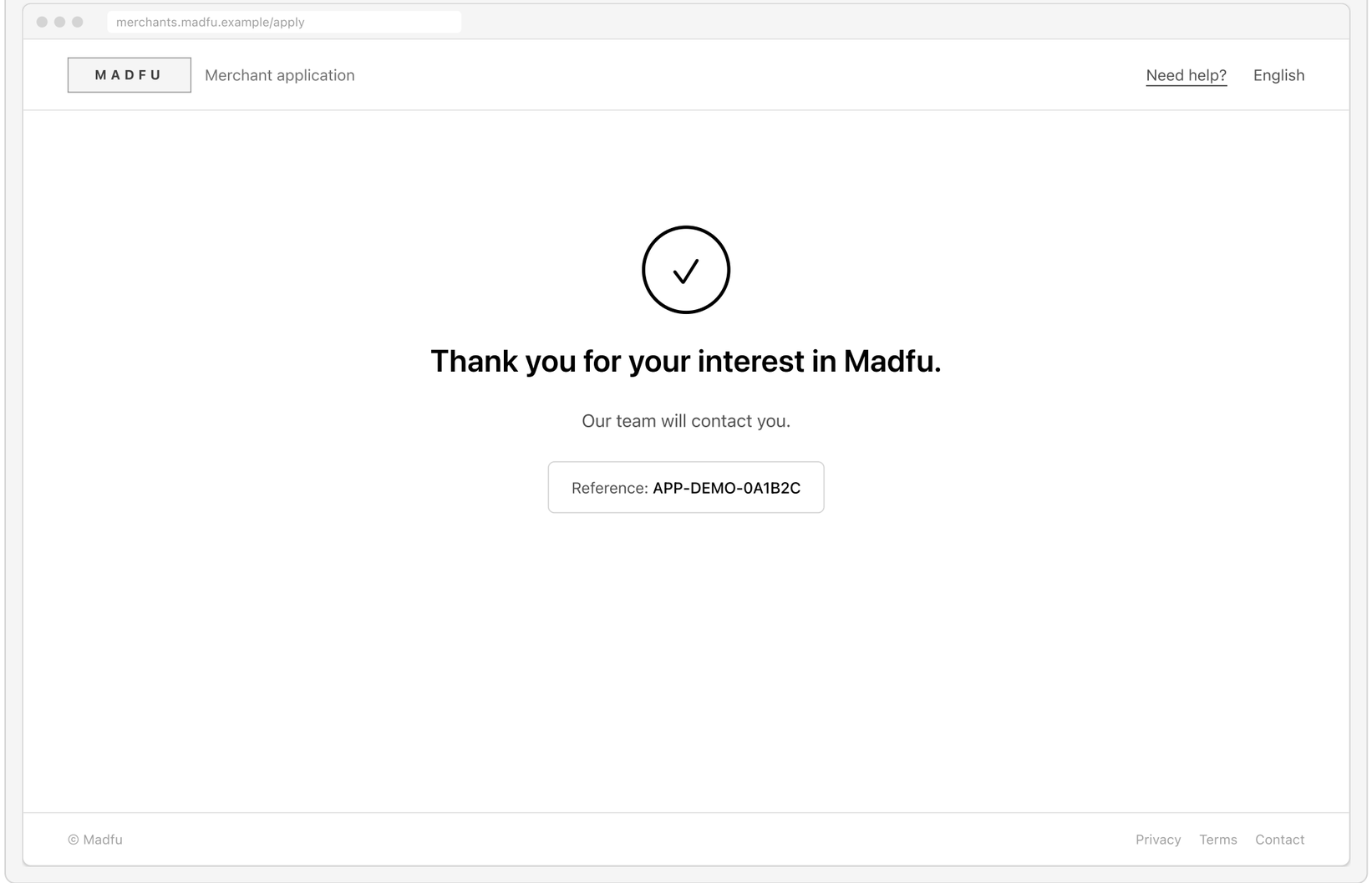
PURPOSE	Neutral confirmation for applications that need manual verification.
WHO SEES IT	Applicants without successful Nafath, or with blocking review flags.
ENTRY CONDITION	S08 submitted and contract not allowed.
REQUIRED INPUTS	• None
PRIMARY ACTION	None (final state)
NEXT SCREEN	Manual verification by Madfu team.
ALTERNATIVE PATHS	—
BACKEND BEHAVIOUR	• Internal: NAFATH_MANUAL_VERIFICATION and/or certificate review, IBAN pending, APPLICATION_SUBMITTED.
BUSINESS RULES	• Do not disclose internal reasons.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

Cross-references

Journey maps: Map A (p. 4) · Map E (p. 23) · Map F (p. 25)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/O-B--received

O-C Outcome C — Interest received (sales lead) · Sales lead



O-C Outcome C — Interest received (sales lead) Both

PURPOSE	Confirm lead capture for applicants routed to Sales.
WHO SEES IT	CR outside the self-onboarding revenue range; restricted / review categories; restricted certificate after Nafath.
ENTRY CONDITION	Revenue routing = sales lead, or category decision = stop.
REQUIRED INPUTS	• None
PRIMARY ACTION	None (final state)
NEXT SCREEN	Sales / business review team contacts the applicant.
ALTERNATIVE PATHS	—
BACKEND BEHAVIOUR	• Internal: SALES_LEAD (+ CATEGORY_RESTRICTED / CATEGORY_REVIEW_REQUIRED / SALES_LEAD_REVENUE). • No authorized persons, Nafath, IBAN, VAT, channels or contract requested on this path.
BUSINESS RULES	• Neutral message — never state the restriction reason.
GLOBAL NOTE	All error, empty, loading, retry and edge states are required — see the global UX/UI handover note (G-UX).

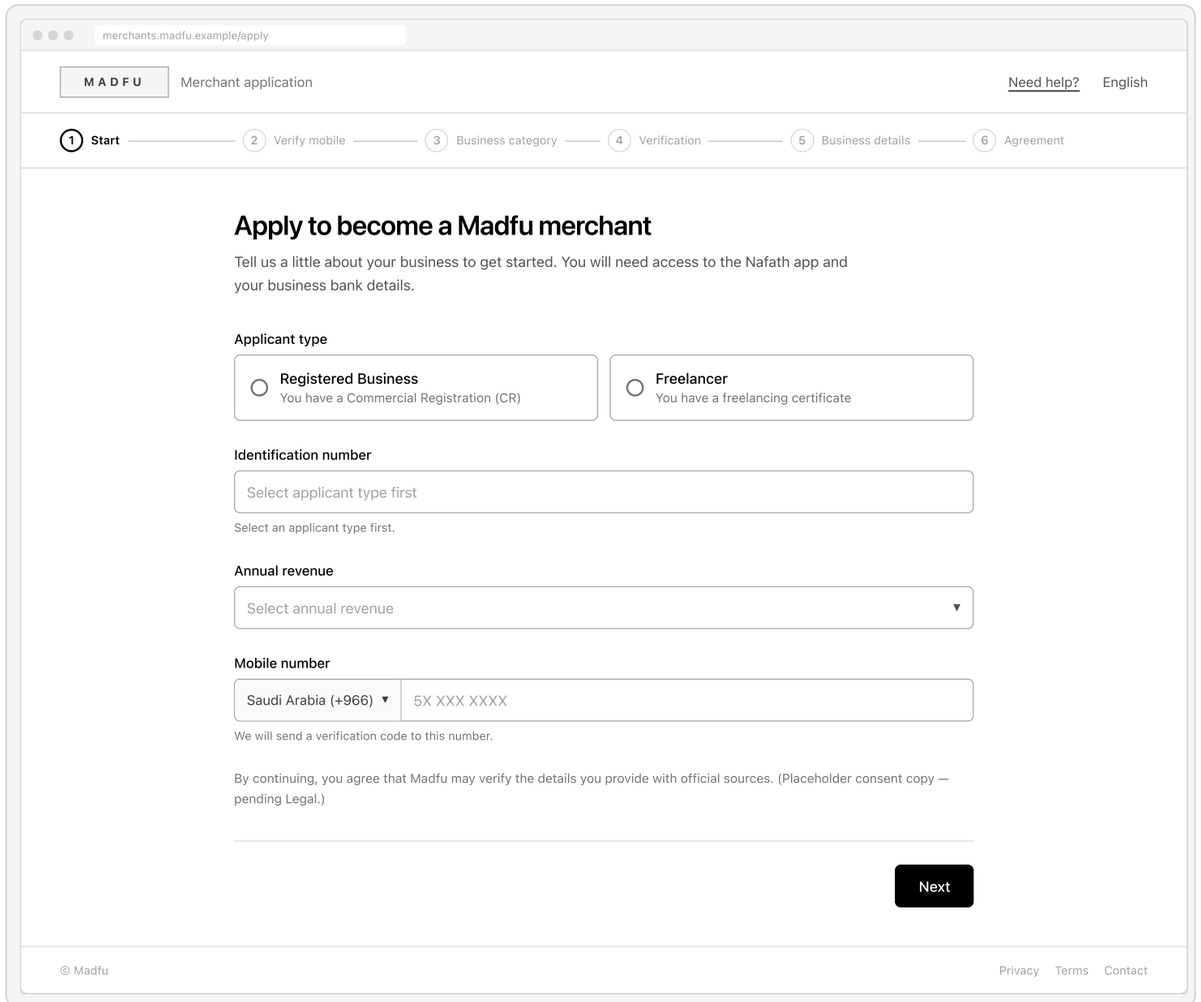
Cross-references

Journey maps: Map A (p. 4) · Map B (p. 5) · Map C1 (p. 13) · Map D1 (p. 18) · Map D2 (p. 19) · Map E (p. 23)

Interactive: https://wearemishkat.github.io/madfu-onboarding-wireframes/#/gallery/O-C--lead

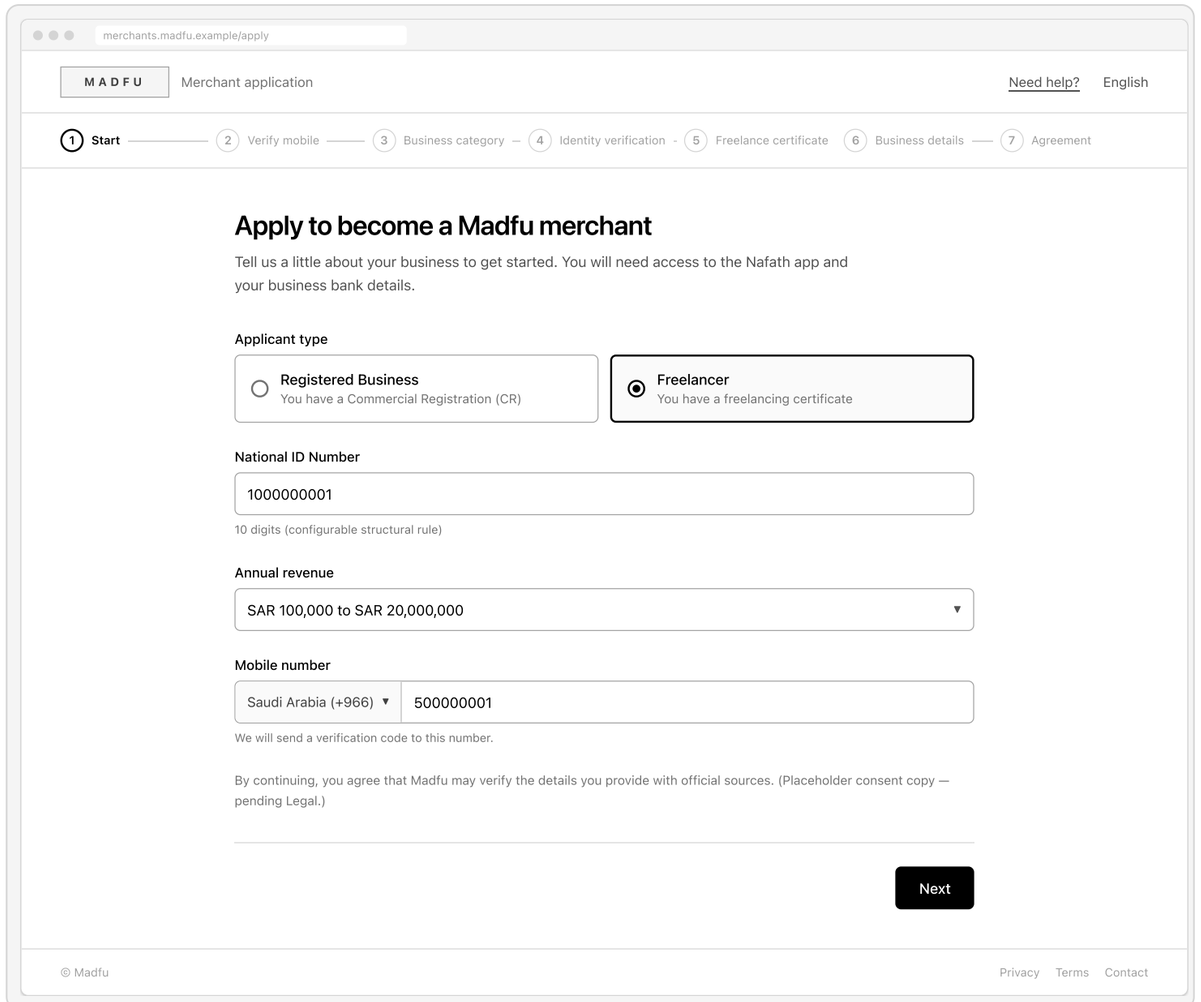
S01 Empty form

Start application (shared entry) · main page p. 7



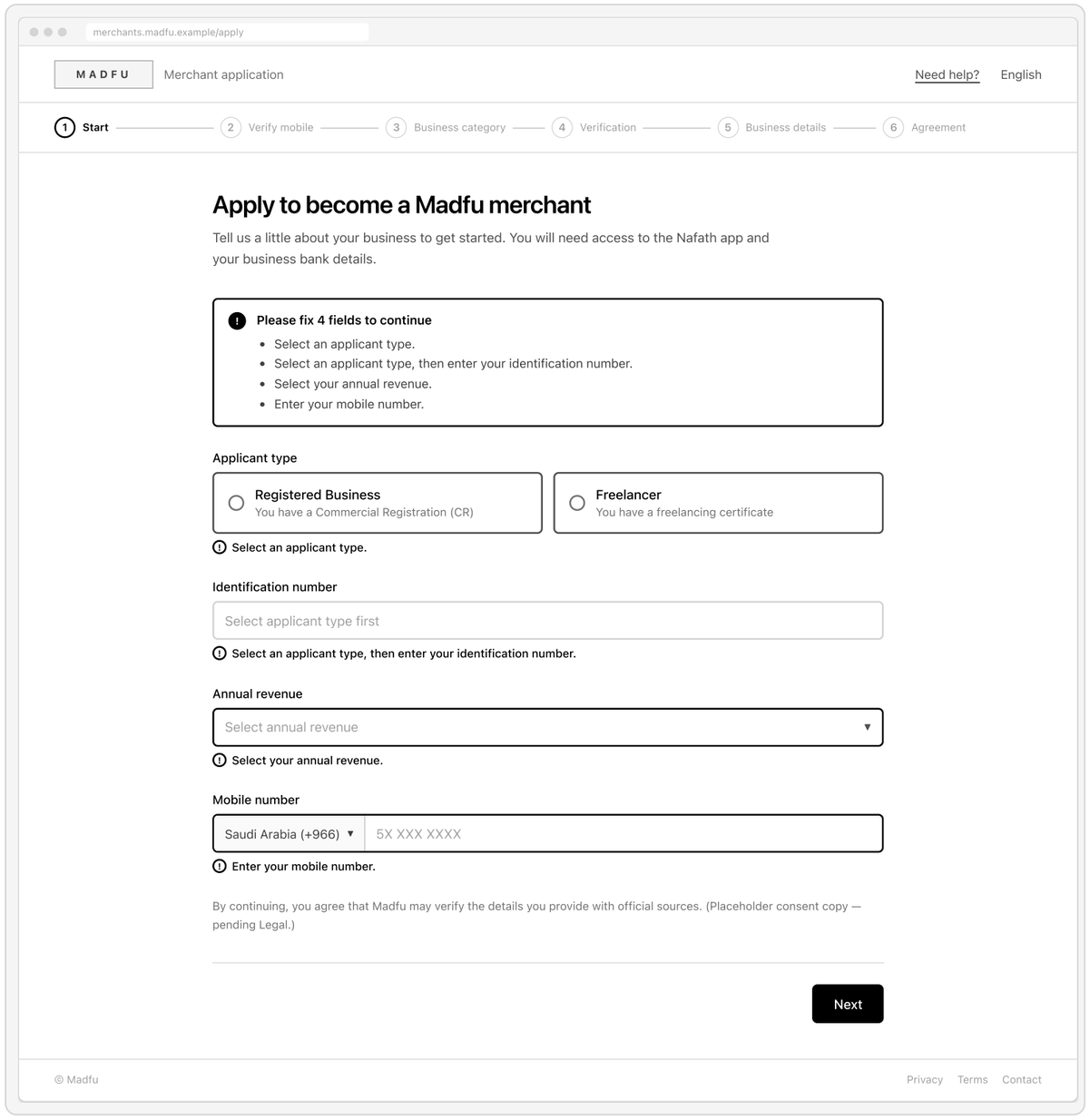
S01 Freelancer — National ID field

Start application (shared entry) · main page p. 7



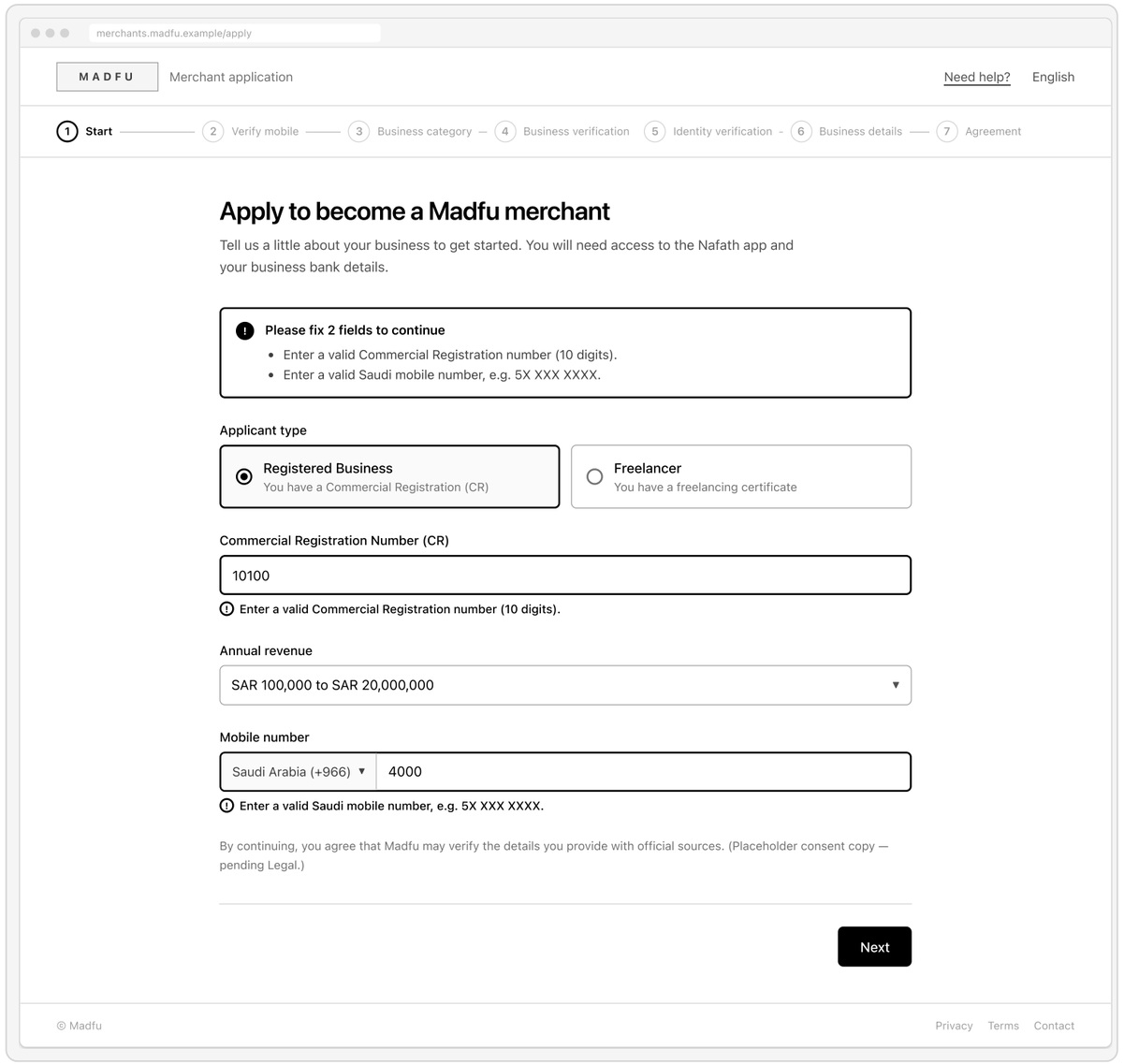
S01 Validation — required fields missing

Start application (shared entry) · main page p. 7



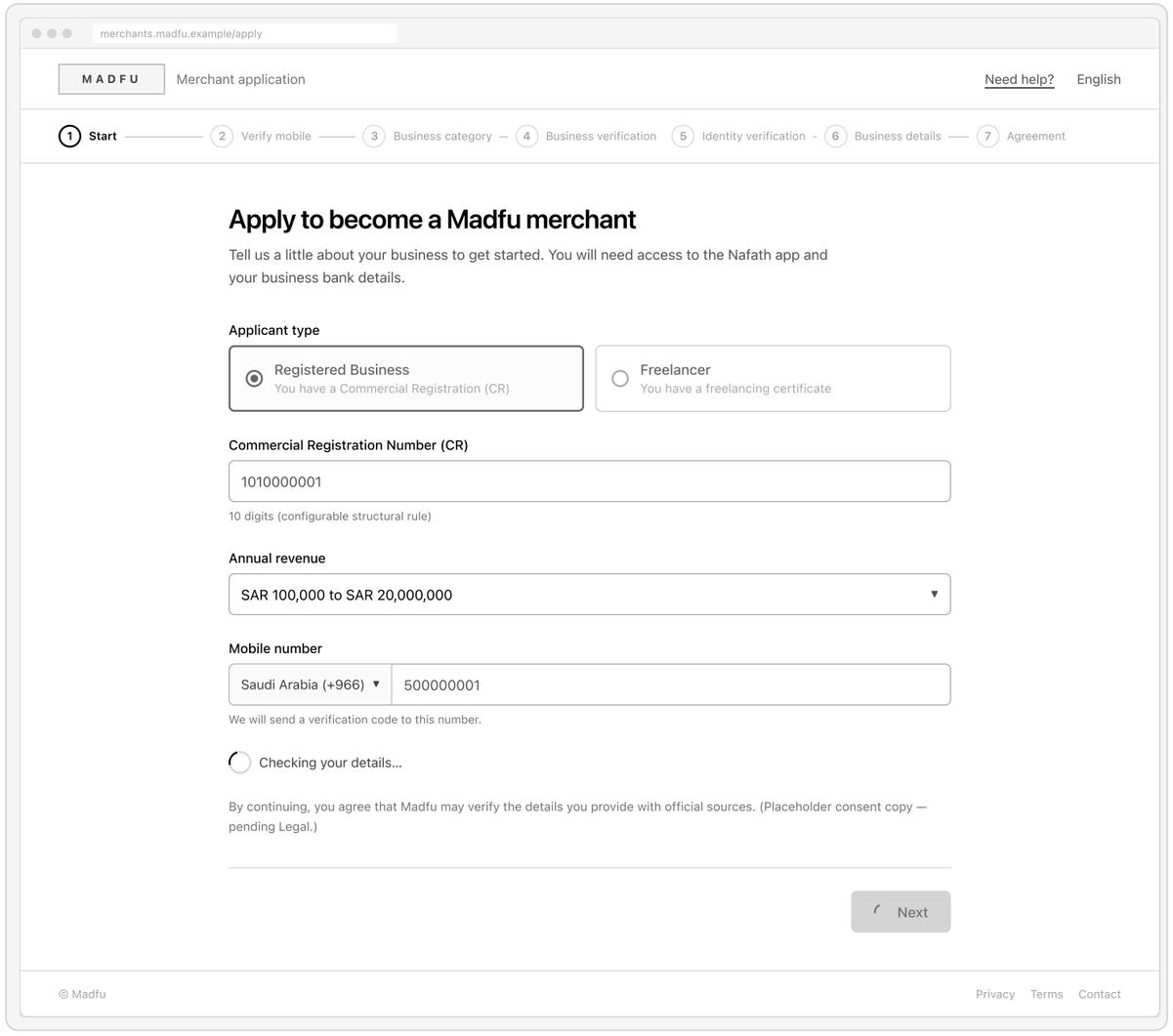
S01 Validation — invalid identification and mobile format

Start application (shared entry) · main page p. 7



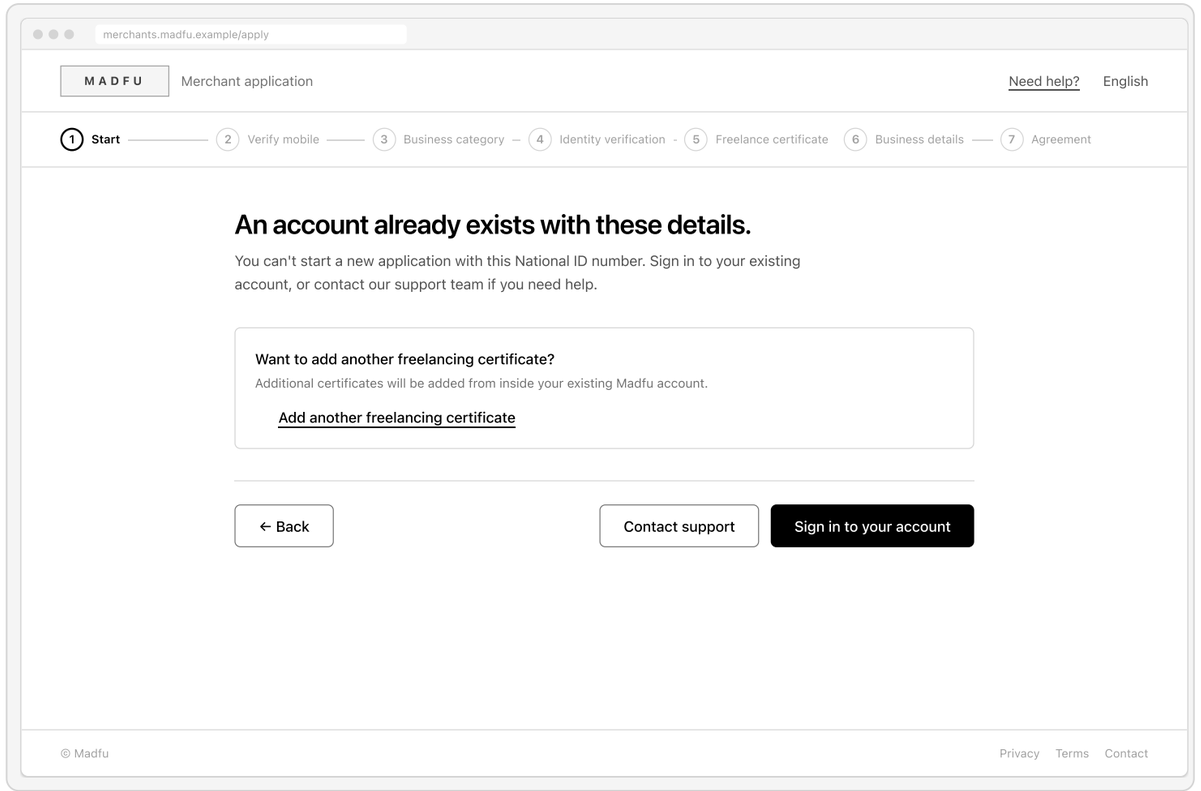
S01 Loading — backend duplicate checks

Start application (shared entry) · main page p. 7



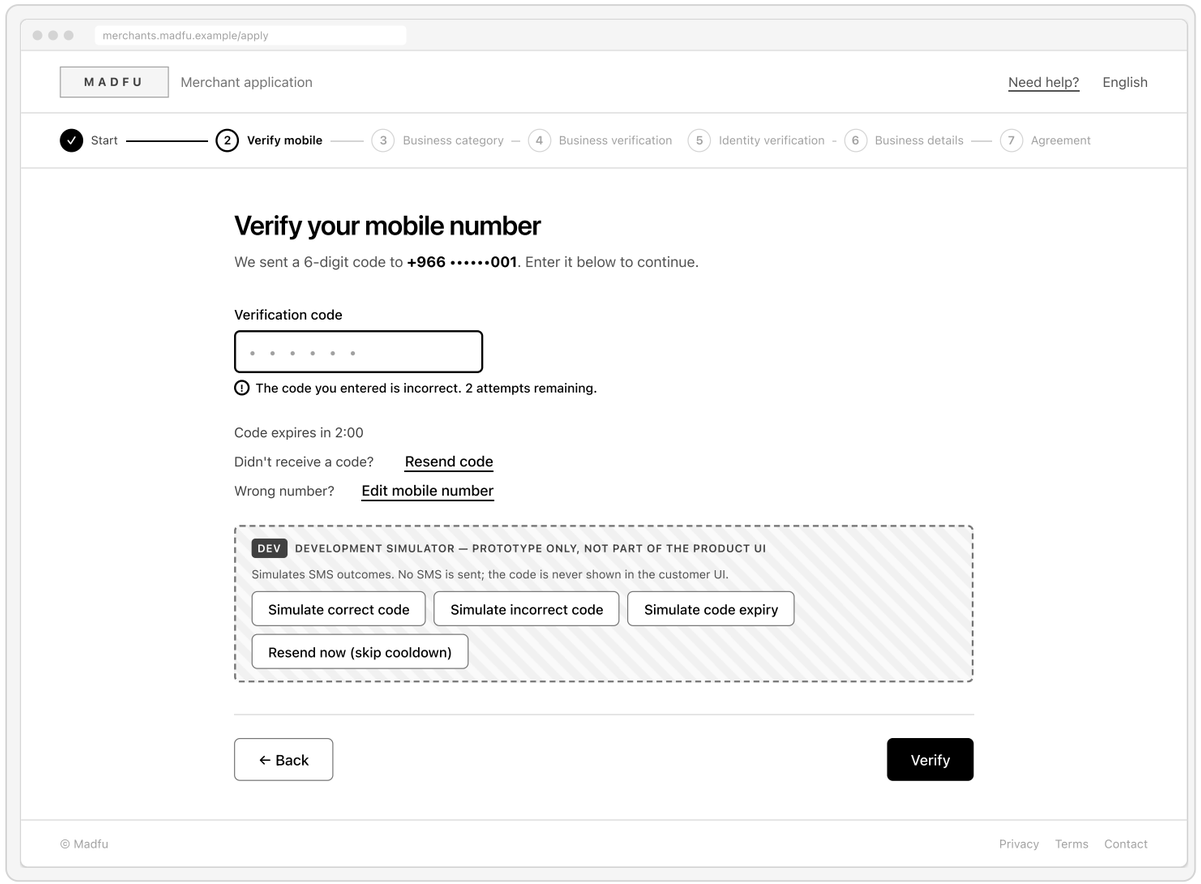
S01-X Existing National ID (with future add-certificate branch)

Existing account found · main page p. 8



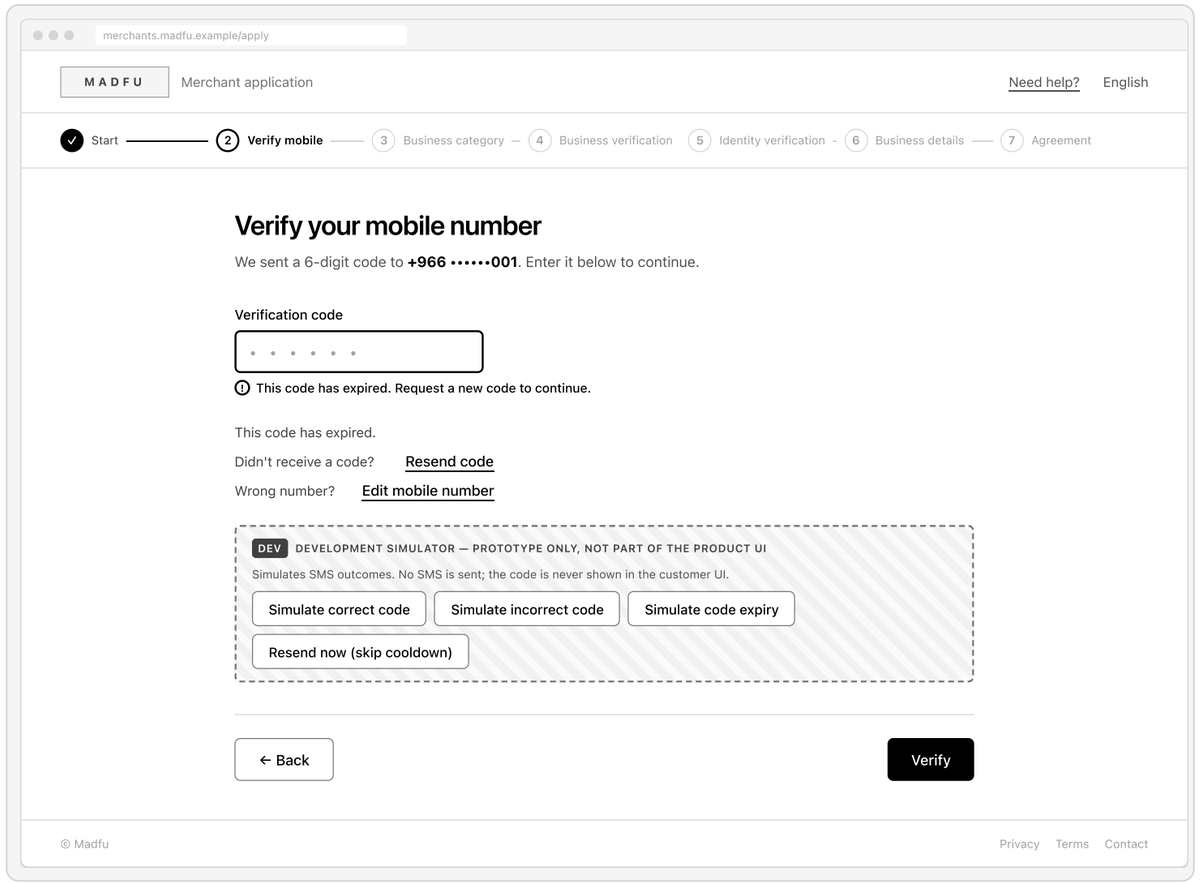
S02 Incorrect code (2 attempts remaining)

Mobile OTP verification · main page p. 10



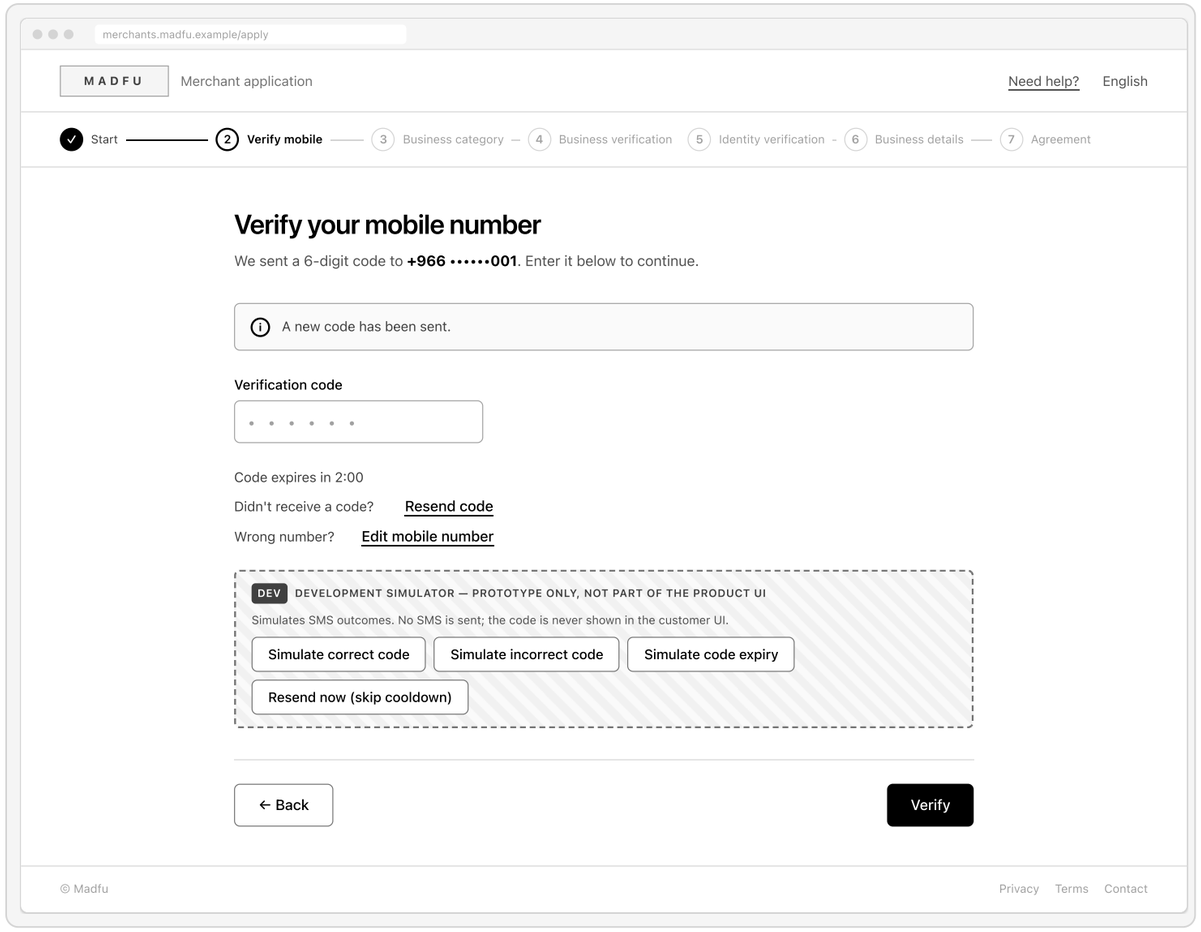
S02 Expired code (no attempt consumed)

Mobile OTP verification · main page p. 10



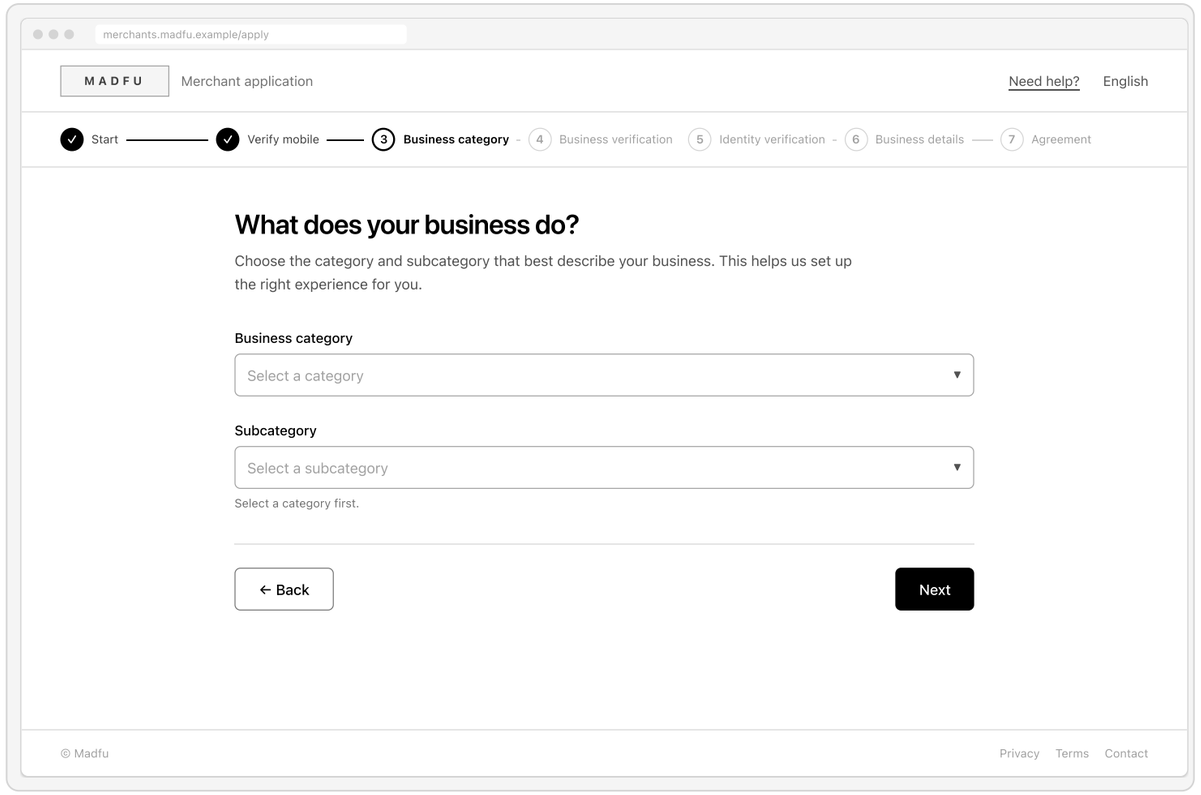
S02 New code sent

Mobile OTP verification · main page p. 10



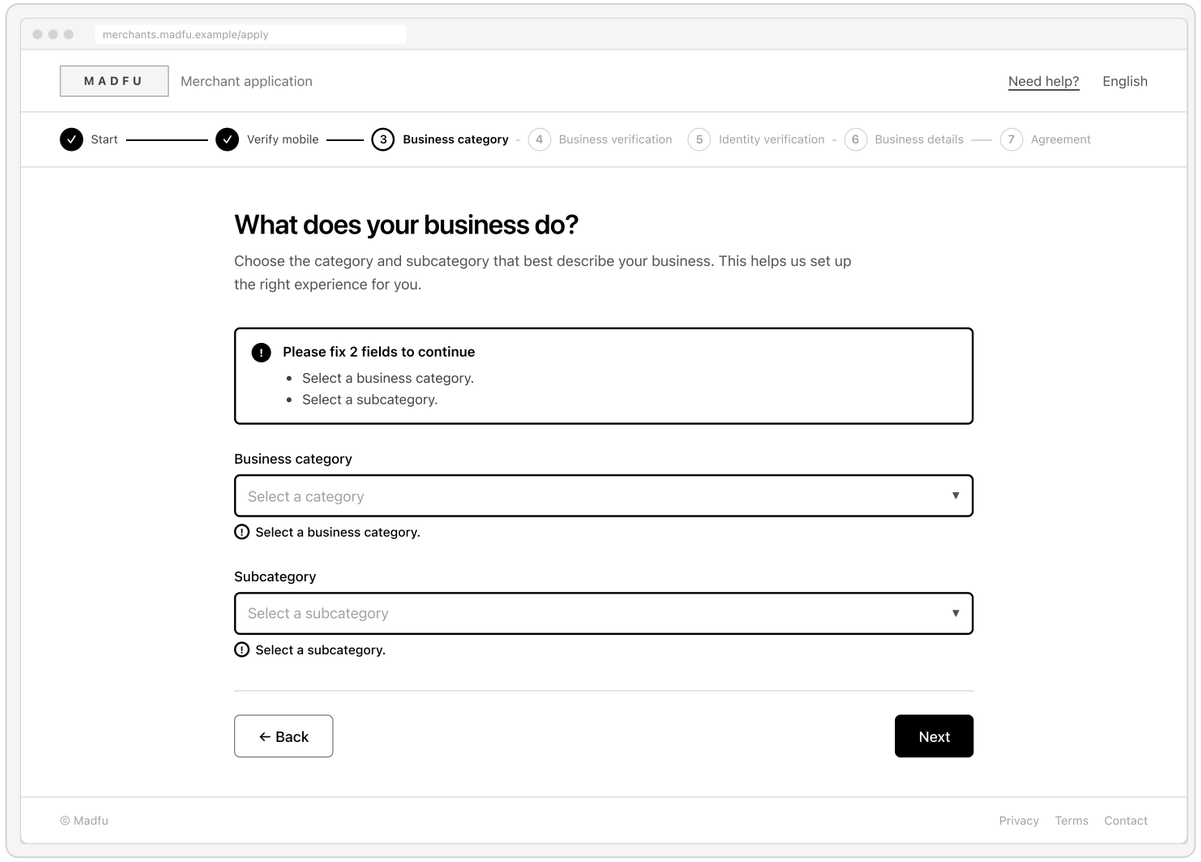
S03 No selection

Business category · main page p. 12



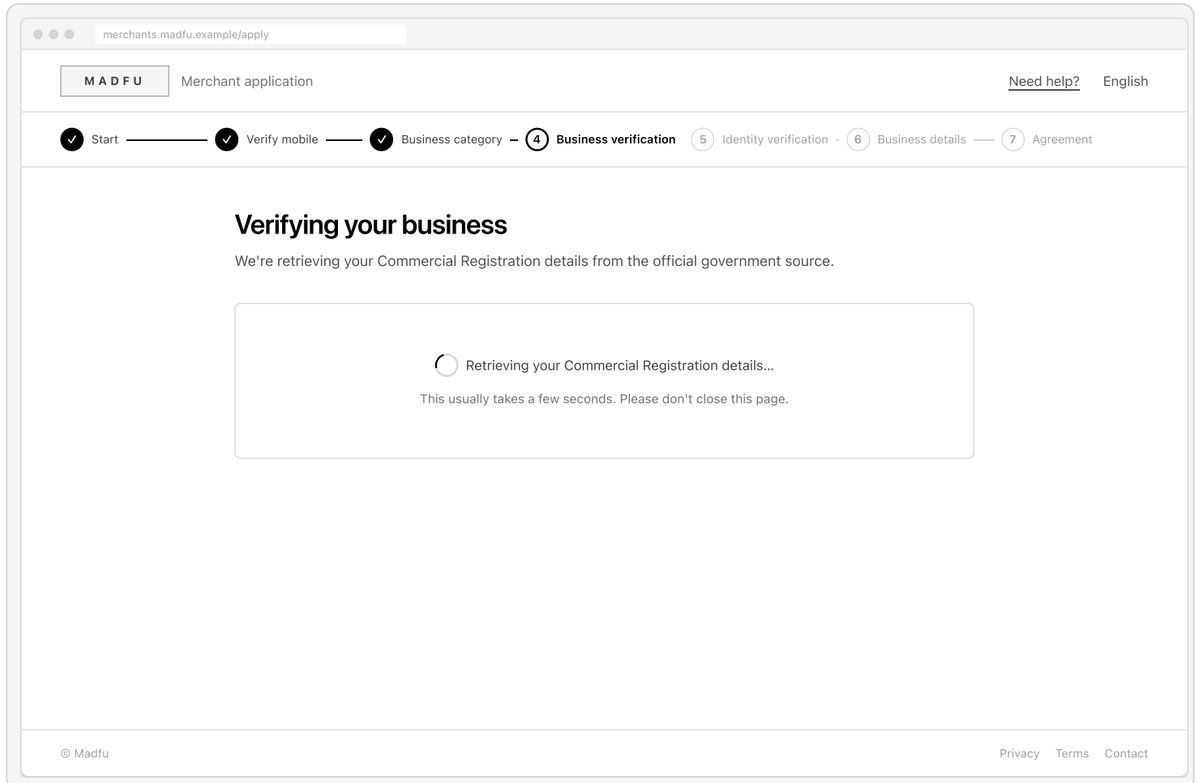
S03 Validation — nothing selected

Business category · main page p. 12



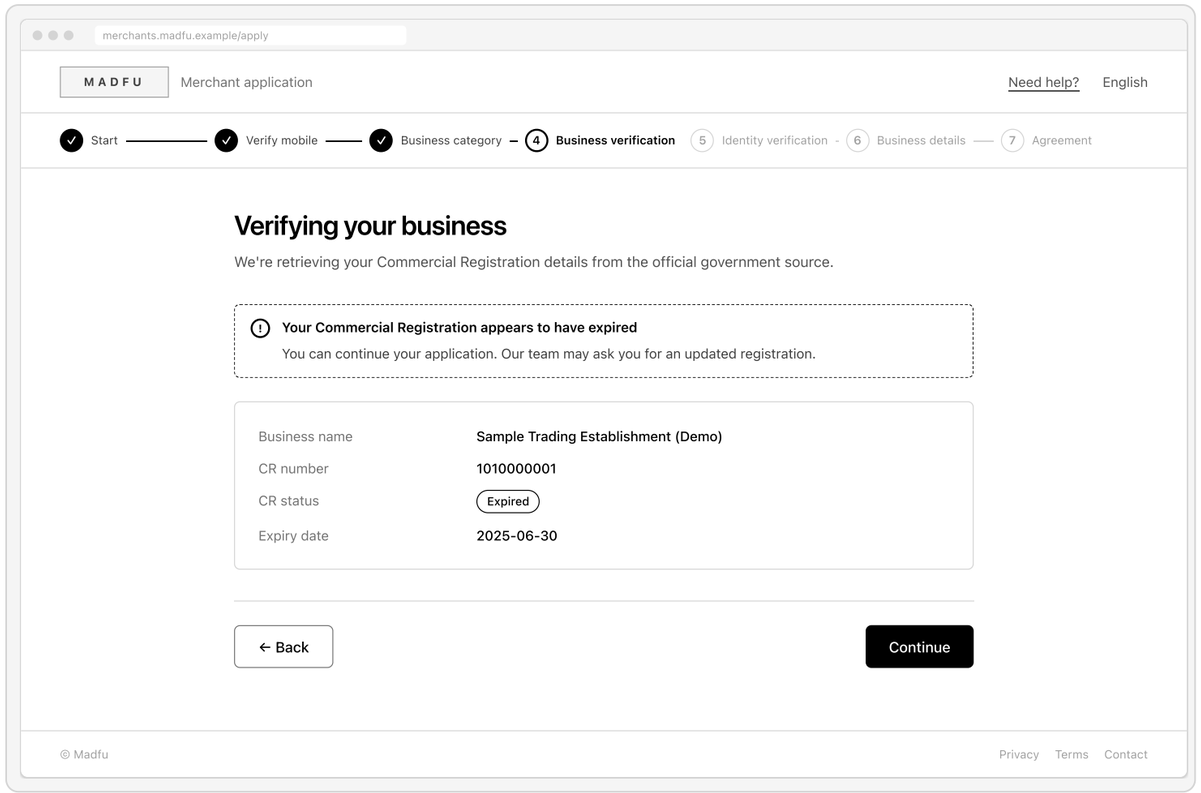
S04 Loading — retrieving CR

Business verification (CR lookup) · main page p. 15



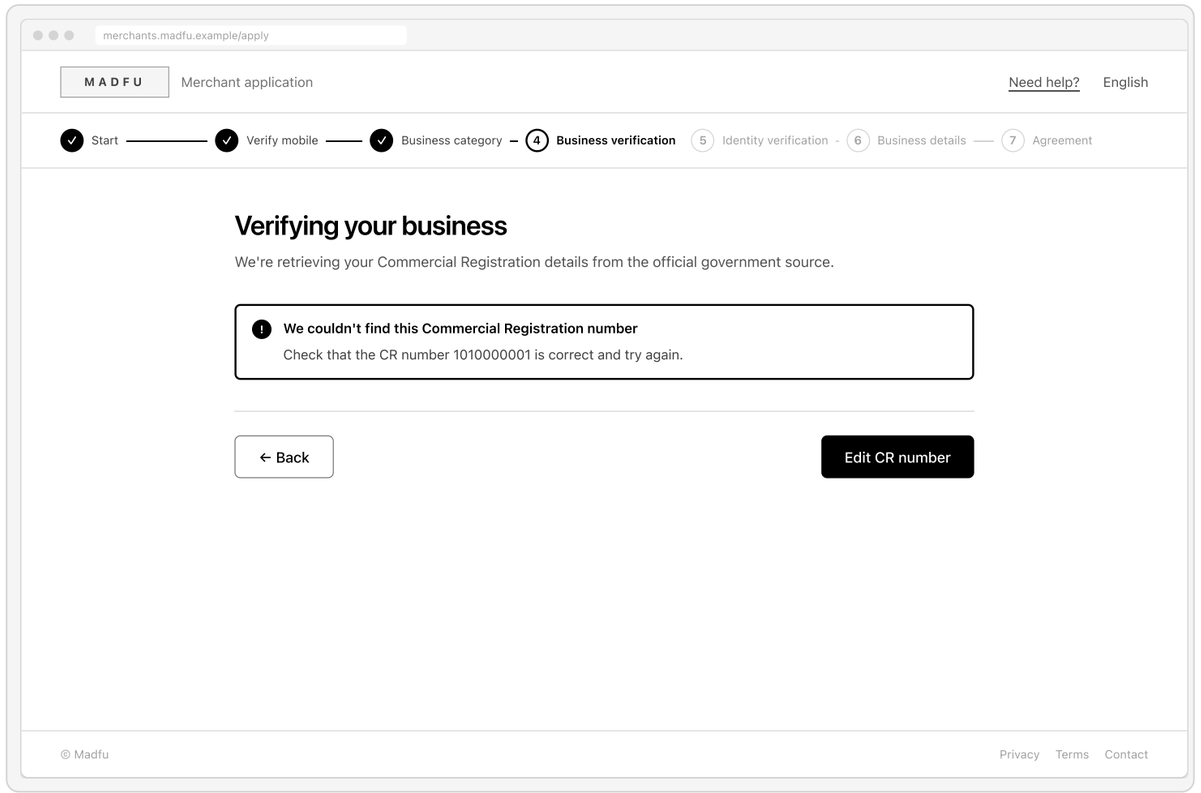
S04 Expired CR (may continue, flagged)

Business verification (CR lookup) · main page p. 15



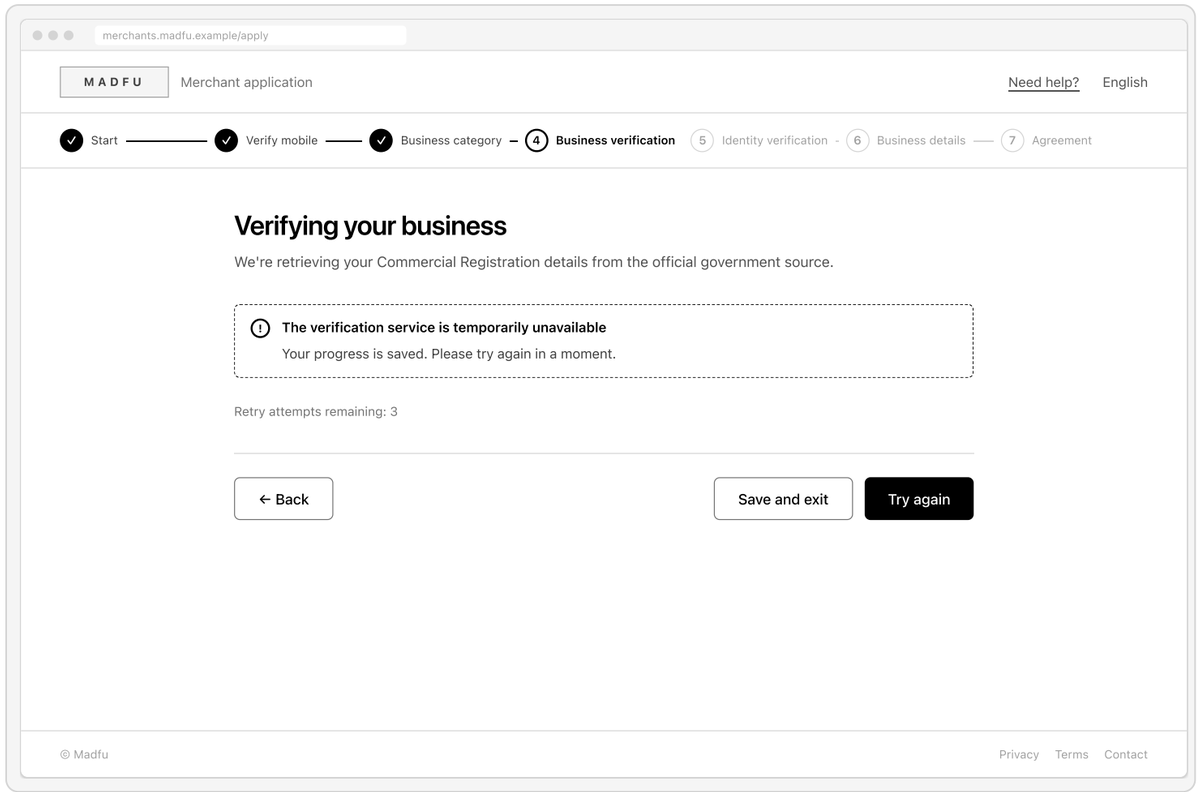
S04 CR not found

Business verification (CR lookup) · main page p. 15



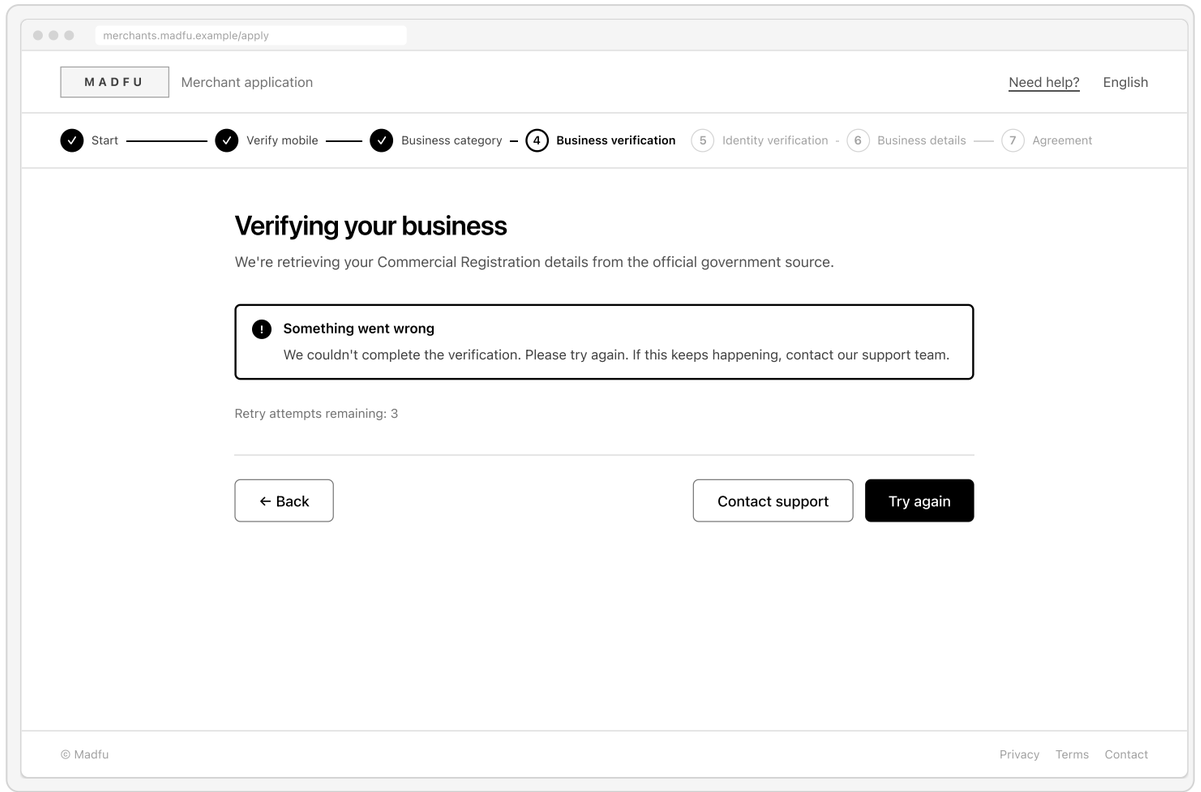
S04 Service unavailable after automatic retries

Business verification (CR lookup) · main page p. 15



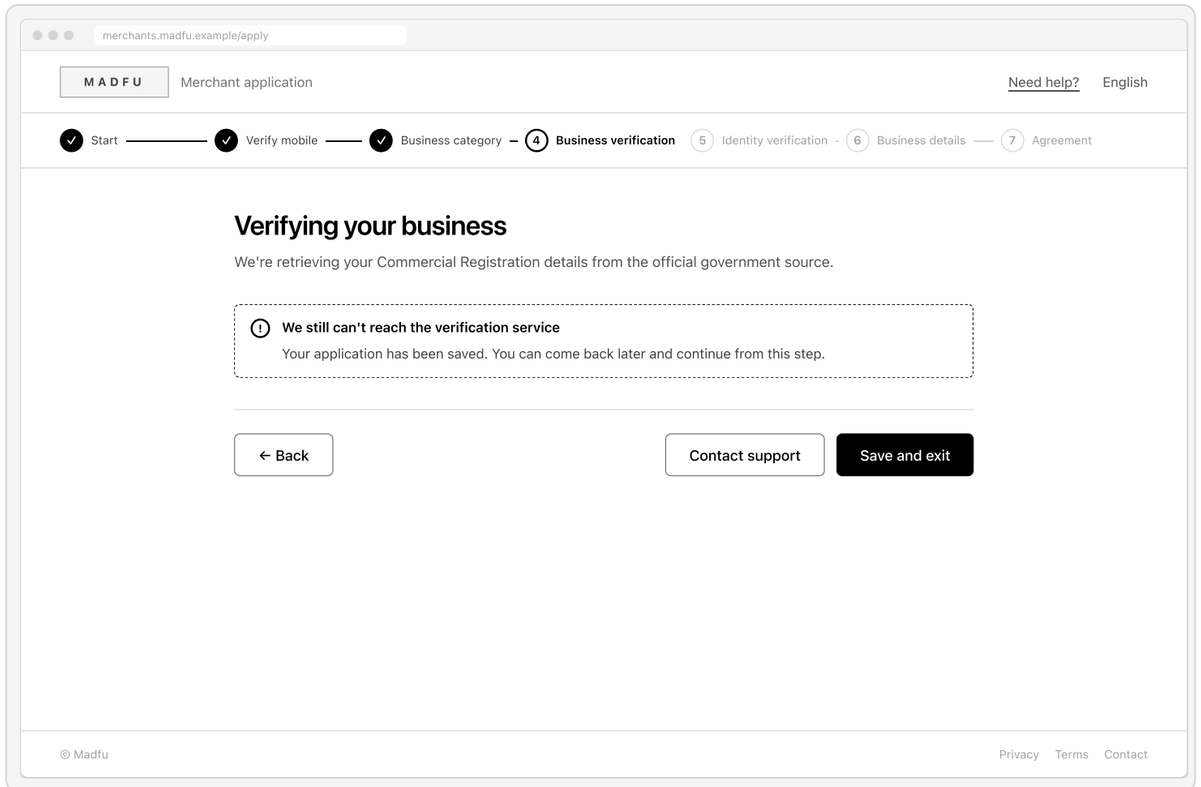
S04 Unexpected provider error

Business verification (CR lookup) · main page p. 15



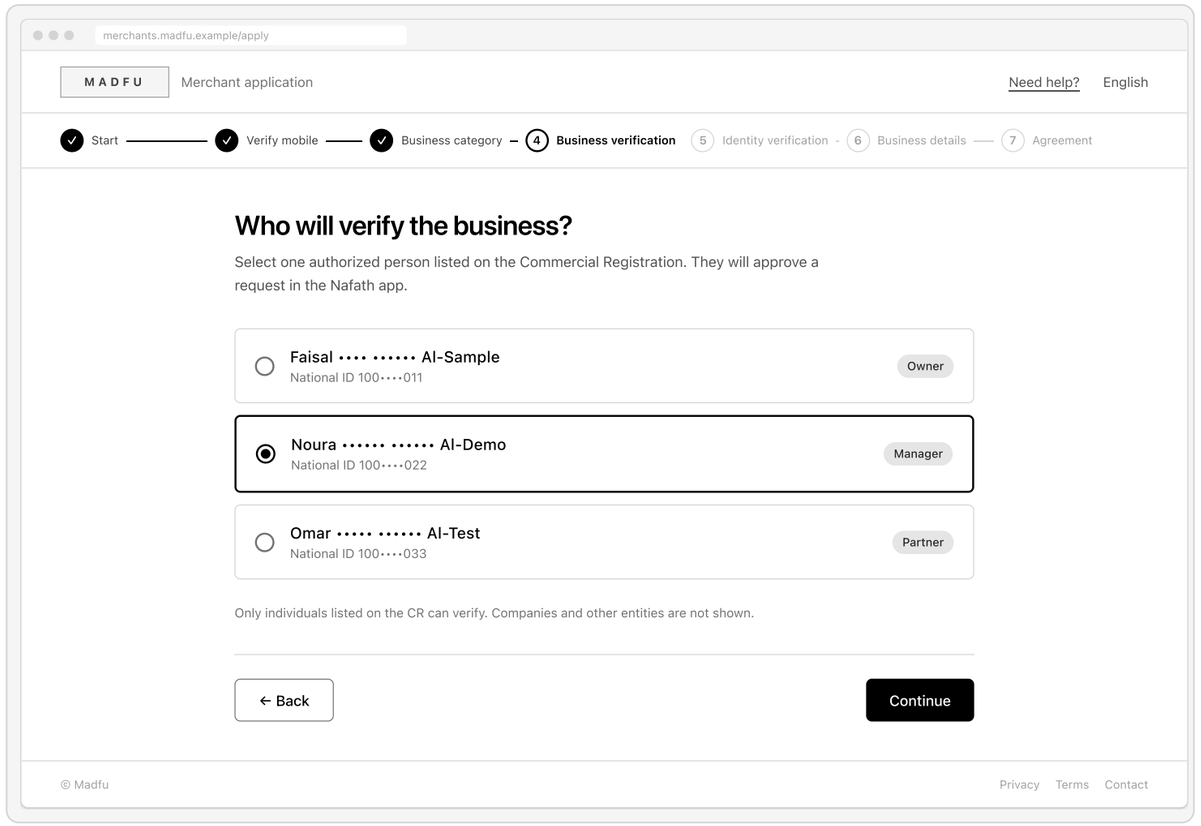
S04 Retry limit reached — progress saved

Business verification (CR lookup) · main page p. 15



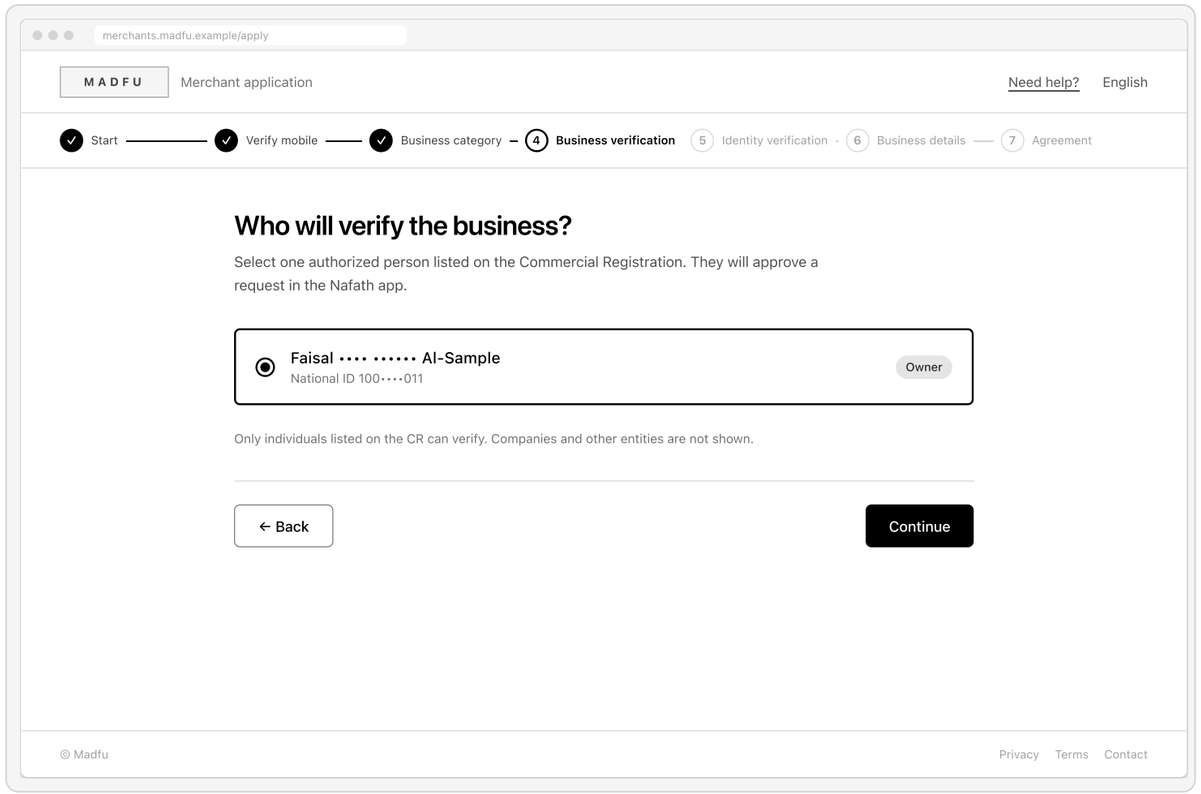
S05 One individual selected

Authorized person selection · main page p. 16



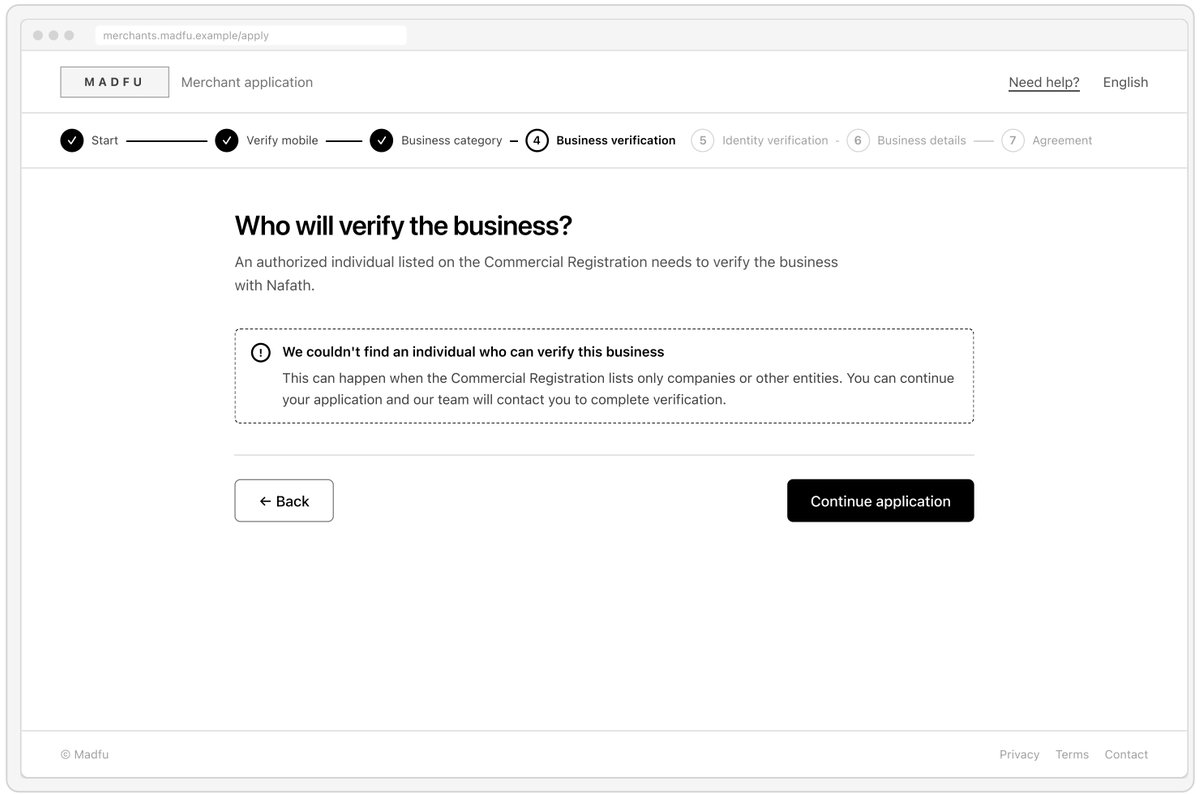
S05 Single individual (pre-selected)

Authorized person selection · main page p. 16



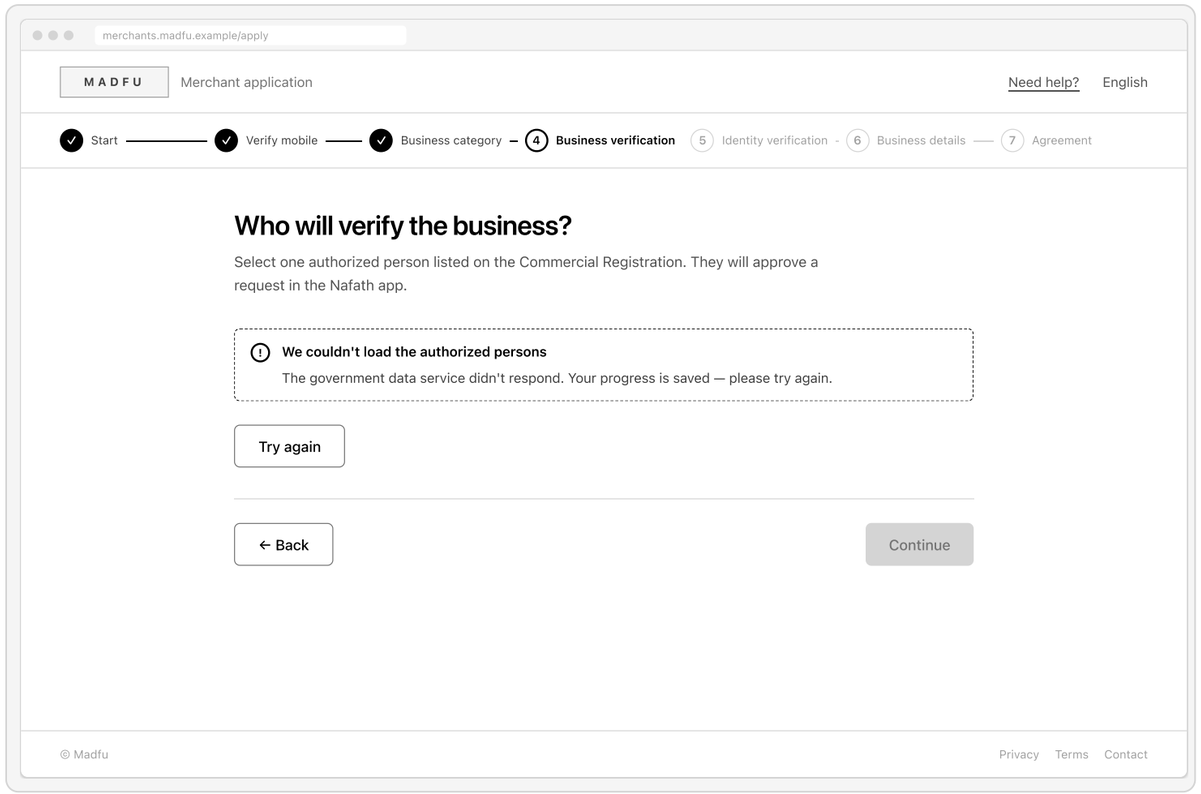
S05 No eligible individual (entities only)

Authorized person selection · main page p. 16



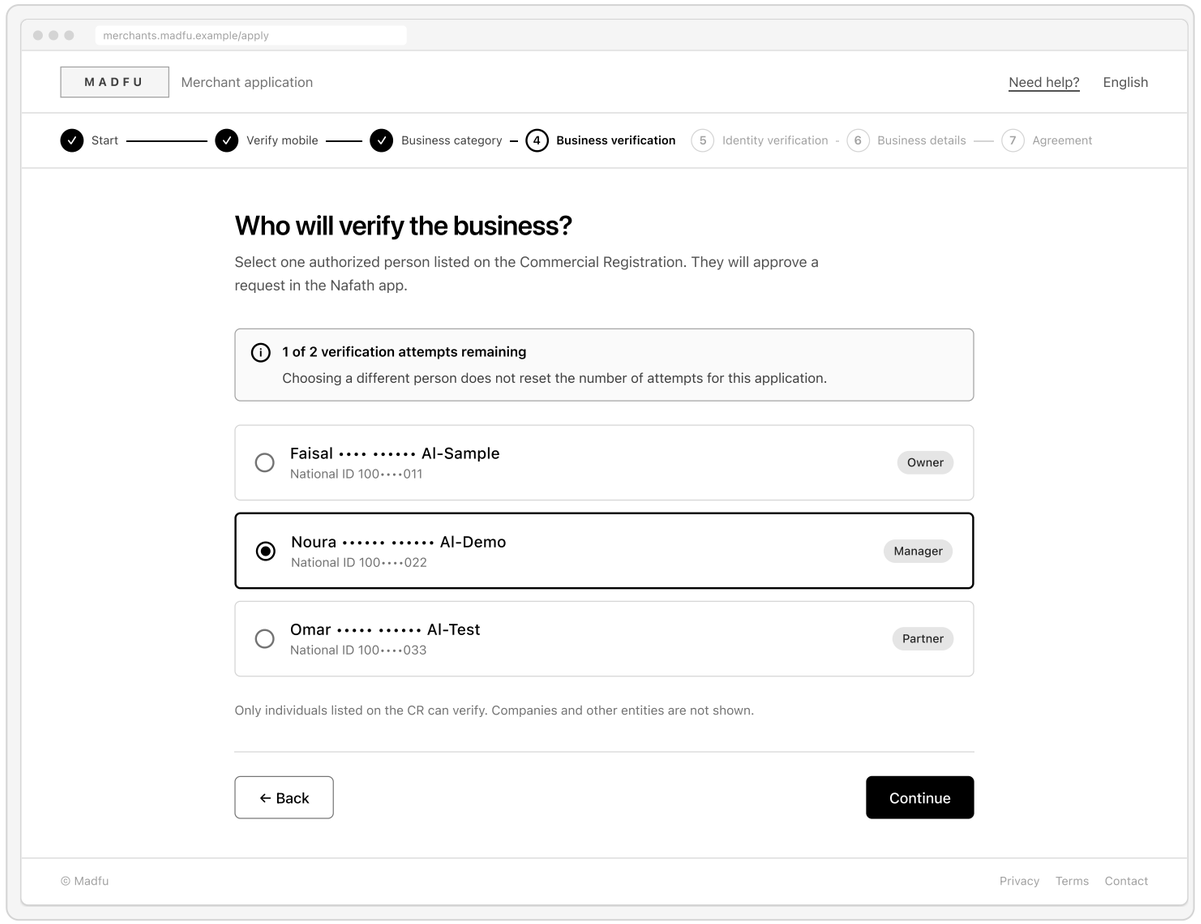
S05 Related-party data unavailable

Authorized person selection · main page p. 16



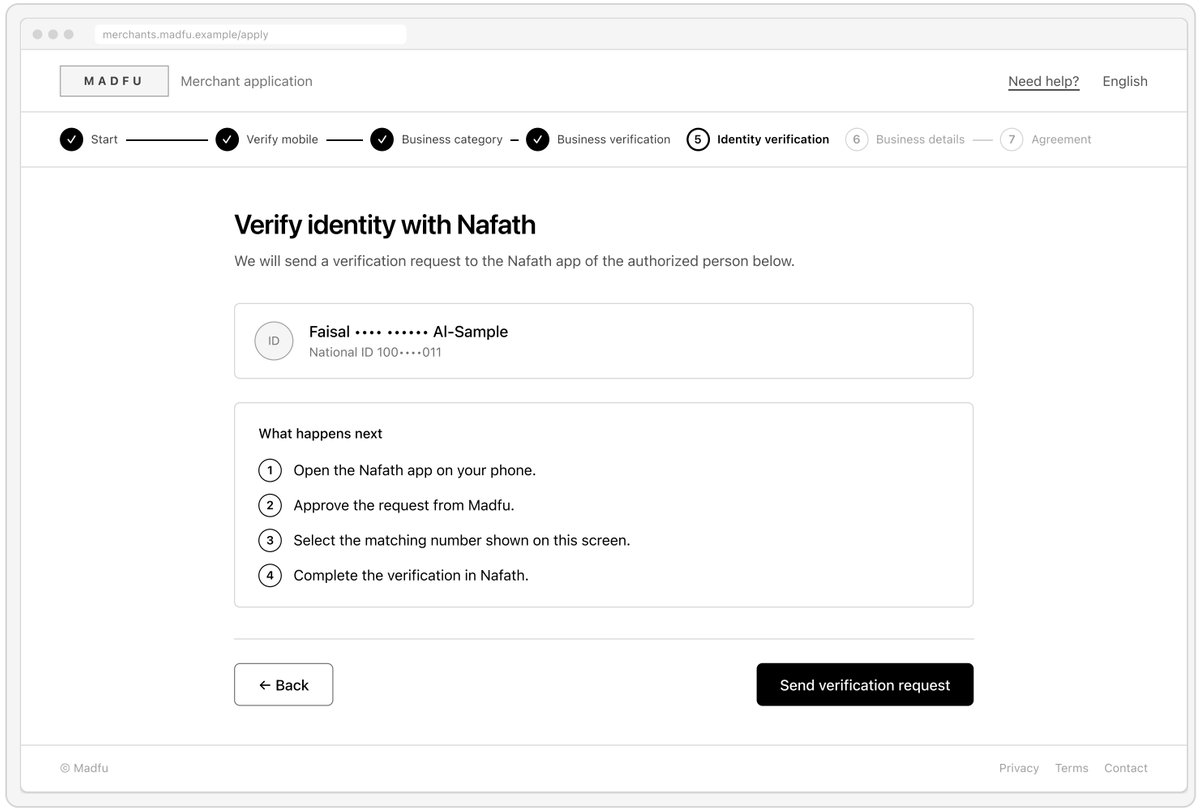
S05 Changing person after a failed attempt (counter preserved)

Authorized person selection · main page p. 16



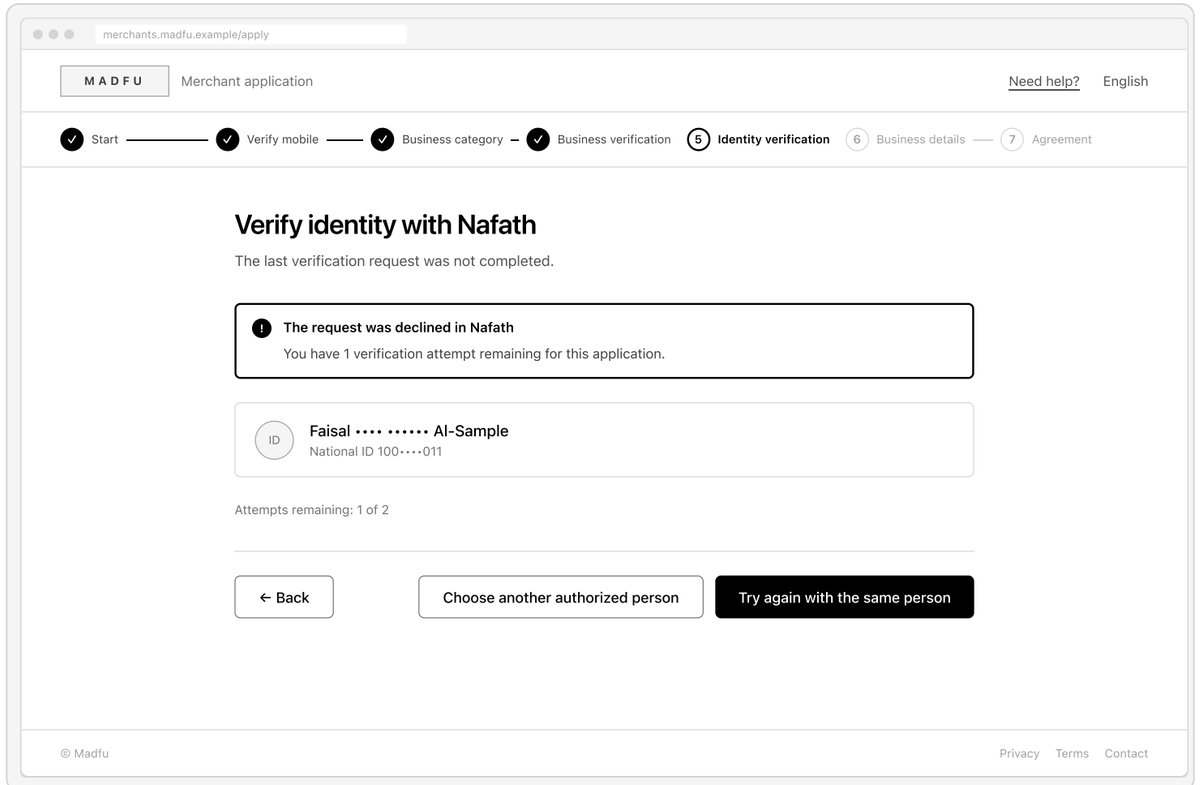
S06 Ready to send (CR, selected person)

Nafath identity verification · main page p. 17



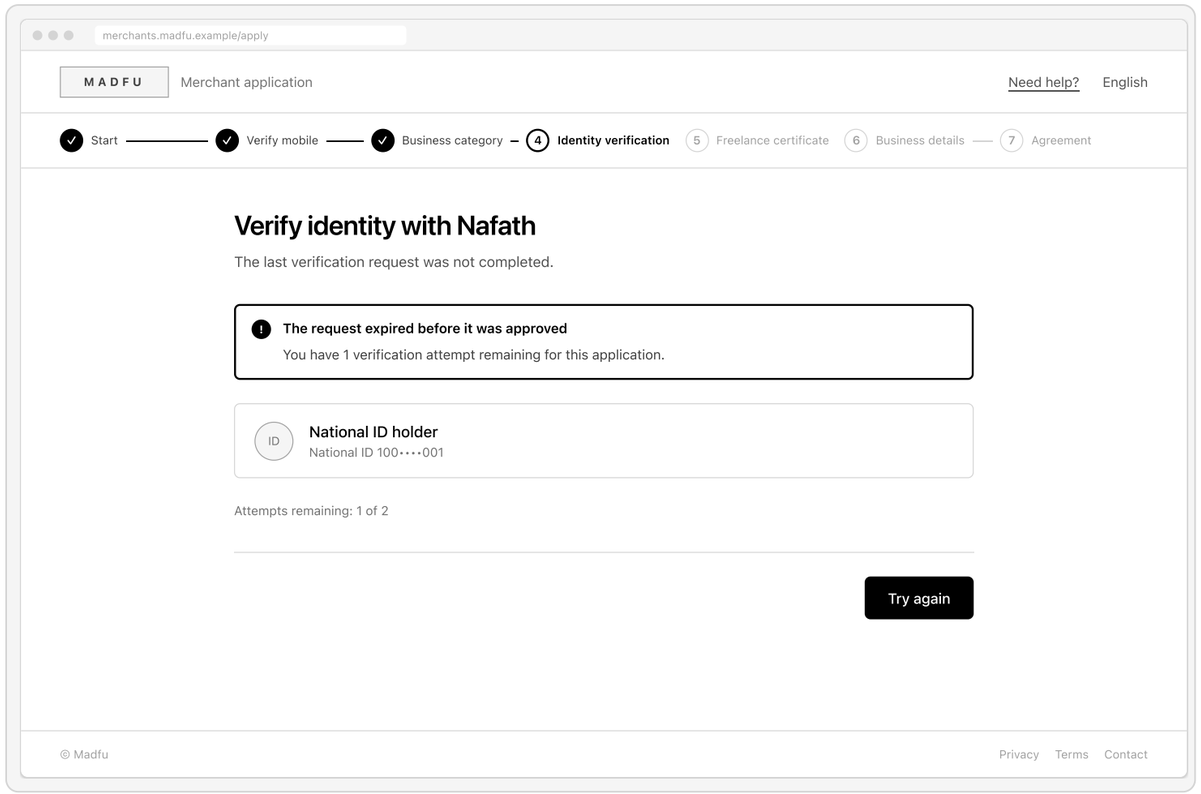
S06 Rejected — 1 attempt remaining (CR)

Nafath identity verification · main page p. 17



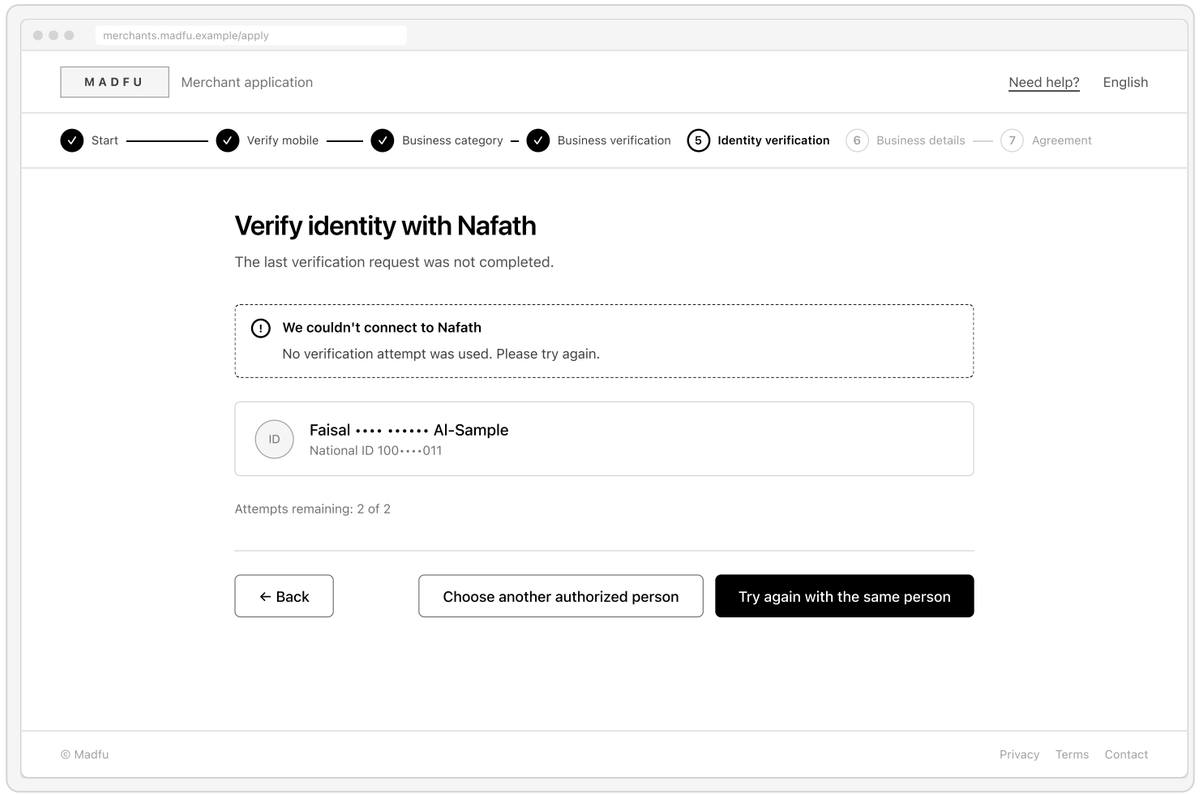
S06 Timed out — 1 attempt remaining (Freelancer)

Nafath identity verification · main page p. 17



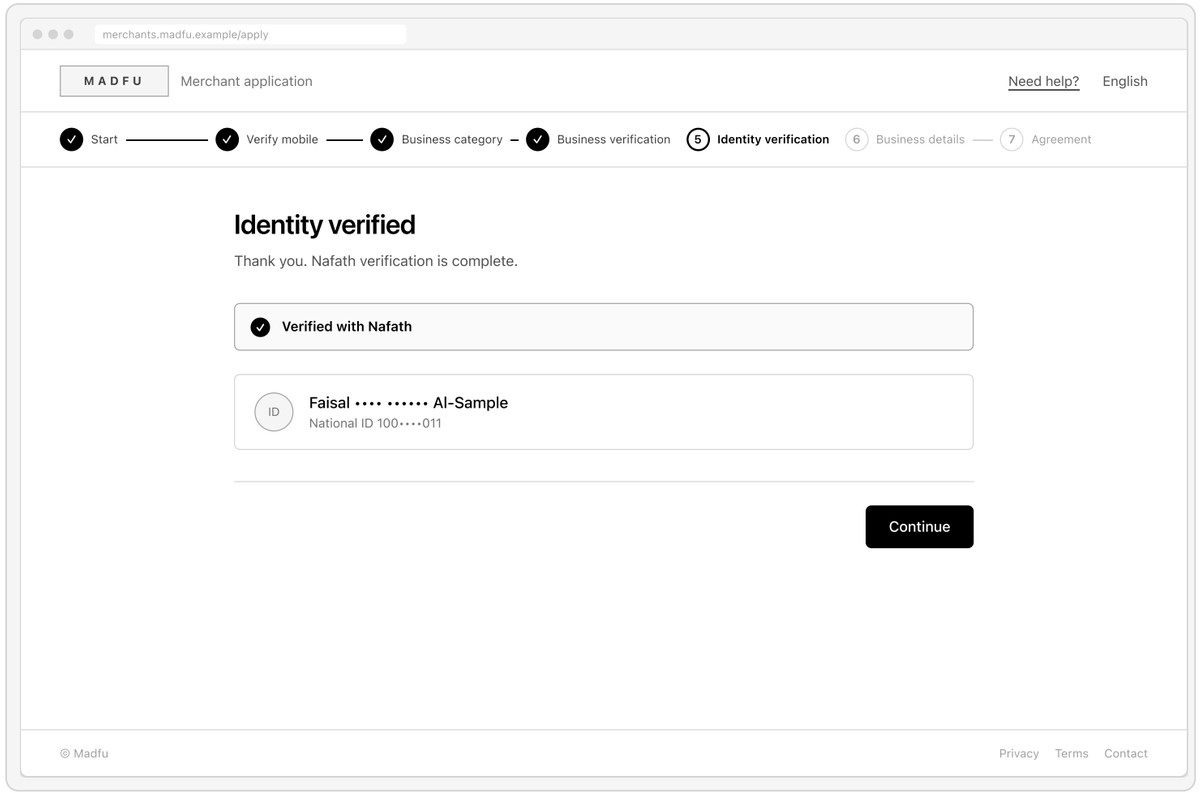
S06 Technical failure — no attempt consumed

Nafath identity verification · main page p. 17



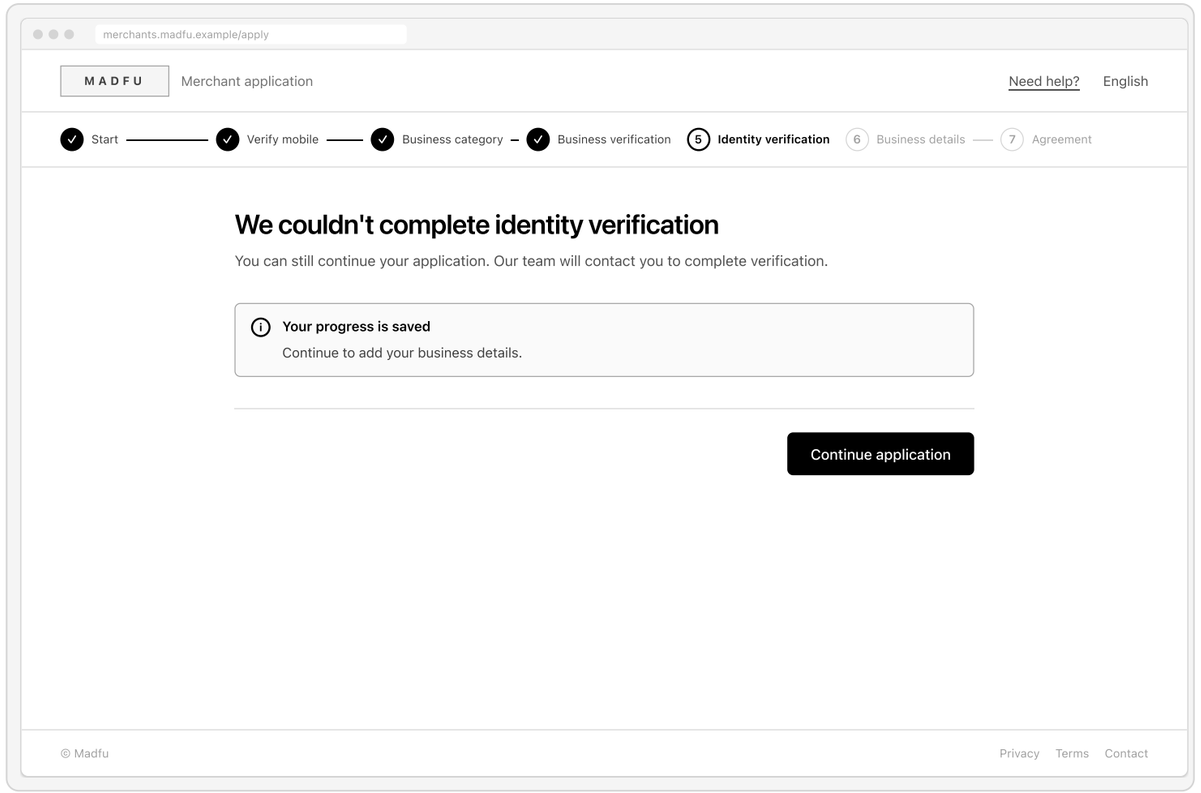
S06 Verified

Nafath identity verification · main page p. 17



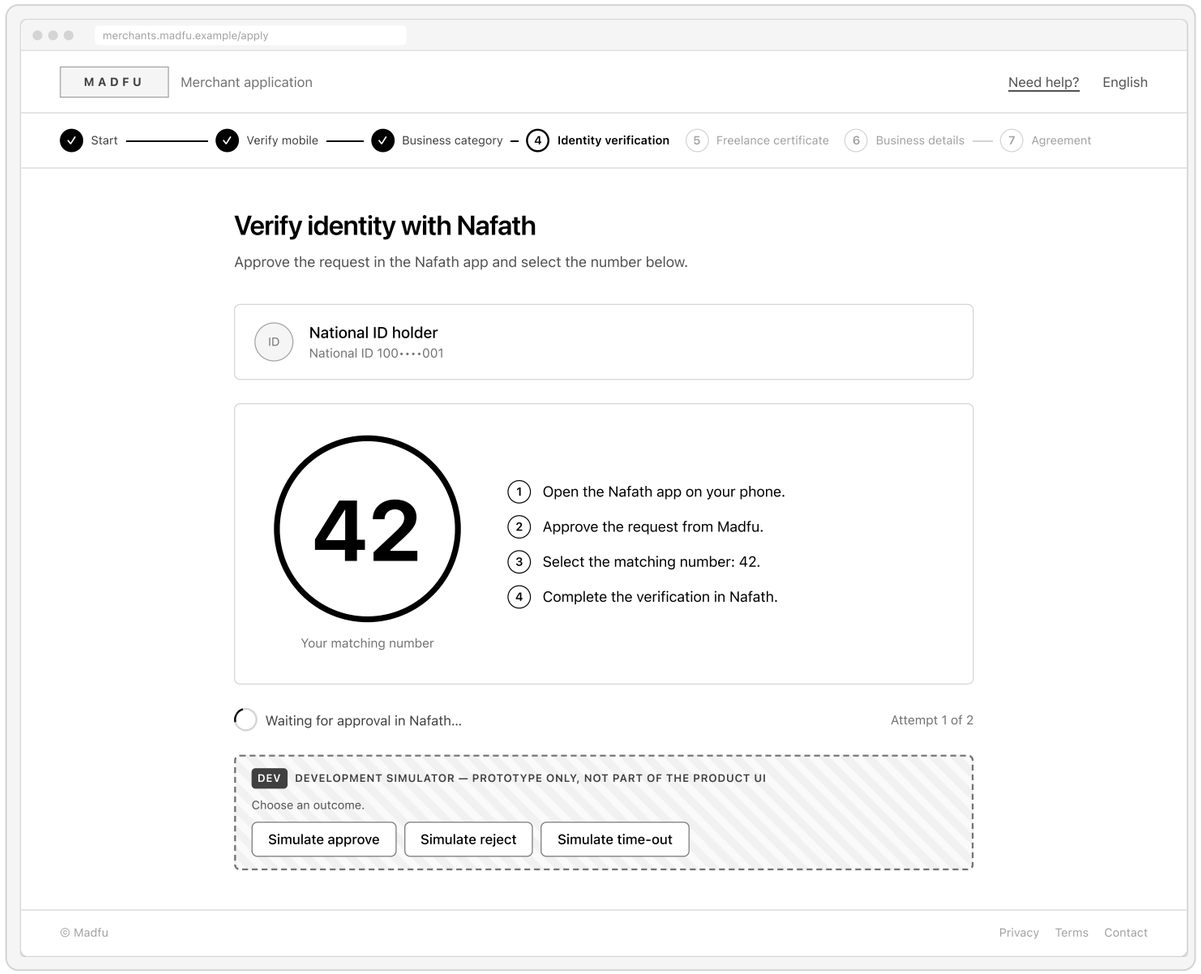
S06 Two unsuccessful attempts — manual path

Nafath identity verification · main page p. 17



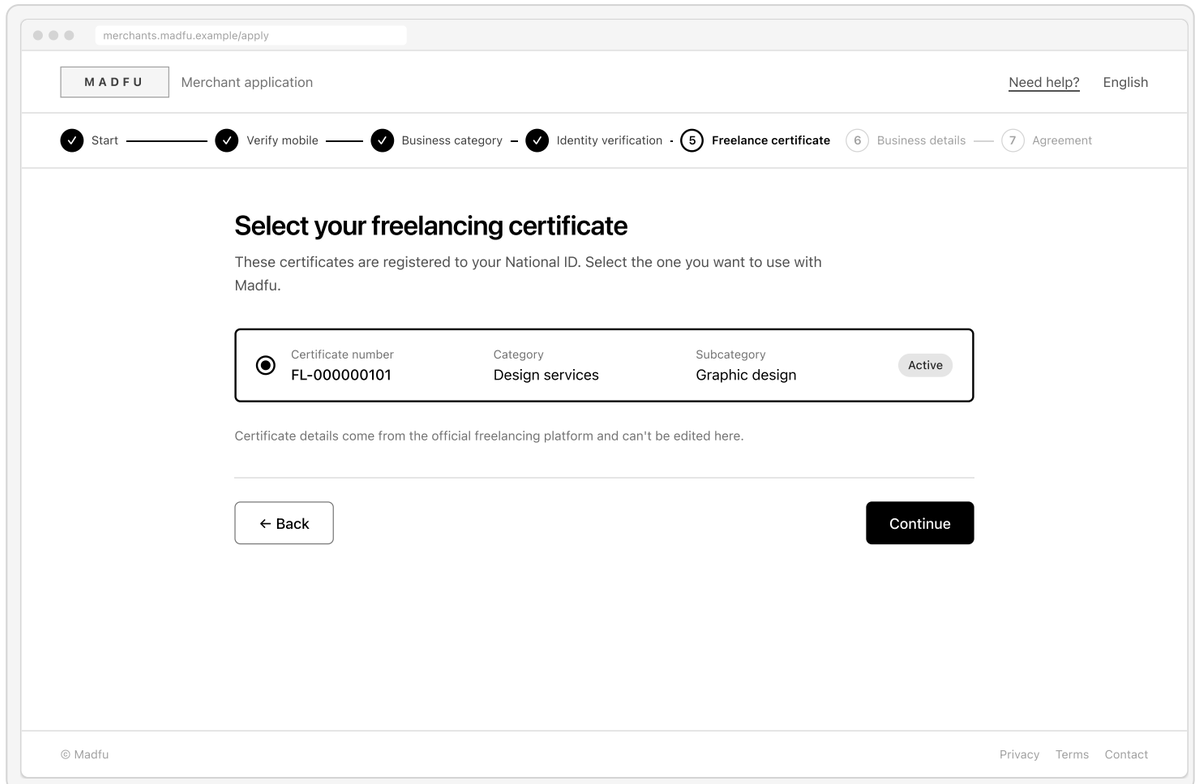
S06 Waiting — Freelancer (National ID from entry)

Nafath identity verification · main page p. 17



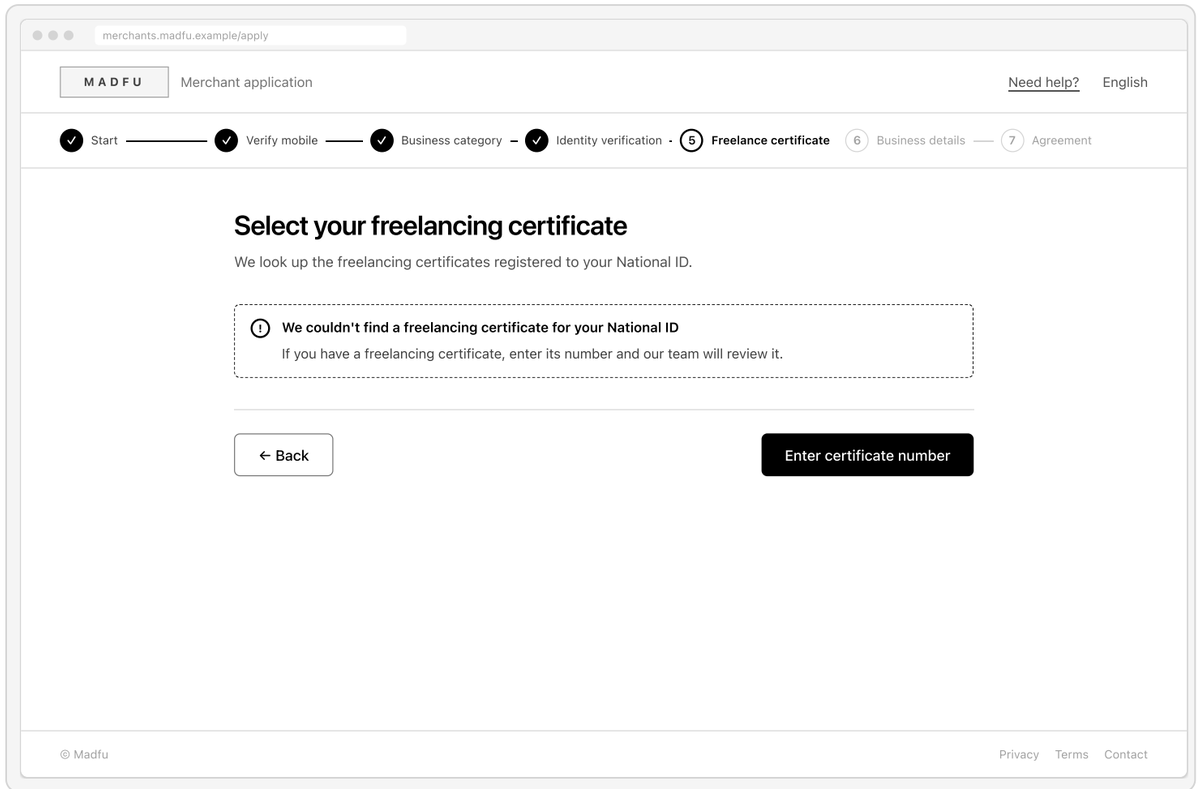
S07 One certificate (pre-selected)

Freelancing certificate selection · main page p. 21



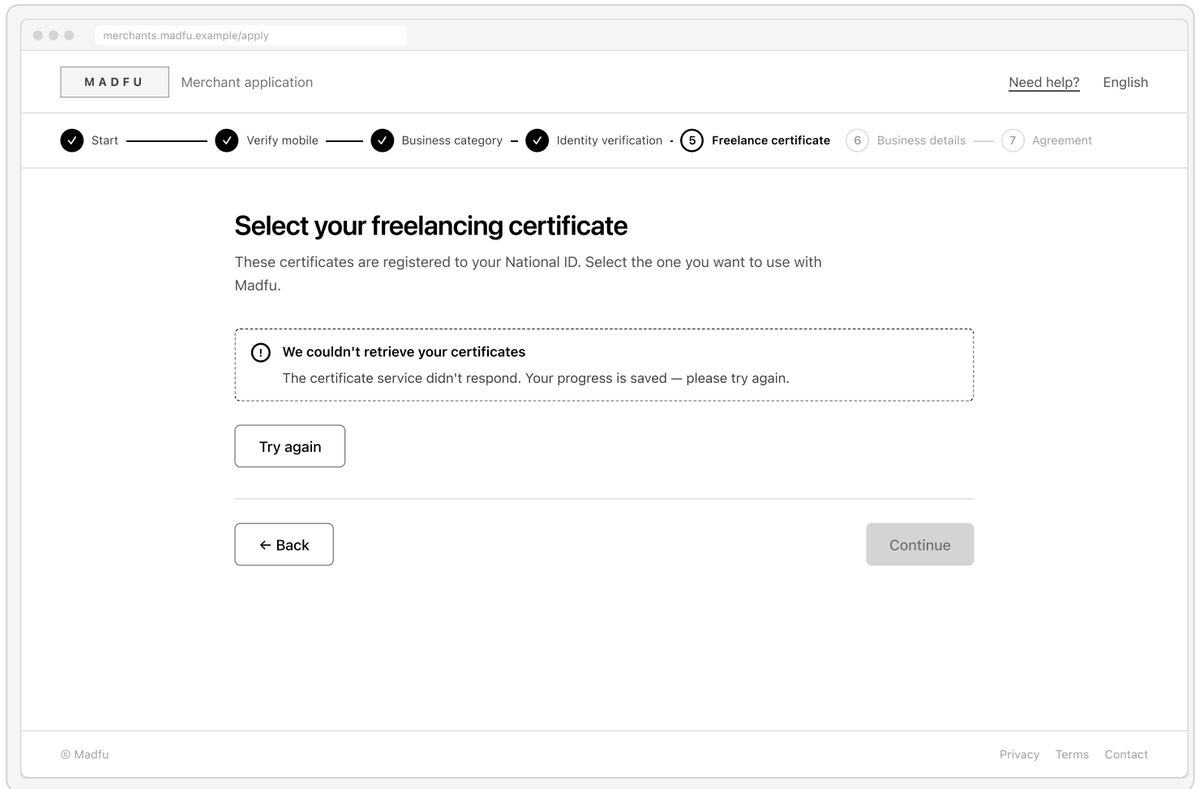
S07 No certificates found

Freelancing certificate selection · main page p. 21



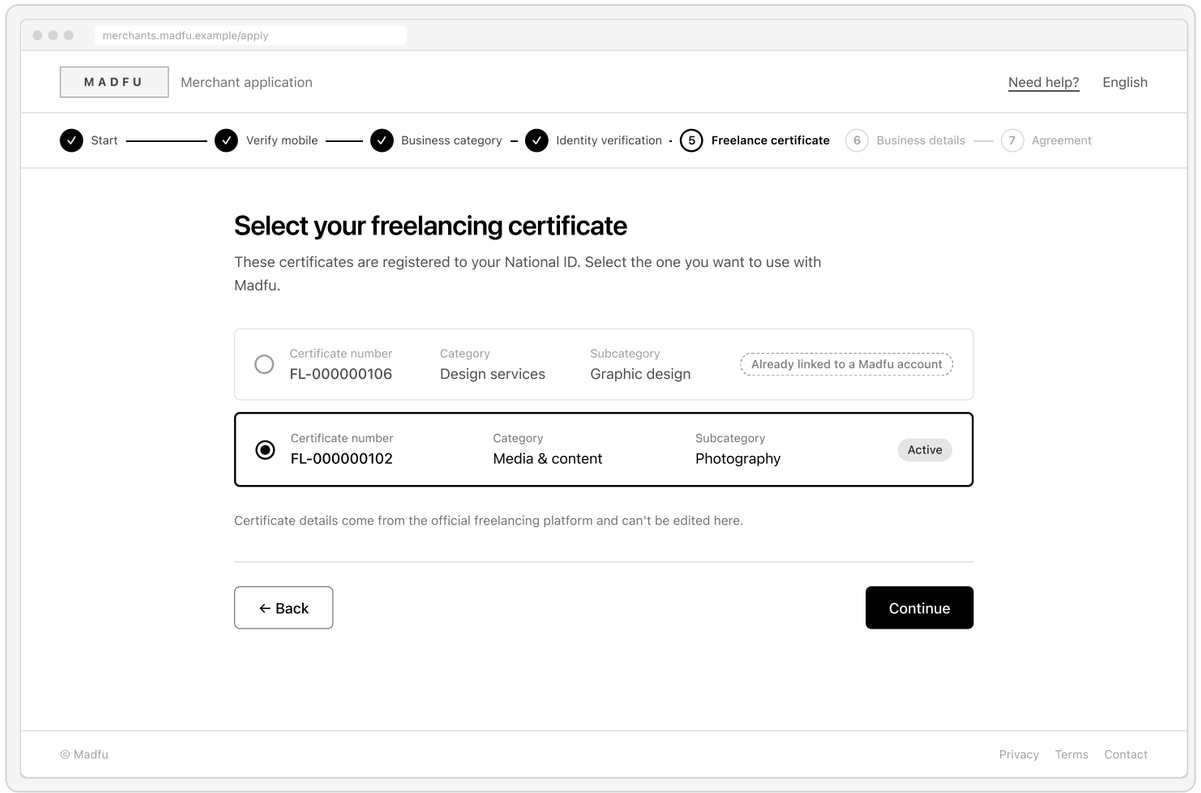
S07 Certificate service unavailable

Freelancing certificate selection · main page p. 21



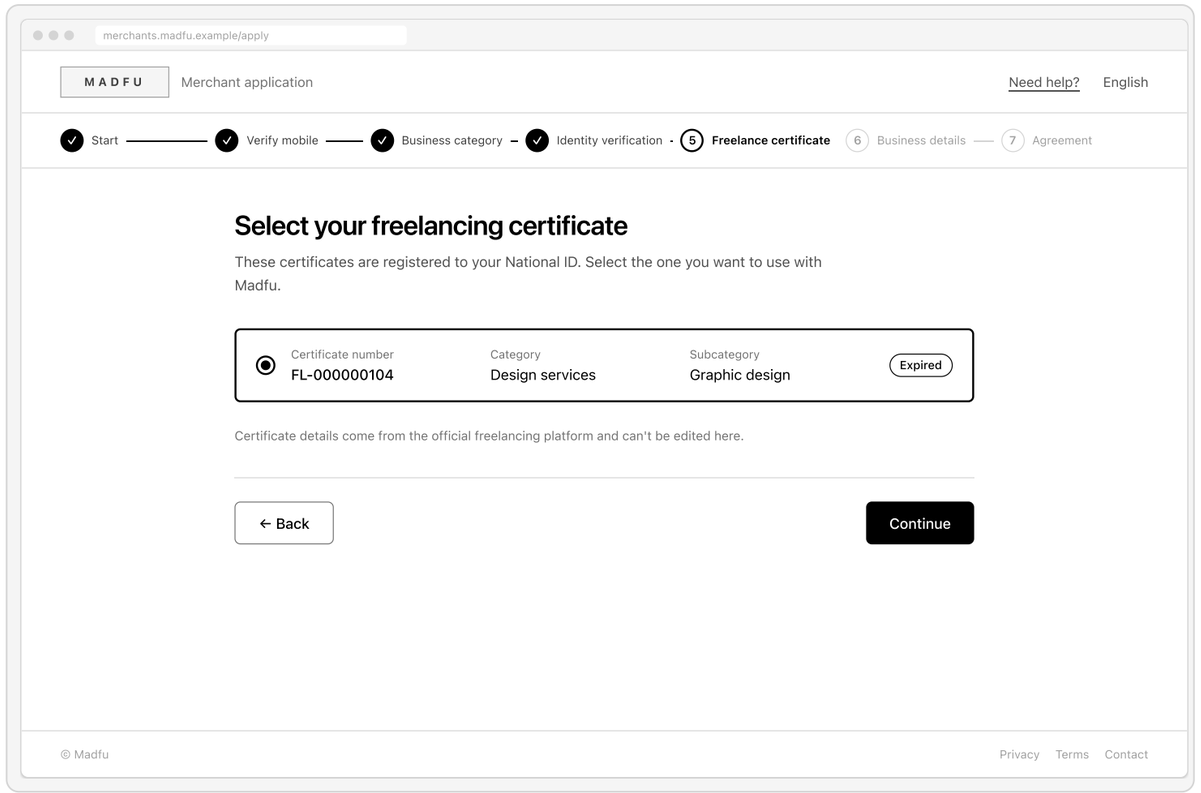
S07 Certificate already attached to an account

Freelancing certificate selection · main page p. 21



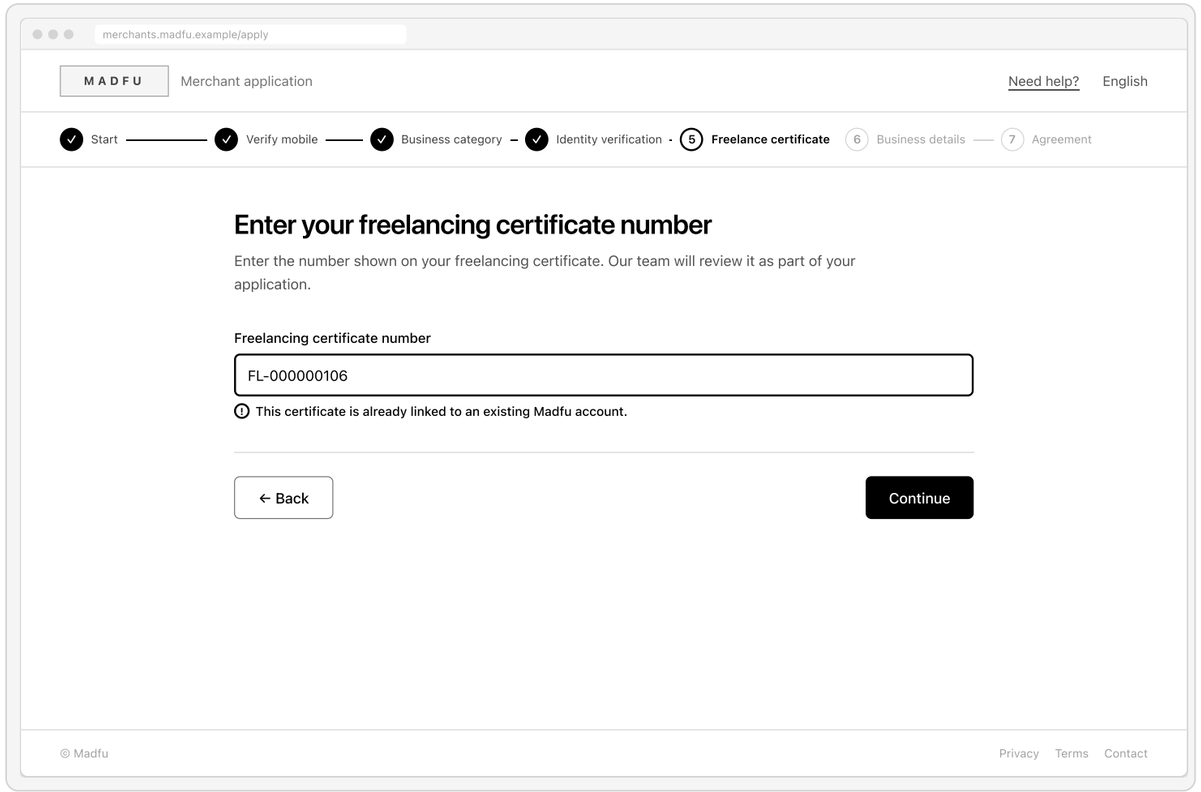
S07 Expired certificate

Freelancing certificate selection · main page p. 21



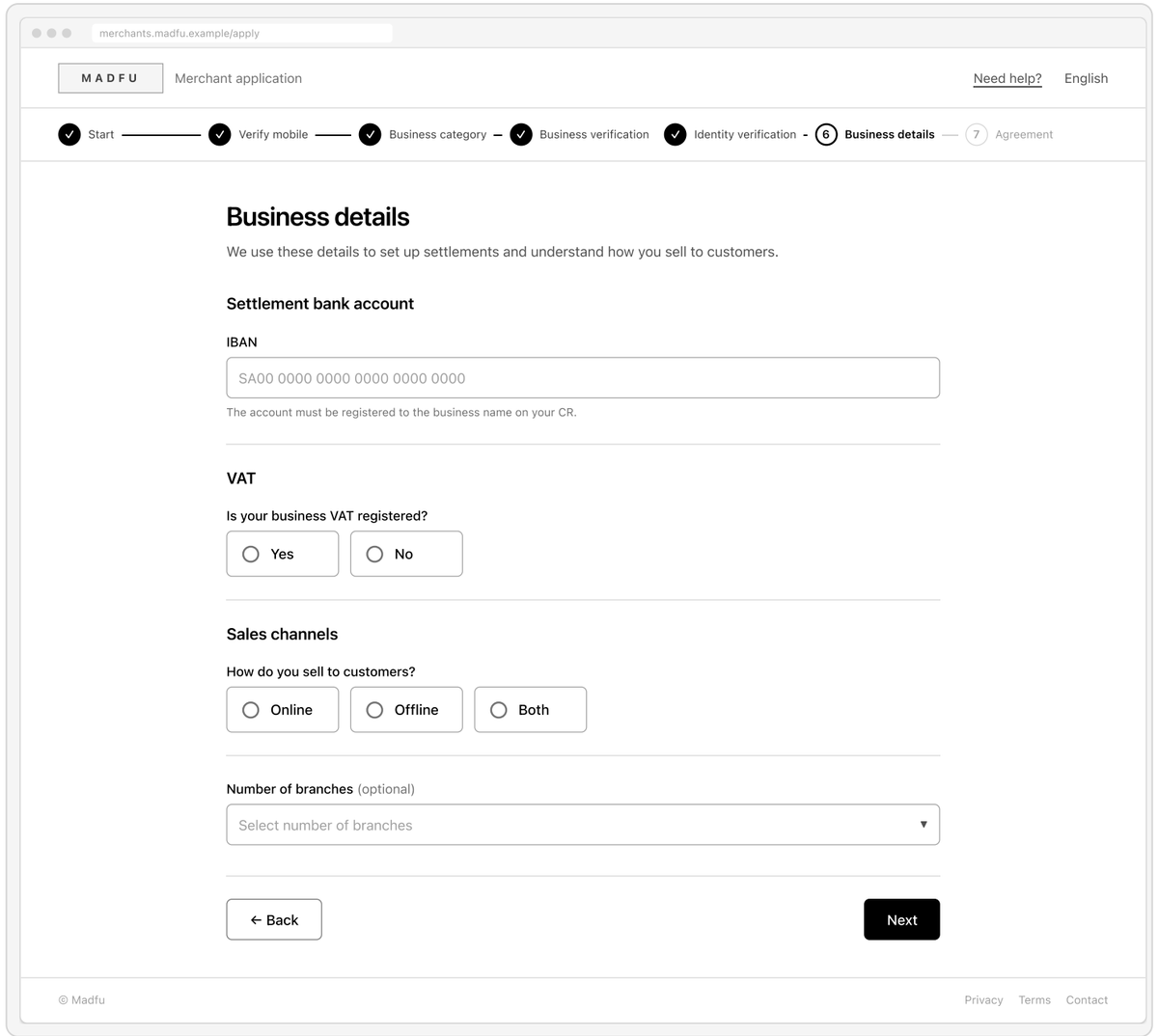
S07-M Certificate already linked to an account

Manual certificate entry · main page p. 22



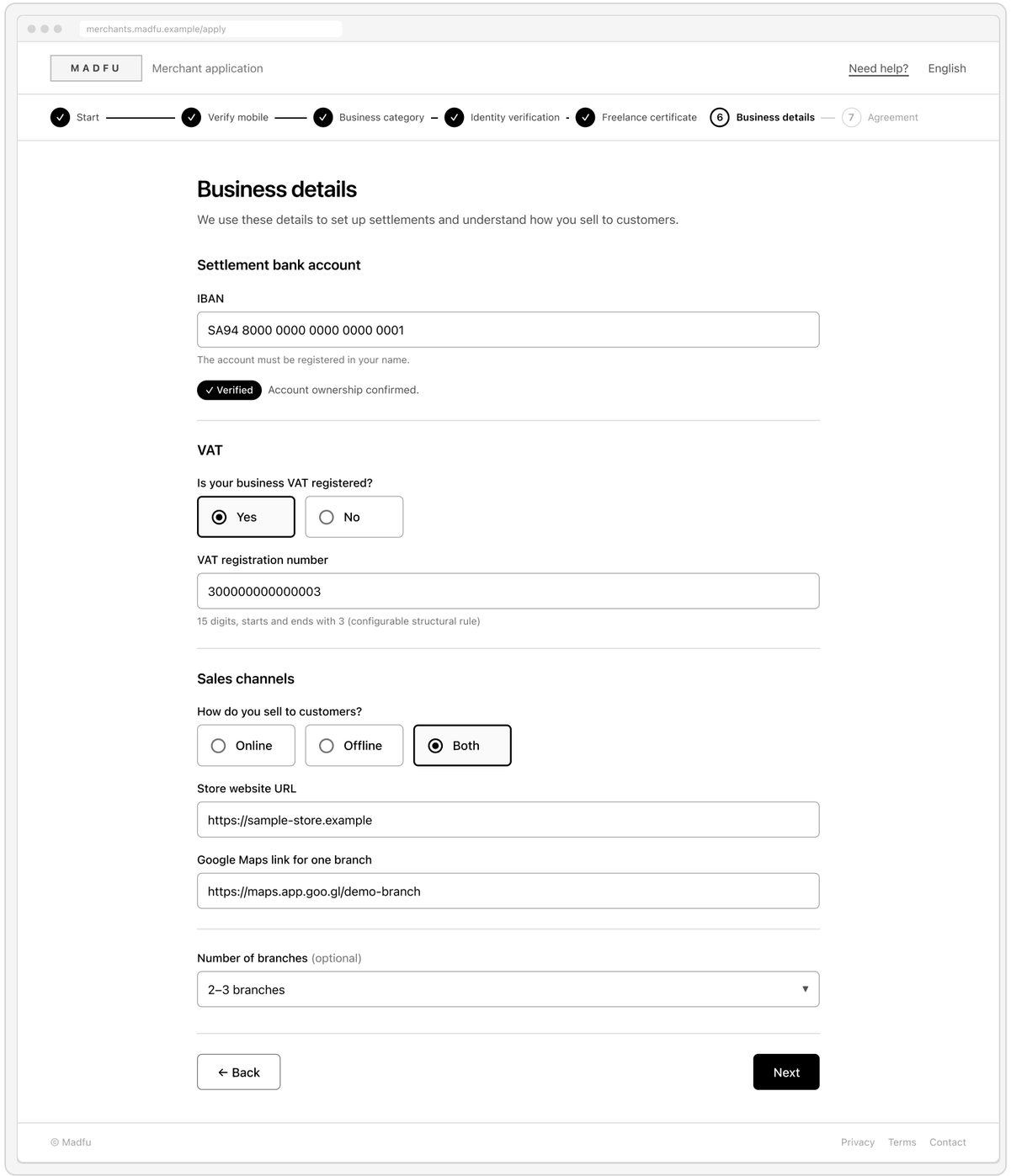
S08 Empty form

Business details (shared) · main page p. 26



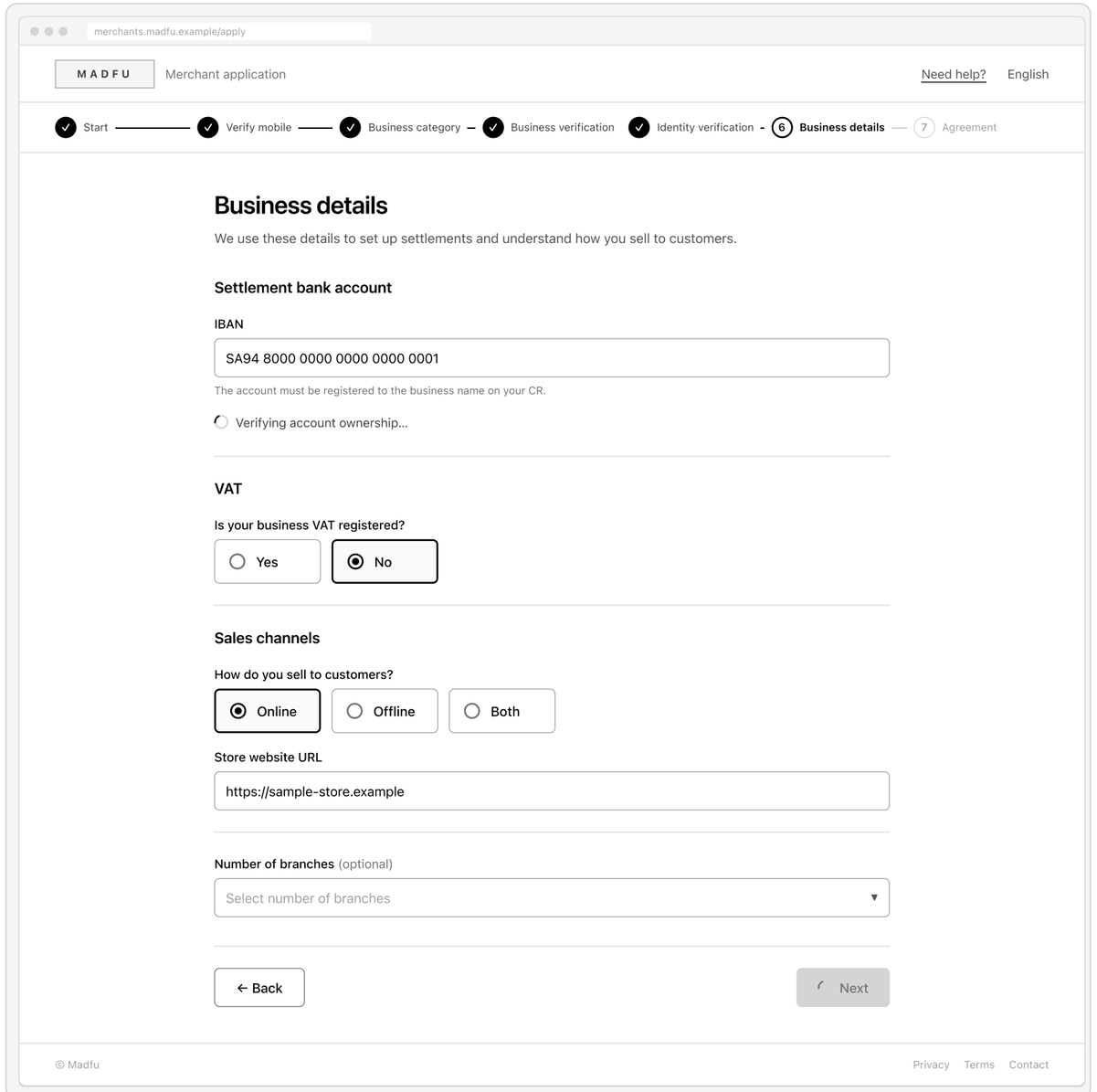
S08 Freelancer — IBAN verified against National ID

Business details (shared) · main page p. 26



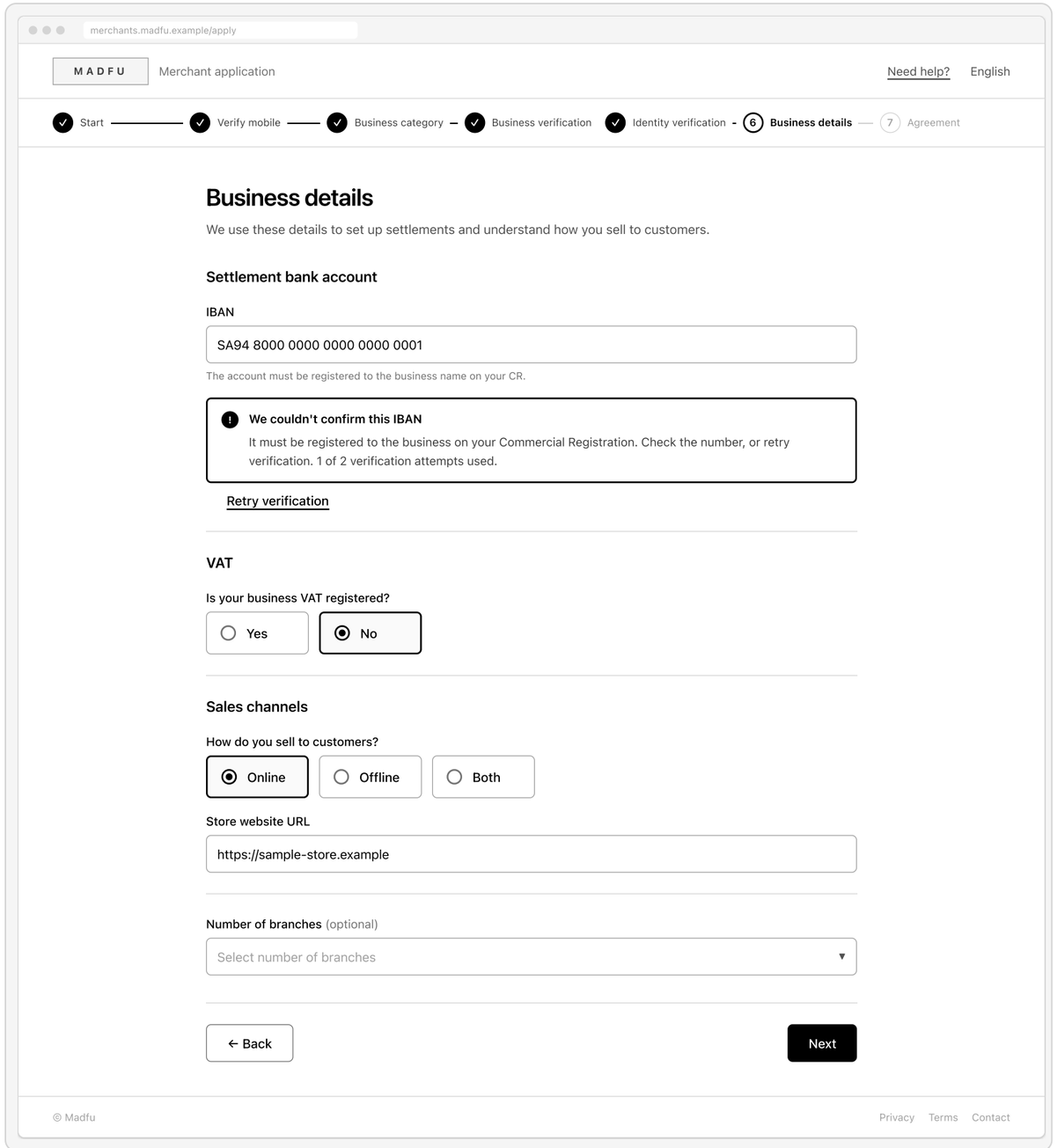
S08 IBAN verifying

Business details (shared) · main page p. 26



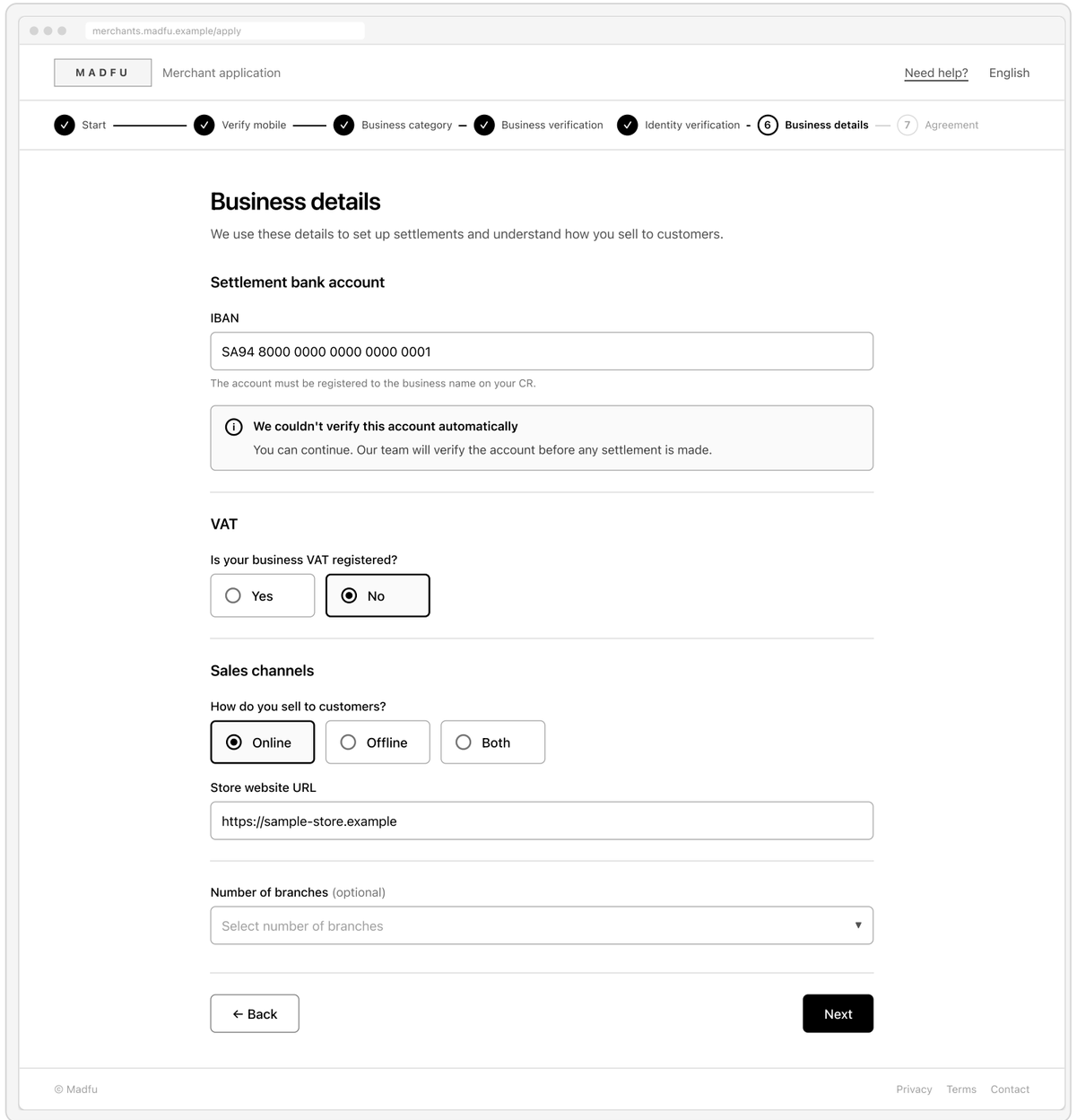
S08 IBAN not verified (1 of 2)

Business details (shared) · main page p. 26



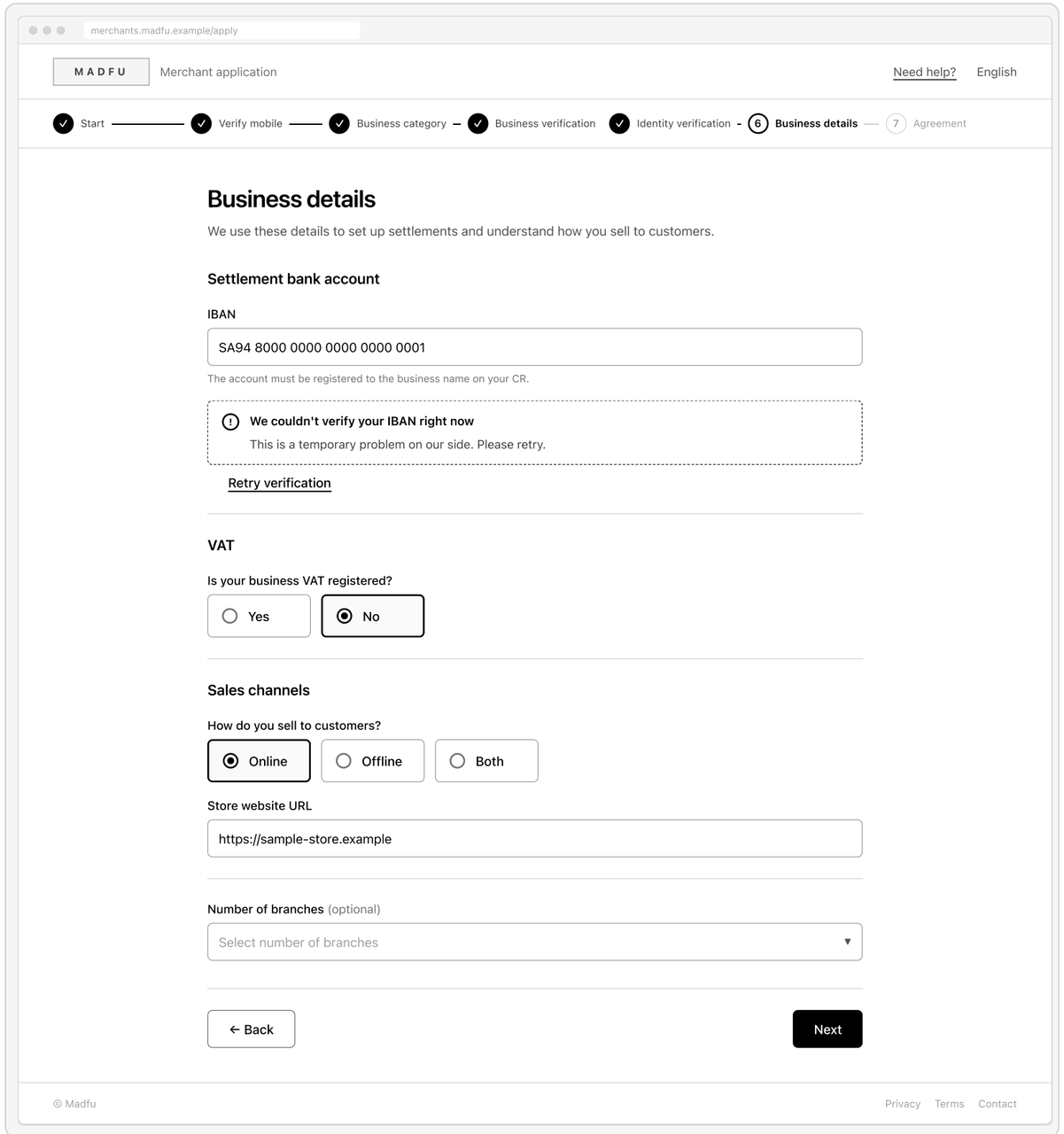
S08 IBAN not verified twice — may continue

Business details (shared) · main page p. 26



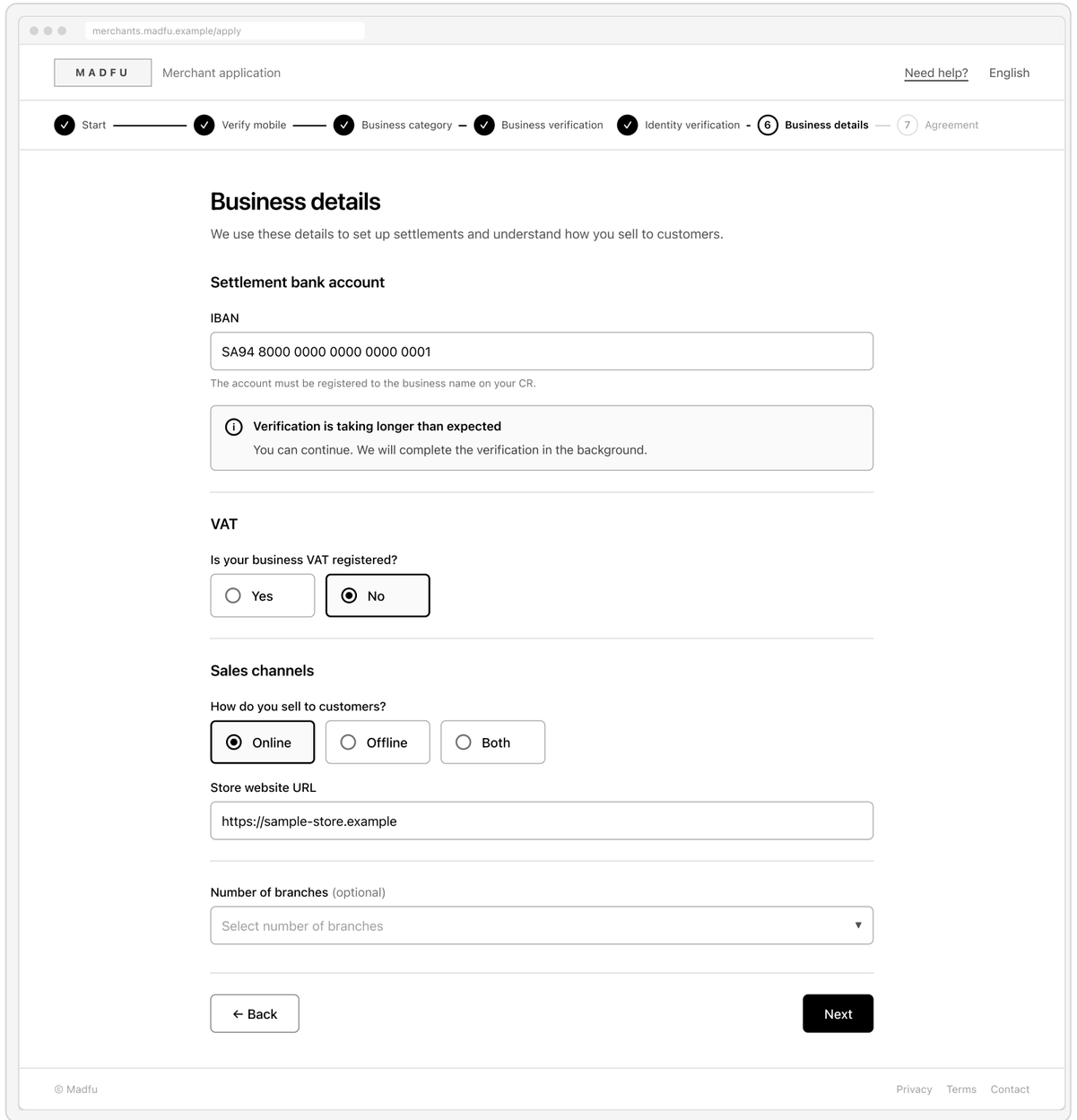
S08 IBAN technical error

Business details (shared) · main page p. 26



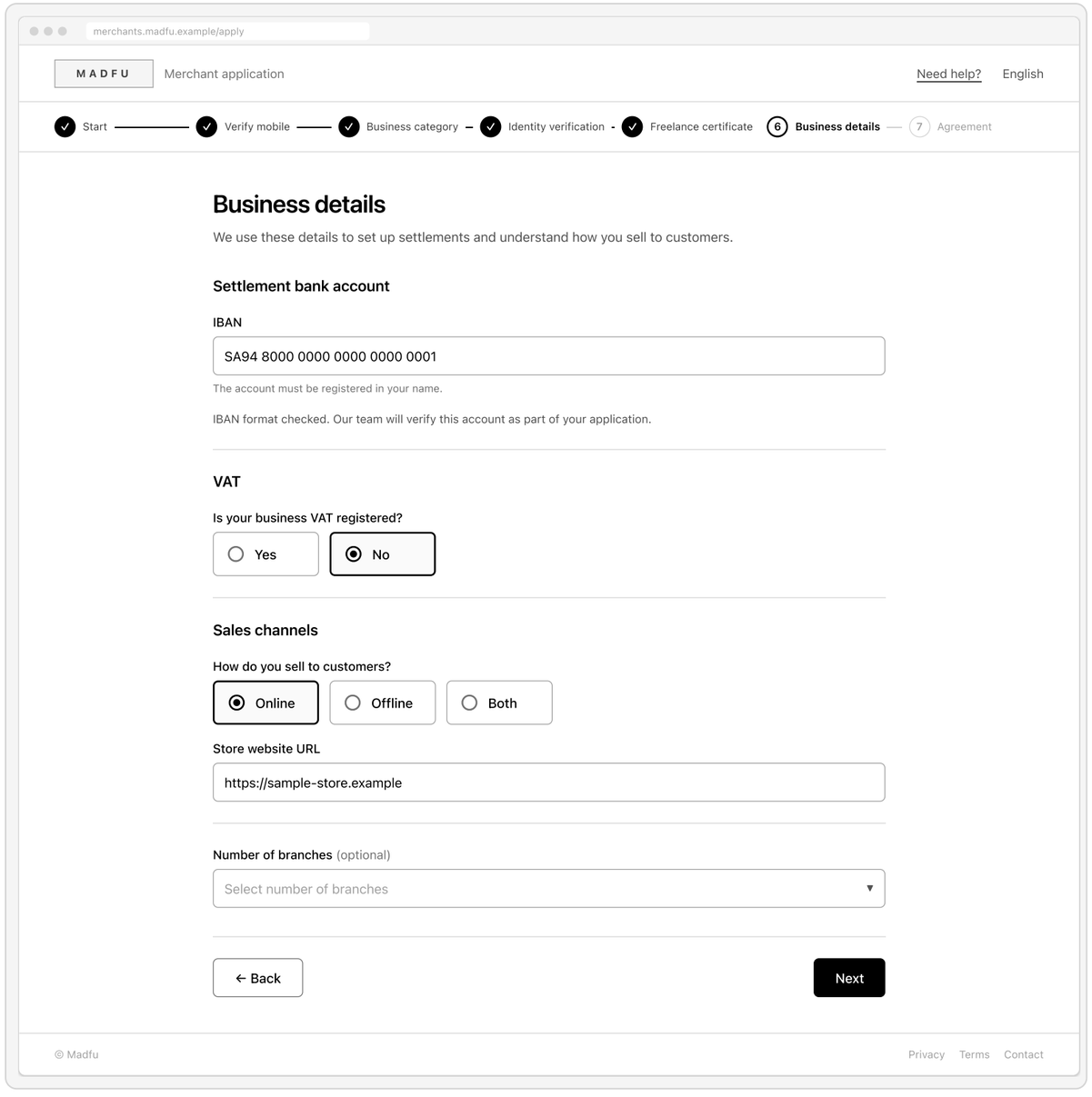
S08 IBAN provider outage — pending (not a failure)

Business details (shared) · main page p. 26



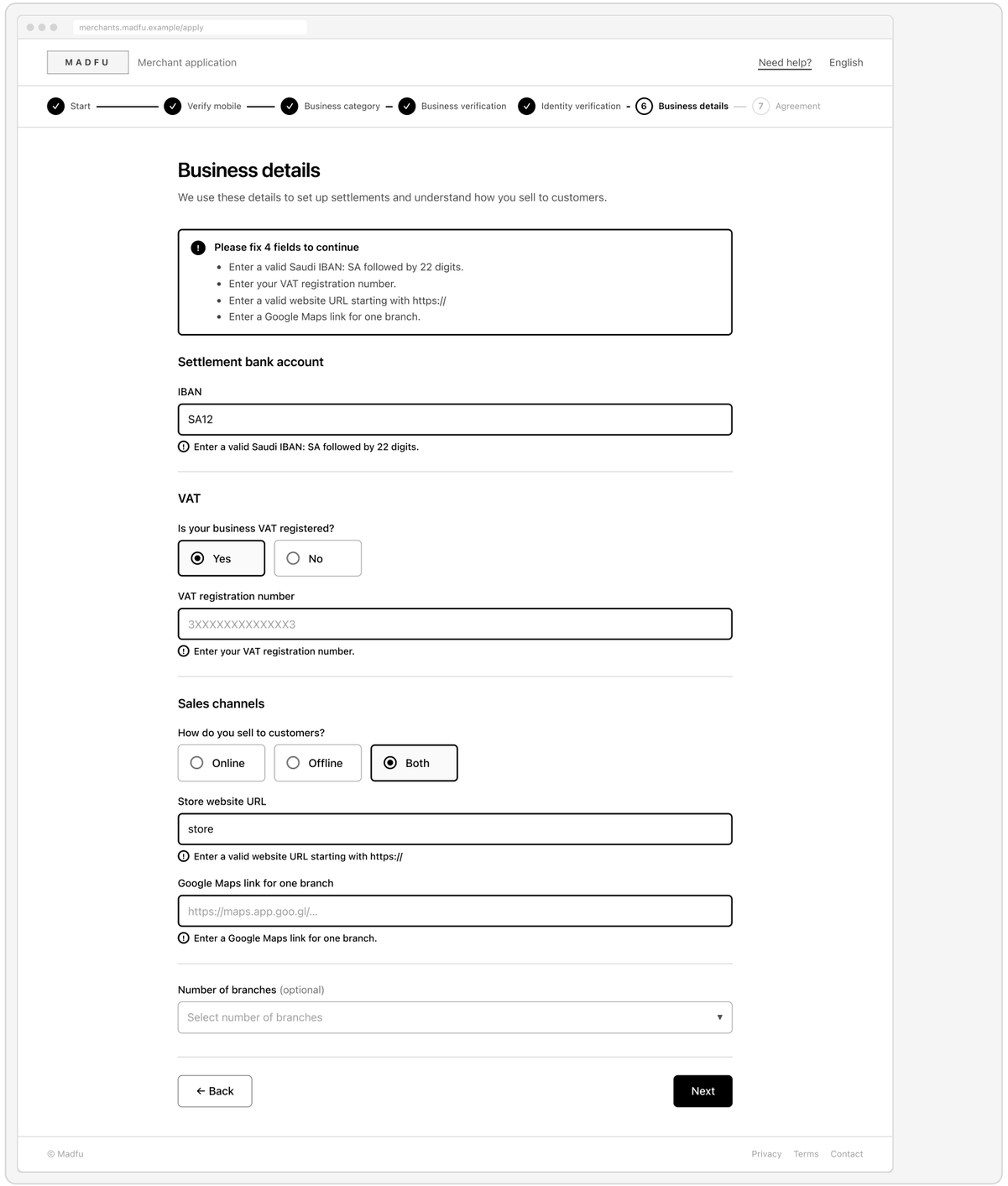
S08 Freelancer, Nafath unverified — IBAN collected only

Business details (shared) · main page p. 26



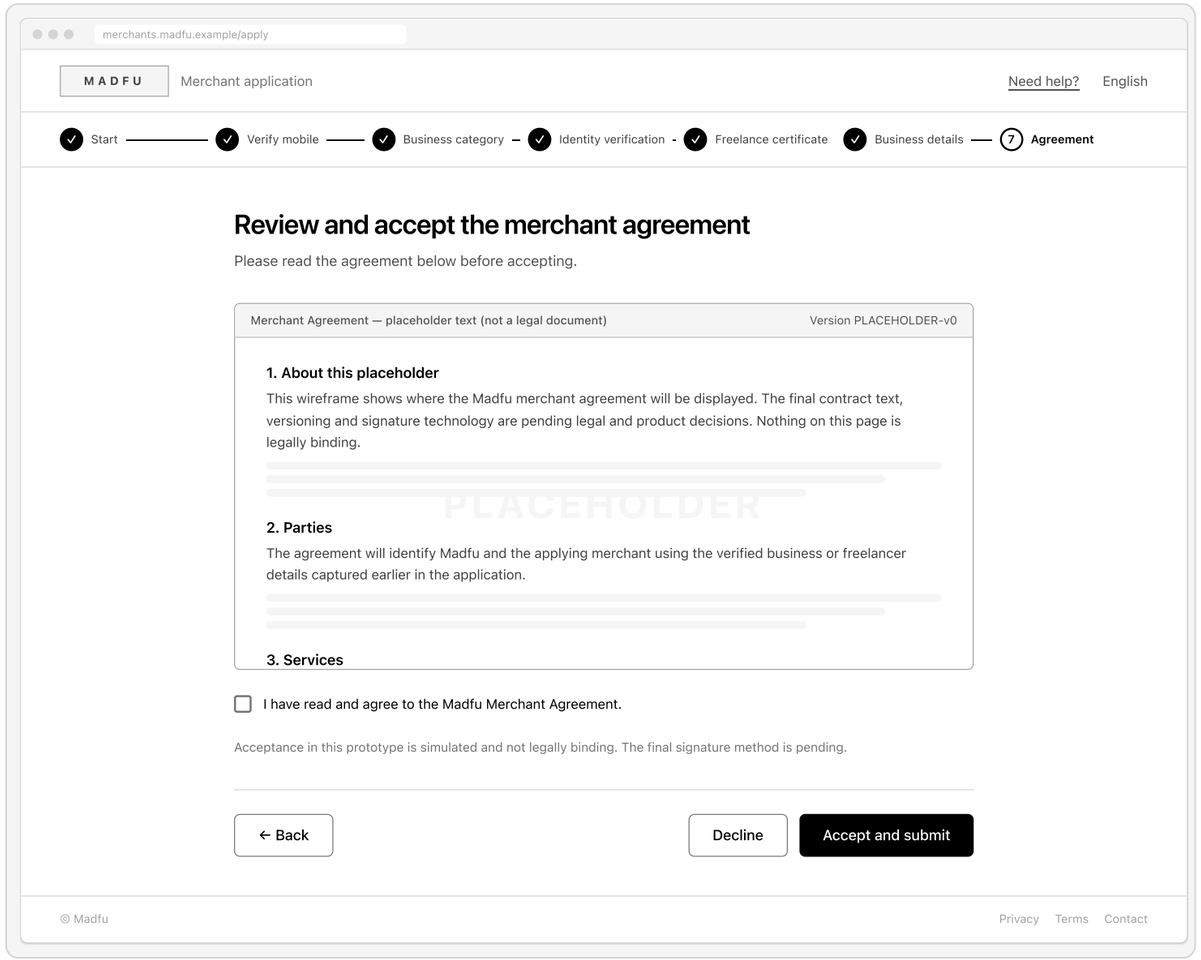
S08 Validation errors

Business details (shared) · main page p. 26



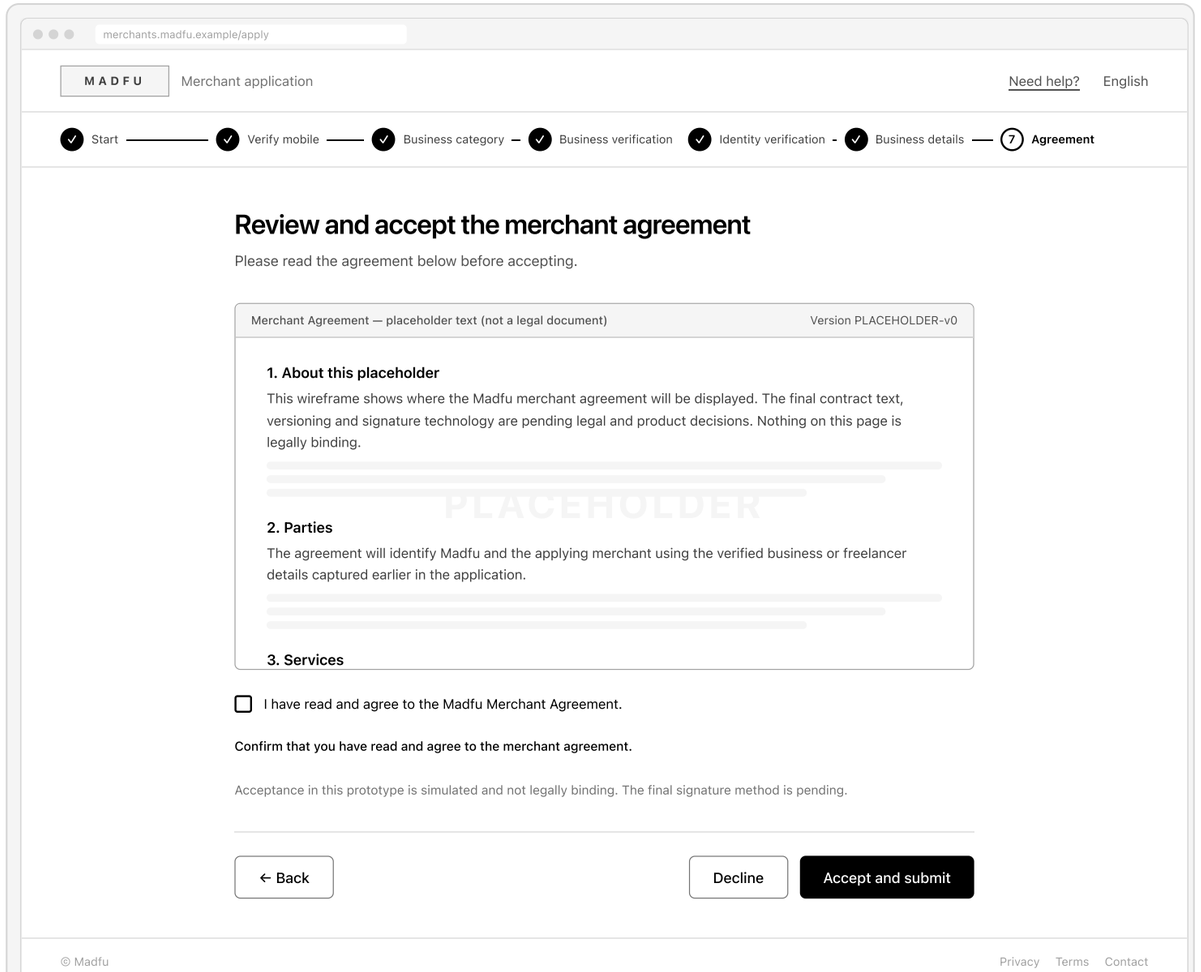
S09 Agreement — Freelancer

Merchant agreement (contract) · main page p. 27



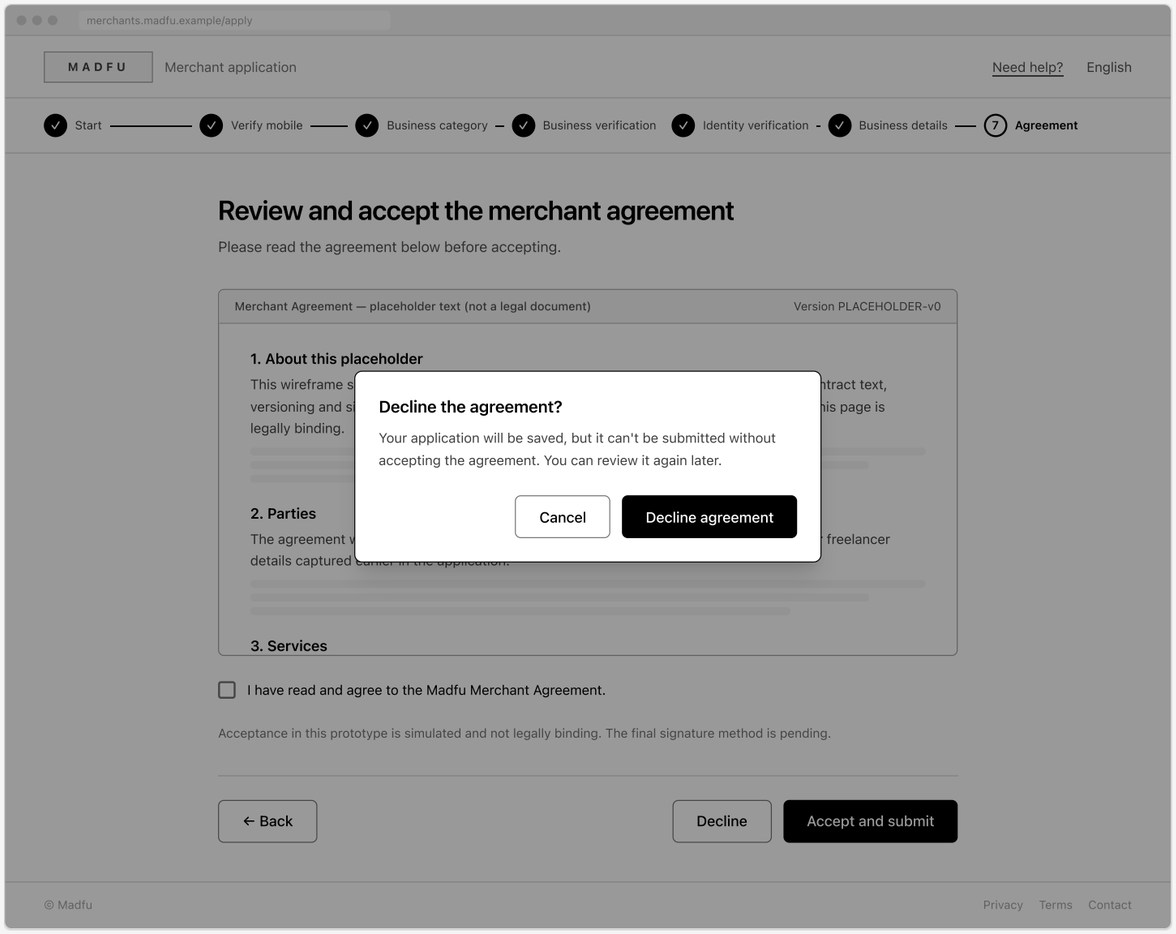
S09 Acceptance without confirmation

Merchant agreement (contract) · main page p. 27



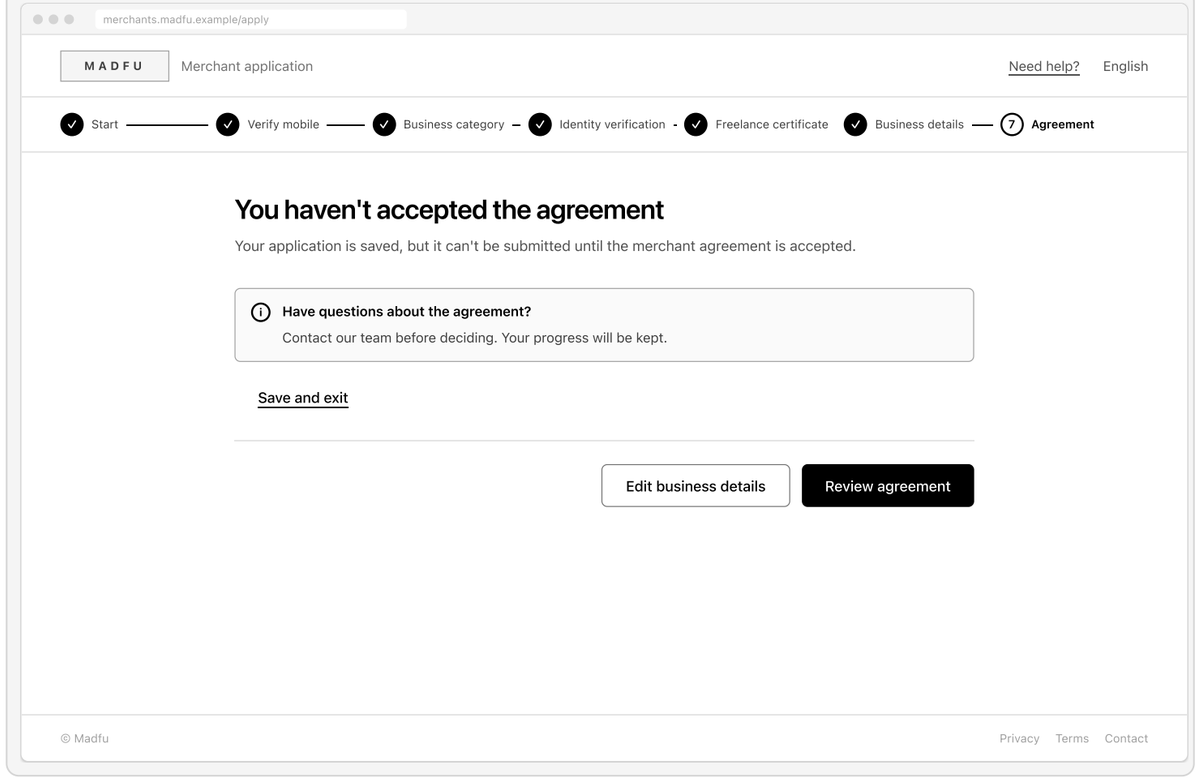
S09 Decline confirmation

Merchant agreement (contract) · main page p. 27



S09-D Agreement declined — Freelancer

Agreement not accepted · main page p. 28



R1 · Revenue routing (after OTP)

Exactly SAR 100,000 and SAR 20,000,000 fall in the middle band (provisional D-01). Configurable: REVENUE_BANDS / REVENUE_ROUTING.

Annual revenue	Numeric range	Registered Business	Freelancer
Below SAR 100,000	< 100,000	Sales lead → O-C	Self-onboarding → S03
SAR 100,000 to SAR 20,000,000	≥ 100,000 and ≤ 20,000,000	Self-onboarding → S03	Self-onboarding → S03
Above SAR 20,000,000	> 20,000,000	Sales lead → O-C	Self-onboarding → S03

R2 · Existing-account detection

Check 1 (identity) always runs first. If the identity exists the application stops and Check 2 (mobile) is skipped. Mobile reuse is informational only.

Example	Result	Reason
Mobile A + CR 1	Allowed	New identity
Mobile A + CR 2	Allowed	Different identity; mobile reuse permitted
Mobile A + National ID 1	Allowed	Different identity type
Same CR again	Not allowed → S01-X	Identity already registered
Same National ID again	Not allowed → S01-X	Identity already registered (freelancer: add-certificate is a future flow)

R3 · Frontend structural validation (configurable, not government validation)

Field	Rule (placeholder)	Message
Commercial Registration Number (CR)	`/^\d{10}$/`	Enter a valid Commercial Registration number (10 digits).
National ID Number	`/^[12]\d{9}$/`	Enter a valid National ID number (10 digits).
Mobile number	`+966 /^0?5\d{8}$/`	Enter a valid Saudi mobile number, e.g. 5X XXX XXXX.
IBAN	SA + 22 digits + ISO 13616 mod-97	Enter a valid Saudi IBAN / This IBAN is not valid
OTP	6 digits	Enter the 6-digit code.

R4 · OTP policy

Values other than the 3-attempt maximum are provisional (D-04).

Rule	Value	Behaviour
Max incorrect attempts	3	Third incorrect code → S02-L locked
Code expiry	120 s	Expired code: message + resend; does not consume an attempt
Resend cooldown	30 s	Resend link disabled with countdown
Max resends	3	Further requests blocked with support message
Resend resets attempts	No	

R5 · Category catalogue and eligibility (FIXTURE — not Madfu's real list)

Category API shape: category ID/name, subcategory ID/name, eligibility, permitted applicant types, manual-review requirement.

Category	Subcategory	ID	Eligibility	Permitted types	Manual review	Behaviour
C10 Retail	Fashion & apparel	C10-01	ALLOWED	CR, FREELANCER	No	Continue
C10 Retail	Electronics & accessories	C10-02	ALLOWED	CR, FREELANCER	No	Continue
C10 Retail	Home & furniture	C10-03	ALLOWED	CR, FREELANCER	No	Continue
C20 Health & beauty	Cosmetics & personal care	C20-01	ALLOWED	CR, FREELANCER	No	Continue
C20 Health & beauty	Medical & dental clinics	C20-02	CR_ONLY	CR	No	CR: continue · Freelancer: continue + flag
C30 Education & training	Private tutoring	C30-01	ALLOWED	CR, FREELANCER	No	Continue
C30 Education & training	Licensed training centres	C30-02	CR_ONLY	CR	No	CR: continue · Freelancer: continue + flag
C40 Professional & creative services	Graphic & digital design	C40-01	ALLOWED	CR, FREELANCER	No	Continue
C40 Professional & creative services	Photography & videography	C40-02	ALLOWED	CR, FREELANCER	No	Continue
C40 Professional & creative services	Business consulting	C40-03	ALLOWED	CR, FREELANCER	No	Continue
C50 Events & hospitality	Event planning	C50-01	ALLOWED	CR, FREELANCER	No	Continue
C50 Events & hospitality	Fixture — subcategory requiring business review	C50-02	MANUAL_REVIEW	CR, FREELANCER	Yes	Stop → review lead (O-C) before paid checks
C90 Fixture — restricted demo category	Fixture — restricted subcategory A	C90-01	BLOCKED	CR, FREELANCER	No	Stop → lead (O-C) before paid checks
C90 Fixture — restricted demo category	Fixture — restricted subcategory B	C90-02	BLOCKED	CR, FREELANCER	No	Stop → lead (O-C) before paid checks

R6 · Certificate activity → Madfu category mapping (FIXTURE)

Government and Madfu taxonomies do not share IDs. Restricted mapping after Nafath → lead with cost recorded; mismatch → flag, preliminary selection preserved.

Certificate activity code	Madfu subcategory
FL-DES-01	C40-01
FL-PHO-01	C40-02
FL-CON-01	C40-03
FL-EDU-01	C30-01
FL-FAS-01	C10-01
FL-RST-01	C90-01
FL-UNK-99	Unmapped → CATEGORY_MISMATCH

R7 · Government CR verification outcomes

Automatic retries: 2; manual retries: 3; then save-and-exit (D-08). Authorized-person data is never fabricated.

Result	Applicant sees	Internal	Next
Valid	Business found + details	—	S05
Expired	Expired notice, may continue	Flag CR_EXPIRED	S05
Not found / invalid	Correction state	—	Edit CR → S01
Temporarily unavailable	Recovery state, progress saved	Retry counters	Retry → S04 / save & exit
Provider error	Generic error + retry + support	Retry counters	Retry → S04

R8 · Nafath attempts

Max 2 attempts per application, shared across individuals. Technical failure consumes an attempt: no (max 3, D-11).

Situation	Registered Business	Freelancer
Verified	S08 Business details	S07 Certificate selection
1st unsuccessful (rejected / timed out)	Retry same person, or choose another person (counter kept)	Retry
2nd unsuccessful	Manual verification; continue to S08; contract disabled	S07-M manual certificate → S08 (IBAN without API) → O-B
Technical failure	Retry without using an attempt	Retry without using an attempt
No eligible individual on CR	Manual verification; continue to S08; contract disabled	—

R9 · IBAN ownership verification

Ownership API runs when the field is left, never twice for an unchanged value. Max 2 unsuccessful checks; 3 technical errors → pending (D-10). Format validation ≠ ownership.

Status	Applicant can continue?	Internal	Contract impact
Verified	Yes	—	None
Not verified (1st)	No — correct or retry	Attempt 1 recorded	—
Not verified (2nd)	Yes	IBAN_UNVERIFIED_MANUAL_FINANCIAL_REVIEW	Allowed if Nafath verified
Technical error (< 3)	No — retry	Technical counter	—
Technical error (3)	Yes	IBAN_VERIFICATION_PENDING_PROVIDER (separate from failure)	Allowed subject to processing rules
Nafath unverified	Yes (format only)	IBAN_NOT_CHECKED_NAFATH_UNVERIFIED	Contract not available anyway
Invalid format	No — must be corrected	—	—

R10 · Contract eligibility matrix

Additional provisional blocking flags (D-06/D-16): CERT_EXPIRED, CERT_INCONSISTENT, CERT_NOT_FOUND_MANUAL_ENTRY, CERT_MANUAL_ENTRY_UNVERIFIED, NO_ELIGIBLE_AUTHORISED_PERSON.

Nafath	IBAN ownership	Contract
Verified	Verified	Allowed
Verified	Failed twice	Allowed
Verified	Pending / technical issue	Allowed subject to required processing rules
Not verified	Any status	Not allowed → O-B
Verified + blocking review flag	Any status	Not allowed → O-B (provisional)

R11 · Internal statuses vs applicant-facing messages

Internal classifications are never shown to applicants.

Internal status	Set when	Applicant sees
SALES_LEAD	Revenue lead, restricted / review category, restricted certificate	O-C “Thank you for your interest in Madfu. Our team will contact you.”
CATEGORY_REVIEW_REQUIRED	Restricted / review / CR-only (freelancer) / mismatch	No change in copy
NAFATH_MANUAL_VERIFICATION	2 unsuccessful attempts or no eligible individual	O-B “Thank you. Your application has been received…”
IBAN_VERIFICATION_PENDING	Failed twice, provider pending, or not checked	Neutral inline note
CR_EXPIRED	CR lookup returned expired	Expired notice on S04
CERTIFICATE_REVIEW_REQUIRED	Expired / inconsistent / manual certificate	No change in copy
CONTRACT_PENDING	Contract offered or declined	S09 / S09-D
CONTRACT_ACCEPTED	Simulated acceptance	O-A
APPLICATION_SUBMITTED	Outcome A or B	O-A / O-B

R12 · Integration adapters (all simulated)

Typed interfaces in src/mocks/services.ts. Real API behaviour is not proven by this prototype.

Adapter	Operation	Simulated outcomes	Paid call (cost ledger)
IdentityRegistryService	identityExists · mobileLinkedAccounts · certificateAttached	exists / new; linked count	No
OtpService	send	code sent (verification simulated locally)	No
GovernmentCrService	lookup · relatedParties	valid · expired · not_found · unavailable · provider_error	Yes (lookup)
CertificateService	listByNationalId	0–3 certificates · unavailable	Yes
NafathService	createRequest · status	verified · rejected · timeout · technical_failure	Yes
IbanOwnershipService	verify	verified · not_verified · technical_error	Yes
CategoryService	catalogue	fixture catalogue	No
ContractService	document · recordAcceptance	placeholder document; simulated acceptance	No

G-UX · GLOBAL UX/UI HANDOVER NOTE — APPLIES TO EVERY SCREEN

All screens require complete UX/UI validation, error-state, empty-state, loading-state, retry, and edge-case designs. The absence of a specific state from the wireframes does not mean it is excluded from the final product requirements. The UX/UI team is responsible for designing comprehensive error and validation experiences for every field, step, and integration.

H4 · Design decisions and open UX questions

- Desktop-first (1440 px); single-column form (max 760 px) under a 7-step progress indicator that adapts to applicant type.
- Back is always bottom-left (secondary); the primary action is bottom-right. Browser Back maps to the same allowed transition.
- Errors: inline per field plus an error summary when two or more fields fail. Error states use weight and borders, not colour.
- Verification results use neutral language; internal classifications (restricted, manual review, flags) are never shown.
- Nafath uses a dedicated screen with a 192 px circular number and numbered instructions; remaining attempts shown after failures.
- Development simulator strips are hatched and labelled “prototype only” — they are not part of the product UI.
- Masking: names (first + last visible), National IDs (first 3 + last 3), mobile (last 3).
- Outcome pages share one layout and never state approval or activation.

ID	Type	Topic	Statement	Prototype behaviour	Owner
D-15	UX/UI decision needed	Resume authentication	Production resume should re-authenticate (e.g. OTP) before showing a saved draft.	Browser-storage resume without re-authentication (demo only).	UX / Security
D-18	UX/UI decision needed	Category picker	Real catalogue size may require a searchable / grouped picker instead of two dropdowns.	—	UX
D-19	UX/UI decision needed	Nafath presentation	Full screen vs modal for the Nafath number; behaviour on mobile web.	Dedicated screen with a very large circular number.	UX

H1 · Screen inventory

Screen IDs match the interactive prototype, the gallery and the journey maps.

ID	Screen	Audience	Entry condition	Primary action	Next	States in gallery
S00	Resume application	Both	Saved application exists, not submitted; applicant opens the onboarding entry point.	Continue application	The last active step (e.g. S08 Business details).	1
S01	Start application (shared entry)	Both	Applicant opens onboarding, or returns via "Edit details" / "Edit mobile number".	Next	S02 OTP verification (after backend duplicate checks).	6
S01-X	Existing account found	Both	Duplicate check 1 (identity) returned "already registered".	Sign in to your account (outside prototype scope)	Existing-account sign-in (not designed here).	2
S01-F	Existing freelancer — add another certificate (future flow)	Freelancer	S01-X for a Freelancer → "Add another freelancing certificate".	None (placeholder)	To be handled inside the existing account — pending business specification.	1
S02	Mobile OTP verification	Both	S01 submitted, identity not already registered, OTP sent.	Verify	Revenue routing → S03 Category (self-onboarding) or O-C (CR sales lead).	4
S02-L	Mobile verification locked	Both	Third incorrect OTP.	Start again	S01 with a new application (demo).	1
S03	Business category	Both	OTP verified and revenue routing = self-onboarding.	Next	CR → S04 CR verification. Freelancer → S06 Nafath.	3
S04	Business verification (CR lookup)	Registered Business	Category decision = continue.	Continue	S05 Authorized person selection.	7
S05	Authorized person selection	Registered Business	S04 result valid or expired.	Continue (disabled until a selection is made)	S06 Nafath verification.	6
S06	Nafath identity verification	Both	CR: person selected (S05). Freelancer: category decision = continue.	Send verification request → (wait) → Continue	Verified: CR → S08; Freelancer → S07 certificate selection.	9
S07	Freelancing certificate selection	Freelancer	Nafath verified.	Continue	S08 Business details.	6
S07-M	Manual certificate entry	Freelancer	Nafath unverified (2 failures) or "Enter certificate manually" from S07.	Continue	S08 Business details (IBAN collected without ownership check when Nafath failed).	2
S08	Business details (shared)	Both	CR: Nafath finished (verified or exhausted) or no eligible person. Freelancer: certificate selected or entered.	Next	Contract allowed → S09. Not allowed (Nafath unverified / blocking review flag) → O-B.	10
S09	Merchant agreement (contract)	Both	S08 submitted and contract eligibility = allowed.	Accept and submit (simulated)	O-A Application submitted.	4
S09-D	Agreement not accepted	Both	Decline confirmed on S09.	Review agreement	S09 Contract.	2
O-A	Outcome A — Application submitted	Both	Contract accepted.	None (final state)	End of journey.	1
O-B	Outcome B — Application received (manual verification)	Both	S08 submitted and contract not allowed.	None (final state)	Manual verification by Madfu team.	1
O-C	Outcome C — Interest received (sales lead)	Both	Revenue routing = sales lead, or category decision = stop.	None (final state)	Sales / business review team contacts the applicant.	1

H2 · Field behaviour

Screen	Field	Required	Behaviour / validation	Error copy (draft)
S01	Applicant type	Yes	Segmented choice; switching resets ID and dependent verifications	Select an applicant type.
S01	CR number / National ID	Yes	Label + rule switch with type; digits only; read-only after Nafath starts	Enter a valid … number (10 digits).
S01	Annual revenue	Yes	3 bands, no gap / overlap	Select your annual revenue.
S01	Mobile	Yes	+966 default; accepts 5X… or 05X…; change resets OTP	Enter a valid Saudi mobile number.
S02	OTP code	Yes	6 digits; 3 wrong → lock; expiry not counted	The code you entered is incorrect. N attempts remaining.
S03	Category / subcategory	Yes	Subcategory enabled after category; classification hidden from user	Select a subcategory.
S05	Authorized person	Yes	Single select; Continue disabled until chosen	—
S07	Certificate	Yes	Single select; attached certificates disabled	Select one freelancing certificate.
S07-M	Certificate number	Yes	Structural pattern; duplicate check against existing accounts	Enter a valid freelancing certificate number.
S08	IBAN	Yes	Format (mod-97) on submit; ownership check on blur; editing invalidates result	Enter a valid Saudi IBAN.
S08	VAT registered + number	Yes / if Yes	Number shown only when Yes	Enter your VAT registration number.
S08	Sales channel + URLs	Yes	Online → website; Offline → Google Maps; Both → both	Enter a valid website URL / Google Maps link.
S08	Number of branches	No	Optional; internal prioritisation only	—
S09	Agreement confirmation	Yes	Required before simulated acceptance	Confirm that you have read and agree…

H3 · Required state designs per integration

Minimum set the UX/UI team must design. See G-UX: absence from the wireframes does not mean exclusion.

Step	Loading	Empty	Error / failure	Retry / recovery	Edge cases
Duplicate checks (S01)	Button spinner	—	Existing account	Edit details	Mobile reused (no block)
OTP (S02)	Sending / verifying	—	Incorrect, expired, locked	Resend with cooldown	Resend limit, edit mobile
CR lookup (S04)	Retrieving CR	—	Not found, provider error	Auto + manual retry, save & exit	Expired CR
Authorized persons (S05)	Loading list	No eligible individual	Data unavailable	Try again	One person, entities filtered, person changed
Nafath (S06)	Sending request, waiting	—	Rejected, timed out, technical	Retry / change person	Attempts exhausted, refresh while waiting
Certificates (S07)	Retrieving	None found	Service unavailable	Try again / manual entry	Expired, inconsistent, attached, mismatch
IBAN (S08)	Verifying	—	Not verified, technical error	Retry verification	Failed twice, provider outage, not checked
Contract (S09)	Recording acceptance	—	Confirmation missing	Review again after decline	Decline confirmation

Each scenario is loaded from Prototype Controls and walked through the real screens by the automated Playwright suite (see docs/QA_REPORT.md for the latest results).

#	Scenario	What it demonstrates	Expected end
1	CR — Self-onboarding, all verified	Happy path: CR in revenue range, allowed category, valid CR, Nafath verified, IBAN verified, contract accepted.	O-A
2	CR — Below-revenue sales lead	Revenue below SAR 100,000 → lead captured after OTP; no self-onboarding steps.	O-C
3	CR — Above-revenue sales lead	Revenue above SAR 20,000,000 → lead captured after OTP.	O-C
4	CR — Duplicate account	CR 1010000099 is already registered → existing-account screen; mobile check skipped.	S01-X
5	CR — Blocked category	Restricted fixture subcategory → lead saved before any CR lookup or Nafath cost.	O-C
6	CR — Expired registration	CR lookup returns EXPIRED → applicant may continue; internal CR_EXPIRED flag.	O-A
7	CR — Nafath rejected twice	First person rejects; reviewer switches to another person (counter preserved); second rejection → manual verification, no contract.	O-B
8	CR — Nafath timeout twice	Both Nafath requests time out → manual verification, contract disabled.	O-B
9	CR — Nafath verified, IBAN failed twice	IBAN ownership fails twice → continue with manual financial verification; contract still allowed.	O-A
10	CR — No individual authorized persons	CR lists only companies/entities → no Nafath candidate → manual-review path, contract disabled.	O-B
11	Freelancer — All verified	Freelancer happy path: Nafath verified, one certificate matching the selected category, IBAN verified, contract accepted.	O-A
12	Freelancer — Multiple certificates	Three certificates returned; exactly one must be selected.	O-A
13	Freelancer — Duplicate National ID	National ID 1000000099 already registered → existing-account screen with the future “add certificate” branch.	S01-X
14	Freelancer — Nafath failed twice	Rejected then timed out → manual certificate entry → IBAN collected without ownership API → Outcome B.	O-B
15	Freelancer — Manual certificate entry	Manual entry screen validation: invalid format, a certificate already attached to another account, then a valid number.	O-B
16	Freelancer — Blocked category	Restricted fixture subcategory → lead before Nafath.	O-C
17	Freelancer — CR-only category flag	Freelancer selects a CR-only subcategory → not rejected; flagged and continues (certificate then also flags a category mismatch).	O-A
18	Freelancer — IBAN verification failed twice	IBAN not matched to the National ID twice → continue; contract still allowed (Nafath verified).	O-A
19	Any applicant — OTP errors	Incorrect code, expired code, resend, and lockout after 3 incorrect attempts.	S02-L
20	Any applicant — API failure and retry	CR service unavailable through 2 automatic retries → recovery state → manual retry succeeds; related-party data fails once.	O-A
21	Any applicant — Resume incomplete application	A saved application at Business details is resumed from the welcome-back screen; values are preserved after refresh.	O-A
22	Eligible applicant — Contract declined	Eligible applicant declines the agreement → saved, not submitted; can review again.	S09-D
23	Freelancer — No certificates returned	Nafath verified but no certificate found → manual entry → manual review (contract disabled).	O-B
24	Freelancer — Certificate/category mismatch	Preliminary category Retail/Fashion vs certificate Photography → mismatch flagged, preliminary selection preserved, journey continues.	O-A
25	Freelancer — Certificate maps to restricted category	Eligibility problem found after Nafath: cost already incurred is recorded, no further paid checks, lead outcome.	O-C
26	Any applicant — IBAN provider outage	Three technical errors → modelled as “pending”, separate from a failed ownership check; contract allowed.	O-A
27	Any applicant — Nafath technical failure then verified	Nafath request cannot be created (no attempt used) → retry succeeds.	O-A
28	Any applicant — Category pending business review	Subcategory classified “pending manual business review” → saved as review lead before paid checks (configurable).	O-C
29	Any applicant — Mobile reused across identities	Mobile already linked to 2 other accounts → allowed (informational flag only).	O-A

ID	Type	Topic	Statement	Prototype behaviour	Owner
C-01	Confirmed requirement	Applicant types	Two applicant types (Registered Business, Freelancer) share one entry page; ID field switches with type; certificate number is not requested at entry.	—	—
C-02	Confirmed requirement	Duplicate order	Identity (CR / National ID) is checked first; if it exists the application stops before the mobile check. Mobile reuse across different identities is allowed.	—	—
C-03	Confirmed requirement	OTP	Maximum 3 incorrect OTP attempts; real codes are never displayed in normal UX.	—	—
C-04	Confirmed requirement	Revenue routing	CR below SAR 100K or above SAR 20M → sales lead after OTP. CR within range → self-onboarding. Freelancers → self-onboarding for all bands.	—	—
C-05	Confirmed requirement	Category before paid checks	Madfu category/subcategory is selected immediately after OTP, before CR retrieval and Nafath. Restricted categories stop self-onboarding without paid calls.	—	—
C-06	Confirmed requirement	CR-only for freelancer	A freelancer choosing a CR-only category is flagged, not rejected; final eligibility is a business decision.	—	—
C-07	Confirmed requirement	Authorized persons	Only individuals are shown; one selection; name (first/last visible) and ID (first 3/last 3 visible) masked.	—	—
C-08	Confirmed requirement	Nafath attempts	Maximum 2 Nafath attempts per application, shared across individuals; changing person does not reset the counter.	—	—
C-09	Confirmed requirement	IBAN	Max 2 unsuccessful ownership checks; after that the applicant continues and the IBAN is queued for manual financial verification. IBAN failure never blocks the contract when Nafath succeeded.	—	—
C-10	Confirmed requirement	Contract eligibility	Successful Nafath is the deciding factor; without it the contract is not available.	—	—
C-11	Confirmed requirement	Freelancer without Nafath	After 2 unsuccessful attempts: manual certificate number, business details, IBAN without ownership API, no contract, manual verification.	—	—
C-12	Confirmed requirement	User-facing outcomes	Three neutral outcomes (submitted / received / interest received); internal statuses are never disclosed.	—	—

ID	Type	Topic	Statement	Prototype behaviour	Owner
D-01	Provisional assumption	SAR 100,000 boundary	Exactly SAR 100,000 is treated as self-onboarding (middle band). Copy: “Below SAR 100,000 / SAR 100,000 to SAR 20,000,000 / Above SAR 20,000,000” — no gap or overlap.	Configurable in rules/config.ts (REVENUE_BANDS, REVENUE_ROUTING).	Product / Risk
D-02	Product decision needed	Contract signing technology	Final signature/acceptance technology (click-wrap, e-signature provider, Nafath-signing) is not decided.	Generic checkbox + “Accept and submit” placeholder; acceptance simulated and not legally binding.	Legal / Product
D-04	Product decision needed	OTP resend & expiry	Resend cooldown, max resends, code expiry, lockout duration and whether resends reset the attempt counter.	Expiry 120 s, cooldown 30 s, 3 resends, resend does not reset attempts; lockout requires a new application.	Product / Security
D-05	Product decision needed	Category restriction catalogue	Madfu’s actual categories, restricted list, CR-only list and review list are not provided.	Clearly labelled fixture catalogue in rules/categories.ts.	Risk / Compliance
D-06	Product decision needed	Certificate / category reconciliation	Mapping between government freelancing activities and Madfu categories, and treatment of expired / inconsistent certificates.	Fixture mapping table; mismatches flagged (journey continues); expired / inconsistent certificates flagged and contract disabled (manual review).	Risk / Product
D-07	Product decision needed	Existing freelancer — add certificate	How an existing freelancer adds another certificate inside their account.	Documented branch from S01-X to S01-F placeholder marked FUTURE FLOW. No implementation.	Product
D-08	Product decision needed	API failures after retries	What happens when the CR service stays unavailable after retries (notify, auto-resume, convert to lead?).	2 silent auto-retries, then 3 manual retries, then “save and exit”; resuming grants a fresh retry allowance.	Product / Engineering
D-09	Product decision needed	Contract decline handling	Follow-up after a decline (reminders, expiry of the application, sales contact).	Application saved, not submitted, status CONTRACT_PENDING with CONTRACT_DECLINED flag; applicant can review again.	Product / Sales
D-10	Product decision needed	IBAN technical errors	A persistent provider outage is not a verified failure. Policy needed for whether contract processing waits for verification.	3 technical errors → status “pending (provider)”, separate flag; contract allowed subject to processing rules.	Finance / Product
D-11	Provisional assumption	Nafath technical failure	A request that could not be created does not consume one of the 2 attempts; bounded at 3 technical failures, then manual verification.	NAFATH_POLICY.technicalFailureConsumesAttempt = false (configurable).	Product / Risk
D-12	Product decision needed	IBAN check for CR without Nafath	The brief specifies no ownership API for freelancers without Nafath; CR without Nafath is not specified.	Not called (cost avoidance; application goes to manual review). Configurable per applicant type.	Product / Finance
D-13	Provisional assumption	“Pending manual review” category	Whether a review-required category stops before paid checks or continues with a flag.	Stops and saves a review lead (Outcome C) before any paid call. Configurable: STOP_AS_LEAD | CONTINUE_WITH_FLAG.	Risk
D-14	Provisional assumption	Identity lock	Applicant type and identification cannot be edited once a Nafath request has been sent (prevents resetting the shared counter).	Fields read-only; a new application is required. Abuse via restarting needs a server-side control.	Product / Risk
D-16	Provisional assumption	Certificate not found after Nafath	A verified freelancer with no certificate may enter a number manually; treated as manual review with contract disabled.	Flag CERT_NOT_FOUND_MANUAL_ENTRY (blocking).	Product / Risk
D-17	Provisional assumption	Lead resubmission	Submitted applications (A/B) register the identity so a repeat application hits the duplicate check; leads (C) do not.	Demo registry in browser storage (clearable from Prototype Controls).	Sales / Product
D-23	Product decision needed	VAT number rule	Exact VAT number validation and whether VAT certificate upload is required.	15 digits starting and ending with 3 (structural placeholder).	Finance / Compliance

ID	Type	Topic	Statement	Prototype behaviour	Owner
D-03	Engineering integration question	Government validation patterns	Exact CR and National ID format rules and the government API contracts are not confirmed.	Structural regex placeholders (10 digits) kept separate from simulated government verification.	Engineering
D-15	UX/UI decision needed	Resume authentication	Production resume should re-authenticate (e.g. OTP) before showing a saved draft.	Browser-storage resume without re-authentication (demo only).	UX / Security
D-18	UX/UI decision needed	Category picker	Real catalogue size may require a searchable / grouped picker instead of two dropdowns.	—	UX
D-19	UX/UI decision needed	Nafath presentation	Full screen vs modal for the Nafath number; behaviour on mobile web.	Dedicated screen with a very large circular number.	UX
D-20	Engineering integration question	Nafath status delivery	Polling vs callback for Nafath status, request expiry duration, and idempotency on refresh.	Waiting state persists across refresh; resolution simulated by timer or dev simulator.	Engineering
D-21	Engineering integration question	Related-party data	Whether owners/managers come from the same CR call or a separate endpoint, and how entity types are identified.	Separate mock call; entity type field drives the individuals-only filter.	Engineering
D-22	Engineering integration question	IBAN ownership provider	Provider, matching rules (exact / fuzzy name), and cost per call.	Typed adapter returns verified / not_verified / technical_error.	Engineering / Finance